

A CRITICAL COMPILATION

Low-Energy Nuclear Reactions

What the peer-reviewed record actually says about nuclear anomalies in metal–hydrogen systems — the 1989 crisis, the strongest evidence, the modern reassessment, the theoretical gap, and where a breakthrough would plausibly come from. Something to play with in every chapter.

10 chapters · in browser · self-contained

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- I **The Yardstick: Fusion and the Coulomb Barrier** Tunneling, the Gamow factor, and the canonical D+D numbers every later claim must be measured against.
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- II **1989: Palladium, Heavy Water, and a Scientific Crisis** What Fleischmann and Pons claimed, what the replications found, and what two DOE reviews concluded.
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- IV **The Claimed Nuclear Evidence** Helium correlated with heat, tritium, transmutation, and energetic particles — claim by claim, critique by critique.
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IX **Theories and the Explanatory Gap** Screening, phonons, Widom-Larsen — every proposed mechanism, how far short it falls, and the results that outrun all of them.

X **Where the Breakthroughs May Come From** Informed speculation, clearly labeled: the materials, detection, and program designs most likely to settle the question.



Conventions used throughout: energies in eV/keV/MeV; cross sections σ in barns ($1 \text{ b} = 10^{-28} \text{ m}^2$); astrophysical factor $S(E)$ in keV · b with $\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta}$; Sommerfeld parameter η ; screening energy U_e in eV; Gamow energy E_G ; loading as the atomic ratio $x_{\text{D/Pd}}$ (D per Pd); excess power P_{xs} in watts. Nuclides are written ^4He ; "D" is the deuterium atom, "d" the deuteron. Useful constants: $e^2/4\pi\epsilon_0 = 1.44 \text{ eV} \cdot \text{nm}$; D + D Q-values 4.03 MeV (t+p), 3.27 MeV ($^3\text{He}+\text{n}$), 23.85 MeV ($^4\text{He} + \gamma$). Audience: a motivated polymath comfortable with calculus and introductory physics; every domain term is defined where it first appears. Chapters IX-X clearly separate established results, contested claims, and speculation. **Appendix: how this book was researched.**

CHAPTER I

The Yardstick: Fusion and the Coulomb Barrier

Before we can weigh any extraordinary claim, we need the ordinary physics it is claimed against — and in nuclear fusion, the ordinary physics is a wall.

BEFORE WE START

This book is about a family of claims: that nuclear reactions — fusion, transmutation, the release of nuclear-scale energy — can happen inside ordinary metals at ordinary temperatures. To judge those claims you need to know what standard physics says should happen, and how confidently it says it. That is this chapter's job. We build the yardstick: the electrostatic barrier that keeps two nuclei apart, the quantum tunneling that occasionally lets them through, and the precisely measured reaction rates and by-products of deuterium–deuterium fusion.

Two numbers from this chapter run through the whole book. The first is the *Gamow factor*, the exponential that suppresses fusion at low energy — at room temperature it suppresses it by dozens of orders of magnitude, which is why "cold fusion" sounds impossible. The second is the expected *nuclear ash*: known fusion reactions produce neutrons, tritium, and gamma rays in fixed proportions, so a watt of fusion power should announce itself unmistakably — and lethally. Every experimental claim in later chapters either fights the first number or is interrogated with the second.

TERMS YOU'LL MEET

nucleus	The tiny positively charged core of an atom, made of protons and neutrons; about 100,000 times smaller than the atom itself.
nucleon	A proton or a neutron — the two building blocks of nuclei.
nuclide	A specific nuclear species, identified by its proton count and total nucleon count, written like ${}^4\text{He}$.
isotope	Nuclides of the same element (same proton count) with different neutron counts — hydrogen, deuterium, and tritium are isotopes of hydrogen.
deuterium (D)	Heavy hydrogen: one proton plus one neutron. About one in 6,400 hydrogen atoms in seawater.
deuteron (d)	The bare nucleus of a deuterium atom.
tritium (T, t)	Hydrogen with two neutrons; radioactive, half-life 12.3 years. A key fusion by-product, and hence a key piece of forensic evidence.
fusion	Two light nuclei merging into a heavier one, releasing energy because the product is more tightly bound.
binding energy	The energy needed to pull a nucleus apart into its nucleons; equivalently, the energy released when it forms.
Q-value	The net energy released by a nuclear reaction, fixed by the mass difference between reactants and products.
electronvolt (eV)	The energy an electron gains crossing one volt: 1.6×10^{-19} joules. Chemistry lives at eV; nuclear physics at MeV — a million times more.
Coulomb barrier	The electrostatic repulsion between two positively charged nuclei that must be overcome (or tunneled through) before the strong force can grab them.
strong nuclear force	The short-range force that binds nucleons; overwhelmingly stronger than electrostatics, but

	only within a few femtometers.
femtometer (fm)	10^{-15} meters — the size scale of nuclei. Atoms are ~100,000 fm across.
quantum tunneling	The quantum-mechanical ability of a particle to pass through an energy barrier it classically could not surmount, with a probability that falls exponentially with barrier size.
Gamow factor	The exponential tunneling-suppression factor $e^{-2\pi\eta}$ in fusion cross sections, from George Gamow's 1928 tunneling theory.
Sommerfeld parameter (η)	The dimensionless number measuring how "classically forbidden" a nuclear collision is; large η means heavy suppression.
cross section (σ)	The effective target area a reaction presents, measured in barns; encodes the reaction probability per collision.
barn (b)	10^{-28} m^2 , roughly the cross-sectional area of a uranium nucleus — jokingly "as big as a barn" to a neutron.
astrophysical S-factor	The cross section with the tunneling exponential and the trivial $1/E$ divided out — the slowly varying part that captures pure nuclear physics.
branching ratio	When a reaction has several possible outcomes, the fraction going to each channel.
reduced mass	The effective mass $\mu = m_1 m_2 / (m_1 + m_2)$ that governs the relative motion of a two-body system.
center-of-mass energy	The collision energy available in the frame where total momentum is zero — the energy that matters for tunneling.
screening	The shielding of nuclear charge by nearby electrons, which effectively lowers the Coulomb barrier by an energy U_e .
muon-catalyzed fusion	Genuine room-temperature fusion in which a heavy cousin of the electron shrinks a hydrogen molecule

~200-fold; real, replicated, and instructive (Chapter IX).

neutron activation

Making surrounding material radioactive by neutron bombardment — one reason a true watt of D+D fusion is hazardous.

WKB approximation

A standard semiclassical method for computing tunneling probabilities through smooth barriers.

§1 Why fusion pays: the million-fold energy gap

Chemistry rearranges the electrons of atoms; nuclear reactions rearrange nucleons. The energy scales differ by a factor of about a million. Burning a hydrogen molecule in oxygen releases a few eV. Fusing two deuterons releases a few MeV. This gap is the entire allure of the field: a nuclear process running quietly in a jar of heavy water would carry an energy density no battery or fuel could approach.

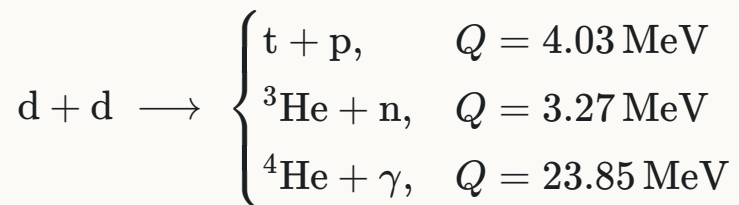
The gap is set by binding energies. The binding energy of a nucleus climbs steeply for light nuclei — the deuteron is bound by 2.22 MeV, while ${}^4\text{He}$ is bound by 28.3 MeV. Fusing light nuclei into helium therefore releases MeV-scale energy, quantified by the reaction's Q-value:

$$Q = \left(\sum_{\text{reactants}} m_i - \sum_{\text{products}} m_f \right) c^2.$$

WHERE

Q	Scalar, MeV. The energy released per reaction, appearing as kinetic energy of the products (and photons, if any).
m_i, m_f	Rest masses of each reactant and product nucleus (constants, MeV/c^2).
c	Speed of light (constant); mc^2 converts mass to energy.

For deuterium–deuterium fusion the ledger is precisely known. Two deuterons can react three ways:



WHERE

$\text{t}, \text{p}, \text{n}$	Triton (tritium nucleus), proton, neutron — the light products of the two dominant channels.
γ	A gamma-ray photon, which must carry away the energy in the third channel because a lone ${}^4\text{He}$ cannot conserve both energy and momentum.

Hold on to those three lines: they are the fingerprint of d+d fusion, and Chapter IV is essentially a forensic investigation into whether any of those fingerprints appear in LENR experiments — and in what proportions.

§2 The wall: the Coulomb barrier

The catch is that both deuterons are positively charged. Before the strong force can bind them — it reaches only a few femtometers — electrostatic repulsion must be overcome. Two point charges $Z_1 e$ and $Z_2 e$ separated by r repel with potential energy

$$V(r) = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 r} = Z_1 Z_2 \frac{1.44 \text{ eV} \cdot \text{nm}}{r},$$

WHERE

$V(r)$ Scalar field of separation r : the electrostatic potential energy of the pair.

Z_1, Z_2 Proton numbers of the two nuclei (both 1 for deuterons).

$e^2/4\pi\epsilon_0$ The Coulomb constant in convenient units: $1.44 \text{ eV} \cdot \text{nm} = 1.44 \text{ MeV} \cdot \text{fm}$.

r Separation between the nuclear centers (nm or fm).

Evaluate it where the strong force takes over, around $r \approx 4 \text{ fm}$ for two deuterons, and $V \approx 360 \text{ keV}$ — the conventional statement that the d+d Coulomb barrier is a few hundred keV high. Room temperature, for comparison, is $k_B T \approx 0.025 \text{ eV}$. The barrier towers about *ten million times* above the thermal energy of anything in a beaker. Classically, nothing happens. Ever.

WHY HOT FUSION IS HOT

Mainstream fusion attacks the barrier head-on: heat a plasma to $\sim 10^8 \text{ K}$ so that keV-scale collisions become common, or slam accelerated beams together. The Sun does it with temperature plus enormous density plus patience. The entire question of this book is whether condensed matter offers any third path.

§3 The crack in the wall: tunneling and the Gamow factor

Quantum mechanics leaves a crack. A particle's wavefunction penetrates a classically forbidden barrier, so there is a finite probability of tunneling through rather than over. George Gamow computed this for alpha decay in 1928; the same physics governs fusion. In the WKB approximation, the probability of two nuclei with center-of-mass energy E penetrating their Coulomb barrier is dominated by the exponential

$$P(E) \sim e^{-2\pi\eta}, \quad 2\pi\eta = \pi \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 \hbar} \sqrt{\frac{2\mu}{E}} = \sqrt{\frac{E_G}{E}},$$

WHERE

η The Sommerfeld parameter (dimensionless): the ratio of the Coulomb interaction strength to the collision velocity. Large η = deeply forbidden.

μ Reduced mass of the pair, $\mu = m_1 m_2 / (m_1 + m_2)$; for d+d, $\mu c^2 = 937.8$ MeV (half the deuteron mass).

E Center-of-mass collision energy (keV) — for a lattice at temperature T , of order $k_B T$.

E_G The Gamow energy (constant): $E_G = 2\mu c^2 (\pi\alpha Z_1 Z_2)^2 \approx 986$ keV for d+d, with $\alpha \approx 1/137$ the fine-structure constant.

The numbers are merciless. At a hot-fusion energy of $E = 50$ keV, $2\pi\eta = \sqrt{986/50} \approx 4.4$ and the suppression is a manageable $e^{-4.4} \approx 10^{-2}$. At $E = 1$ keV, the suppression is $e^{-31.4} \approx 10^{-14}$. And at room temperature — $E \sim 0.025$ eV — the exponent is $\sqrt{986\,000/0.025} \approx 6280$, a suppression of order

10^{-2728} . Numbers like that are why the claims of Chapter II were received the way they were.

The honest statement of the "miracle" required is best made with a real molecule. In gaseous D_2 , two deuterons sit 74 pm apart, their wavefunctions overlapping the barrier's far tail. Koonin and Nauenberg computed the resulting fusion rate: about 3×10^{-64} fusions per molecule per second — one fusion in a liter of D_2 gas would take vastly longer than the age of the universe. To produce a single watt of d+d fusion in a mole of deuterium pairs, that rate must rise to roughly 3×10^{-12} per pair per second: an enhancement of *about fifty-two orders of magnitude*. That is the yardstick.

NOTE

Tunneling is stupendously sensitive to the inputs: the rate scales exponentially in $\sqrt{\mu}$ and in the effective barrier width. Halve the separation, or substitute a heavier "electron" (Chapter IX's muons), and the rate can jump tens of orders of magnitude. This sensitivity cuts both ways — it is what makes exotic mechanisms conceivable, and what makes casual estimates worthless.

§4 Measuring the crack: cross sections and the S-factor

Fusion probability is quantified as a cross section $\sigma(E)$, in barns. Because the raw cross section plunges exponentially at low energy, nuclear physicists factor out the parts they understand — the tunneling exponential and a kinematic $1/E$ — and write

$$\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta},$$

WHERE

$\sigma(E)$ The fusion cross section at center-of-mass energy E (barns).

$S(E)$ The astrophysical S-factor (keV·b): the slowly varying residue containing the genuinely nuclear physics. For d(d,p)t, $S(0) \approx 56 \text{ keV} \cdot \text{b}$; the d(d,n) ^3He branch is nearly equal.

The S-factor formalism matters for this book for one big reason: $S(E)$ is *measured*, down to a few keV, by accelerator experiments — and it is smooth. The standard parametrizations (Bosch–Hale) reproduce the world's d+d data to a few percent. Any claim that fusion proceeds at appreciable rates in a metal at milli-eV energies is therefore a claim that something — the barrier, the screening, the effective mass, the many-body physics — differs radically from the vacuum two-body problem. It cannot hide in uncertainty about nuclear physics; the nuclear physics is nailed down. Whatever is contested in this field, it lives in the exponent, not in $S(E)$.

One measured modification is real and important: electron screening. Electrons around and between the colliding nuclei cancel part of the repulsion, which acts like a small energy boost U_e to the collision:

$$\sigma_{\text{screened}}(E) \approx \sigma_{\text{bare}}(E + U_e), \quad f(E) = \frac{\sigma_{\text{screened}}}{\sigma_{\text{bare}}} \approx \exp\left[\pi\eta \frac{U_e}{E}\right] \text{ for}$$

WHERE

U_e The screening energy (eV): tens of eV for atomic targets, and — remarkably — hundreds of eV inside some metals (Chapter VI).

$f(E)$ The enhancement factor: how many times larger the screened cross section is than the bare one at the same energy. It grows explosively as E falls toward U_e .

Screening is the one experimentally certified sense in which a metal lattice is friendlier to fusion than vacuum. Chapter VI gives it a full treatment; the widget below lets you feel its (in)adequacy directly.

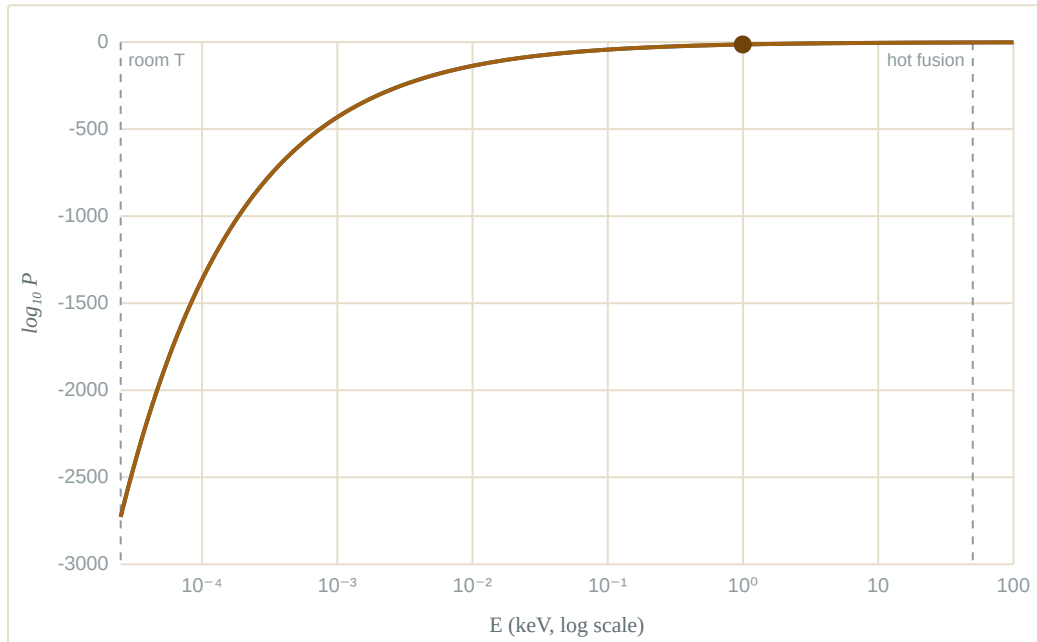
INTERACTIVE · THE GAMOW WINDOW

slide energy · add screening

collision energy E



screening U_e (eV)



$$E = 995 \text{ eV} \cdot 2\pi\eta = 31.5 \cdot P = 10^{-14} \cdot \sigma \approx 10^{-12} \text{ b}$$

$$\text{screening } U_e = 0 \text{ eV} \rightarrow \text{enhancement } f \approx 1\times$$

— bare $P(E)$ — screened $P(E+U_e)$ | room temp · hot fusion

The d+d penetration factor $e^{-2\pi\eta}$ with $2\pi\eta = \sqrt{E_G/E}$, $E_G = 986 \text{ keV}$, and the cross section $\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta}$ with $S(E) = 56 \text{ keV} \cdot \text{b}$, on a log scale from chemistry's meV world up to hot-fusion keV. Drag the energy slider and watch the exponent: between room temperature and 50 keV the probability changes by more than 2,700 orders of magnitude. The screening slider shifts the effective energy to $E + U_e$ — real, measured, and visibly unable to rescue the meV end of the axis.

§5 The expected ashes — and the dead-graduate-student problem

Below the barrier, the two dominant d+d channels fire almost equally: roughly half the fusions yield $\text{t} + \text{p}$, half yield ${}^3\text{He} + \text{n}$. The radiative channel to ${}^4\text{He} + \gamma$ is famously rare — about 10^{-7} of events — because it requires an electromagnetic transition the strong channels bypass.

Now do the bookkeeping for one watt of claimed d+d fusion power. With $\sim 3.65 \text{ MeV}$ released on average per fusion, one watt is about 1.7×10^{12} fusions per second — hence roughly 8×10^{11} neutrons per second at 2.45 MeV, a similar tritium production rate, and prompt gammas from neutron capture in everything nearby. An unshielded experimenter beside such a cell would receive a dangerous dose in minutes. This is the "dead graduate student" problem: Fleischmann and Pons remained conspicuously alive, so either their heat was not nuclear, or the nuclear physics in a lattice differs from vacuum d+d not just in *rate* but in *branching* — a second, independent miracle.

$$R_{\text{fus}} = \frac{P_{\text{xs}}}{\bar{Q}} \approx \frac{1 \text{ W}}{3.65 \text{ MeV} \times 1.6 \times 10^{-13} \text{ J/MeV}} \approx 1.7 \times 10^{12} \text{ s}^{-1}, \quad R_{\text{n}} \approx \frac{1}{2} R_{\text{fus}}$$

WHERE

R_{fus} Fusion events per second implied by an excess power P_{xs} (scalar, s^{-1}).

$\bar{Q}$ Mean energy per fusion for the 50/50 d+d branches, $\approx (4.03 + 3.27)/2 = 3.65 \text{ MeV}$.

R_{n} Expected neutron emission rate: about half of all fusions produce a 2.45 MeV neutron.

Keep this calculation loaded. It converts every wattage claim in Chapters II–VIII into a precise, testable neutron and tritium prediction, and the systematic *absence* of commensurate neutrons is the single most robust experimental fact in the entire LENR literature. Whatever the field has found, it is not vacuum-physics d+d fusion.

§6 What a lattice may and may not plead

A deuteron in palladium is not a deuteron in vacuum, and it is worth stating precisely which pleas are available to it. The lattice provides: conduction electrons (screening, measured, hundreds of eV at best — Chapter VI); confinement and high density (deuterium in Pd approaches the density of liquid hydrogen); phonons and collective modes (the basis of several theories in Chapter IX); and occasional energetic particles from cosmic rays or radioactivity (the seed NASA exploits deliberately in Chapter VII). What the lattice cannot plead is any change to $S(E)$ or to the Q-values: those are properties of the nuclei themselves.

So the honest framing of the whole controversy, and of this book, is a pair of questions with numbers attached:

Rate: can any condensed-matter mechanism raise the d+d (or p+metal) reaction rate by the ~50 orders of magnitude between molecular deuterium and a detectable signal?
Ashes: if it can, why are the products not the ones two-body physics demands? Every theory in Chapter IX is an attempt to answer both at once; every experiment in Chapters III–VIII is an attempt to force the question with data. The *Gamow factor* and the *error bar* — the suppression you must beat and the measurement you must trust — are the two numbers to keep in view on every page that follows.

AHEAD OF THEORY

Nothing in this chapter is contested — it is the yardstick itself. But note the asymmetry it creates: the theoretical prior against room-temperature fusion is so strong that *any* reproducible nuclear signature from a metal-hydride at low energy would be a result ahead of theory. That is exactly the situation claimed by the experiments of Chapters IV, VI, and VIII, and audited in Chapter IX.

FURTHER READING

Krane, K. S. *Introductory Nuclear Physics*, Wiley 1988 — binding energy, Q-values, and barrier penetration; Ch. 4 and Ch. 14 (fusion).

Clayton, D. D. *Principles of Stellar Evolution and Nucleosynthesis*, University of Chicago Press 1983 — the Gamow factor and astrophysical S-factor formalism; §4-2 to §4-4.

Bosch, H.-S. & Hale, G. M. *Improved formulas for fusion cross-sections and thermal reactivities*, Nuclear Fusion 32, 611–631 (1992) — the standard parametrization of $d(d,p)t$ and $d(d,n)^3\text{He}$ cross sections.

Angulo, C. et al. (NACRE collaboration). *A compilation of charged-particle induced thermonuclear reaction rates*, Nuclear Physics A 656, 3–183 (1999) — evaluated low-energy S-factors including $d+d$.

Koonin, S. E. & Nauenberg, M. *Calculated fusion rates in isotopic hydrogen molecules*, Nature 339, 690–691 (1989) — the canonical rate of spontaneous fusion in D_2 , the yardstick's zero point.

Huizenga, J. R. *Cold Fusion: The Scientific Fiasco of the Century*, University of Rochester Press 1992 — the branching-ratio and "missing ashes" arguments, from the co-chair of the 1989 DOE panel.

Assenbaum, H. J., Langanke, K. & Rolfs, C. *Effects of electron screening on low-energy fusion cross sections*, Zeitschrift für Physik A 327, 461–468 (1987) — the screening enhancement formula used in §4.

Chapter production note: drafted directly by the orchestrating lead model (`claude-fable-5`) as the book's conventions-anchor chapter, informed by the baseline-physics research sweep (research agent: `claude-fable-5`); all seven Further-reading citations web-verified by the orchestrating lead (Koonin–Nauenberg, Assenbaum–Langanke–Rolfs, and NACRE during drafting; the remainder in a final pass run after a mid-project model change, so the lead-verification of this chapter was performed as `claude-opus-4-8` — see the methods appendix). See the [methods appendix](#) for the full research process.

CHAPTER II

1989: Palladium, Heavy Water, and a Scientific Crisis

Two accomplished electrochemists announced, at a press conference, that they had tamed the Gamow factor in a jar of heavy water — and the year that followed taught science exactly how much a watt of certainty costs.

BEFORE WE START

On March 23, 1989, Martin Fleischmann and Stanley Pons of the University of Utah told the world's press that a palladium rod, electrolyzing heavy water on a laboratory bench, was releasing more heat than any chemistry could supply — and that the source was nuclear. Chapter I told you why that claim demanded a roughly fifty-order-of-magnitude miracle in the tunneling rate. This chapter tells you what was actually claimed, how the claim was audited, and why nearly every audit failed it. It is the founding episode of the field this book covers, and it is also the founding episode of the field's evidentiary standards: everything in Chapters III–VIII is, one way or another, a response to what went wrong in 1989.

The chapter's spine is bookkeeping. An electrolysis cell is an energy ledger: electrical power in, heat and gas out. "Excess heat" is a claim that the ledger doesn't balance — which means the claim is only as good as the accountant's error bar. We will follow the ledger's most treacherous line items (the thermoneutral potential, gas recombination), then follow the nuclear evidence as it collapsed, the replication wave as it broke, and the two U.S. Department of Energy reviews that bracketed the field's first fifteen years. In the book's running frame: Fleischmann and Pons claimed a Gamow-factor miracle on the strength of an error bar the community could not audit.

TERMS YOU'LL MEET

electrolysis	Driving a chemical reaction with electric current — here, splitting water into hydrogen (or deuterium) and oxygen gas.
electrode	A conductor where current enters or leaves the electrolyte; reactions happen at its surface.
cathode	The negative electrode, where reduction occurs — in these cells, where deuterium is evolved and absorbed. Fleischmann and Pons's cathodes were palladium.
anode	The positive electrode, where oxygen is evolved; usually platinum in these experiments.
electrolyte	The conducting liquid between the electrodes — here heavy water with dissolved LiOD.
heavy water (D₂O)	Water in which both hydrogens are deuterium; chemically near-identical to H ₂ O, ~10% denser.
LiOD	Lithium deuterioxide, the deuterated analogue of LiOH, added to make heavy water conduct.
palladium (Pd)	A noble metal famous for absorbing enormous volumes of hydrogen into its crystal lattice — up to nearly one hydrogen per palladium atom.
loading (D/Pd)	The atomic ratio $x_{\text{D/Pd}}$ of dissolved deuterium to palladium; the variable the field later came to see as decisive (Chapter V).
open cell	An electrolysis cell that vents its gases; the escaping D ₂ and O ₂ carry chemical energy out uncounted unless corrected for.
closed cell	A sealed cell with a catalyst that recombines the gases back into water inside, so all input power must appear as heat — the post-1989 standard.
recombination	D ₂ and O ₂ reacting back into D ₂ O inside the cell, releasing their chemical energy as heat where the bookkeeping may not expect it.
faradaic efficiency	The fraction of the current that goes into the intended electrolysis reaction rather than side processes;

assumed 100% in the simplest accounting.

thermoneutral potential

The cell voltage at which electrolysis neither heats nor cools the cell: exactly the reaction enthalpy per unit charge, 1.527 V for D₂O.

calorimeter

An instrument that measures heat flow; the entire 1989 controversy is, at bottom, a fight about calorimeters.

isoperibolic calorimeter

A design that infers power from the temperature difference between cell and constant-temperature bath through a calibrated heat-leak coefficient.

excess heat

Measured heat output exceeding all accounted inputs; written P_{xs} throughout this book.

current density

Current per unit electrode area (mA/cm²); the knob Fleischmann and Pons said drove the effect.

scintillation detector

A crystal (here sodium iodide, NaI) that flashes light when a gamma ray deposits energy, giving a spectrum of counts versus energy.

Compton edge

A shoulder that must appear in a gamma spectrum below the full-energy peak, from photons that scatter once and escape; its absence marks a spectrum as unphysical.

neutron-capture gamma

The 2.224 MeV photon emitted when a neutron is captured by hydrogen ($n + p \rightarrow d + \gamma$) — the signature by which stray neutrons announce themselves in a water bath.

background

The ever-present counts from cosmic rays and ambient radioactivity that any claimed nuclear signal must rise above.

replication

An independent laboratory reproducing a claimed effect with its own apparatus — the currency in which extraordinary claims are paid for.

null result

A measurement consistent with zero effect, usually stated as an upper bound ("no more than X at 95% confidence").

preliminary note

A short, fast-tracked journal communication of urgent results — the format of the original Fleischmann–Pons

§1 March 23, 1989: the press conference before the paper

The announcement came out of order. On March 23, 1989, the University of Utah convened the press to declare that Fleischmann and Pons had achieved "sustained nuclear fusion" in a test tube — before the supporting paper was available to anyone who might check it. Contemporaneous reporting attributes the haste to two pressures: the university's patent filings, and a priority race with a rival group. Forty-five miles south at Brigham Young University, the physicist Steven Jones was preparing a claim that was superficially similar and quantitatively opposite: neutron emission from deuterium-loaded titanium and palladium at roughly 2×10^{-3} neutrons per second (Jones et al., *Nature* 338, 737, 1989) — a trace nuclear whisper, nearly fifteen orders of magnitude too small to matter energetically, and one that itself failed later replication. The Utah claim was not a whisper. It was watts.

The paper, when it appeared, was M. Fleischmann and S. Pons, "Electrochemically induced nuclear fusion of deuterium," *Journal of Electroanalytical Chemistry* 261, 301 (1989) — a "Preliminary note," submitted March 11 and in print by April 10, followed by an erratum (*ibid.* 263, 187) that among other corrections added the overlooked third author, graduate student Marvin Hawkins. It reported three kinds of evidence: excess enthalpy far beyond the authors' chemical accounting; neutron emission inferred at a rate of order 10^4 per second; and a gamma-ray line attributed to those neutrons being captured by hydrogen in the water bath shielding the cell. It attributed the heat to an unknown nuclear process. Note the shape of the claim from the start: the calorimetry was the discovery, and the nuclear measurements were the corroboration. Within three months the corroboration was gone (§4), and the discovery rested on calorimetry alone.

NOTE

"Press-release science" was the phrase the episode made famous, but the sequence was not lawless: a peer-reviewed paper *was* in press when the cameras rolled. What failed was auditability — the paper was too thin for anyone to reconstruct the error budget, and the authors, swamped and litigious by turns, declined to supply what was missing. The lesson the field drew is the standard this book applies: a claim exists scientifically when an outsider can re-derive its error bar.

§2 The cell, and what "excess heat" meant operationally

The apparatus was almost defiantly simple: a palladium cathode (rods of 0.1–0.4 cm diameter in the first paper) and a platinum anode in a glass cell of heavy water with LiOD electrolyte, driven at current densities from single digits to hundreds of mA/cm², sitting in a constant-temperature water bath. Palladium is the one indispensable ingredient: it absorbs deuterium into its lattice interstices, and weeks of electrolysis push the loading $x_{\text{D}/\text{Pd}}$ toward one deuteron per palladium atom. The cell was open: the D₂ and O₂ bubbled out and left the building.

That venting is where the accounting begins. Part of the electrical power fed to an open cell does not become heat at all — it is stored as the chemical energy of the escaping gases. The dividing line is the thermoneutral potential, the voltage that exactly pays the enthalpy bill of splitting the water:

$$E_{\text{tn}} = \frac{\Delta H}{nF}, \quad E_{\text{tn}}^{\text{H}_2\text{O}} = \frac{285.8 \text{ kJ mol}^{-1}}{2 \times 96\,485 \text{ C mol}^{-1}} = 1.481 \text{ V}, \quad E_{\text{tn}}^{\text{D}_2\text{O}} = 1.5$$

WHERE

- E_{tn} The thermoneutral potential (constant, volts): the enthalpy cost of electrolysis per unit charge.
- ΔH Molar enthalpy of the electrolysis reaction $\text{D}_2\text{O}(l) \rightarrow \text{D}_2 + \frac{1}{2}\text{O}_2$ (constant): 285.8 kJ/mol for H_2O , 294.6 kJ/mol for D_2O .
- n Electrons transferred per water molecule (constant, $n = 2$).
- F The Faraday constant, $96\,485\text{ C mol}^{-1}$: the charge of a mole of electrons.
-

So the heat a well-behaved open cell should produce is not IV but $I(E_{\text{cell}} - E_{\text{tn}})$, and "excess heat" meant, operationally, the measured output minus that expectation:

$$P_{\text{heat}} = I(E_{\text{cell}} - E_{\text{tn}}), \quad P_{\text{xs}} = P_{\text{out}} - I(E_{\text{cell}} - E_{\text{tn}}).$$

WHERE

- I Cell current (scalar, amperes), set by the experimenter.
- E_{cell} Measured cell voltage (scalar, volts), typically 3–6 V in these experiments.
- P_{heat} Electrical input power that should appear as heat in an open cell (W).
- P_{out} Heat output measured by the calorimeter (W).
- P_{xs} The claimed excess power (W): the ledger's unbalanced line.
-

Fleischmann and Pons knew this bookkeeping cold — they used a working value of 1.54 V for D_2O , which folds in the vapor carried off with the gases, and their calorimetric equations (developed fully in their 1990 paper, *J. Electroanal. Chem.* 287, 293) are competent electrochemistry. The vulnerability lies elsewhere: the correction *assumes the gases actually leave*. If a fraction γ of the evolved D_2 and O_2 recombines inside the cell — on the electrode, in the foam, at the surface — its chemical energy comes back as heat the ledger did not expect:

$$P_{\text{heat}} = I [E_{\text{cell}} - E_{\text{tn}}(1 - \gamma)], \quad \Delta P_{\text{fake}} = I E_{\text{tn}} \gamma \leq I E_{\text{tn}}.$$

WHERE

- γ The recombination fraction (dimensionless, 0–1): the share of evolved gas that reacts back to D₂O inside the cell.
- ΔP_{fake} The apparent "excess" (W) an experimenter infers if they assume $\gamma = 0$: real heat, wrongly booked as anomalous.
-

At one ampere, full recombination fakes about 1.5 W of excess — and, more insidiously, a mere 10% recombination fakes the few-percent excess typical of the contested claims. Fleischmann and Pons did measure their recombination and reported it small; their critics' point (Lewis's Caltech group, and later Wilson's GE team and the "faradaic efficiency" papers) was that in an *open, unstirred* cell the experimenter must prove it small continuously, for every run, at an accuracy better than the claimed effect. The widget below lets you run that audit yourself.

THREE VOLTAGES, ONE TRAP

Water electrolysis has three canonical voltages: the reversible potential 1.23 V (the Gibbs free energy per charge — the thermodynamic minimum), the H₂O thermoneutral potential 1.481 V, and the D₂O thermoneutral potential 1.527 V. Using the wrong one silently manufactures or destroys tenths of a watt per ampere. Mistake the thermoneutral value for 1.23 V and the ledger expects 297 mW/A more heat than the cell can actually deliver — an error that buries a real signal; grade a light-water blank with the D₂O value and you manufacture 46 mW/A of phantom excess, while the H₂O value applied to heavy water hides the same amount. Chapter III returns to this ledger line by line.

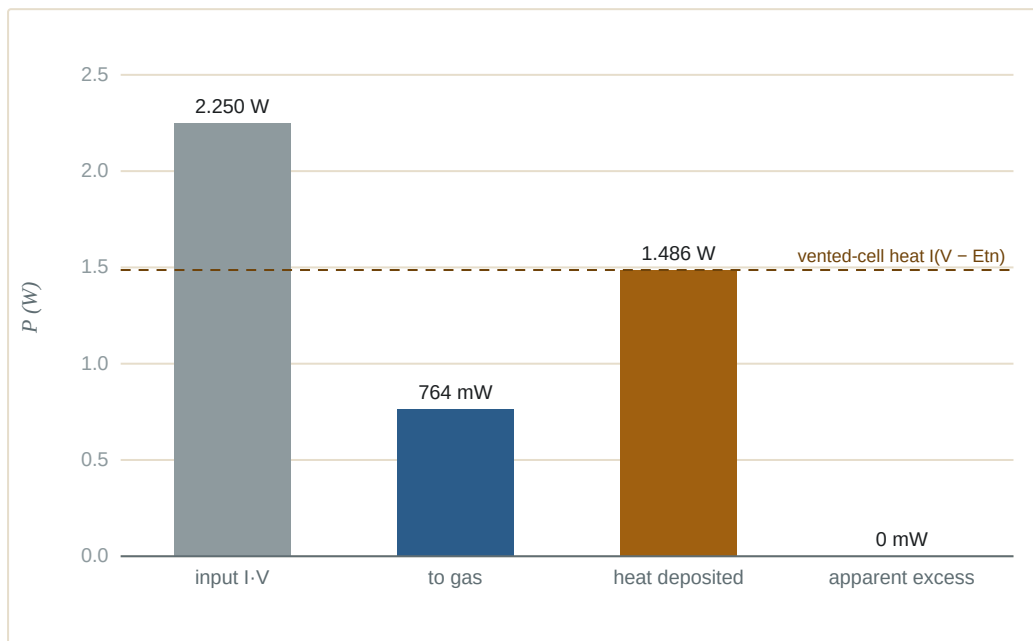
INTERACTIVE · THE OPEN-CELL ENERGY BUDGET

set current · set voltage · dial
recombination

current I (A)

cell voltage V (V)

recombination γ



$I = 0.50 \text{ A} \cdot V = 4.50 \text{ V} \cdot \gamma = 0\% \cdot \text{heat deposited} = 1.486 \text{ W}$
 apparent excess (if γ assumed 0) = 0 mW — the ledger balances

— input $I \cdot V$ — gas stream $I \cdot E_{tn}(1 - \gamma)$ — heat deposited — apparent "excess"

An open-cell audit: input power IV splits into the chemical power leaving with the gas, $IE_{tn}(1 - \gamma)$, and the heat actually deposited, $I[E_{cell} - E_{tn}(1 - \gamma)]$, with $E_{tn} = 1.527 \text{ V}$ for heavy water. At the reference setting ($I = 0.5 \text{ A}$, $E_{cell} = 4.5 \text{ V}$, $\gamma = 0$) the cell deposits 1.487 W and shows zero excess; dial recombination to just 10% and an experimenter who assumes vented gas books a phantom 76 mW — about 5% "excess," the same few-percent scale that ignited a worldwide controversy. Fleischmann and Pons did measure recombination and reported it small; the widget shows why critics demanded closed cells anyway.

§3 The claimed magnitudes

What did they actually claim, in numbers? The 1989 note reported excess heat from rods of palladium at levels the authors tabulated as percentages of input and as watts per cubic centimeter of cathode — of order watts per cm^3 at the higher current densities. The 1990 follow-up paper (Fleischmann, Pons, Anderson, Li and Hawkins, *J. Electroanal. Chem.* 287, 293–348) made the case in earnest across dozens of runs: heavy-water cells in positive excess of hundreds of milliwatts against light-water blanks consistent with zero, with "bursts" during which — the paper's own abstract reports — the thermal output exceeded the enthalpy input "by factors in excess of 40," and specific excess powers reached approximately 100 W cm^{-3} of palladium at current densities near 1 A cm^{-2} . These are the authors' own figures, from their own calorimetry, and they are contested at the root: Chapter III walks through the published reanalyses (Wilson et al., Morrison) that attribute part or — depending on whom you read — all of the effect to calibration systematics.

Two anecdotes gave the claim its folklore. First, the meltdown: the 1989 paper itself recounts an early experiment (1984–85, in most tellings) in which a 1 cm cube of palladium, left electrolyzing overnight, destroyed itself — a substantial portion of the cathode melted, part of it vaporized, and the cell and a section of the fume cupboard were wrecked. Palladium melts at about 1555°C ; the authors argued no chemical process in the cell could concentrate that much energy. Skeptics answered that a cube loaded with deuterium is a store of chemical energy ($\text{D}_2 + \text{O}_2$ recombination, hydride decomposition) entirely capable of violent release, and that a single uninstrumented accident is not a measurement. Second, the projections: taking the burst figures at face value implied integrated energy densities of megajoules per cubic centimeter of palladium — hundreds of eV per palladium atom, far beyond the $\sim 1\text{--}2 \text{ eV/atom}$ ceiling of any chemistry. That arithmetic is exactly why the claim had to be nuclear *if the calorimetry was right* — and why everything turned on whether it was.

§4 The nuclear evidence collapses first

Recall from Chapter I §5 what a watt of d+d fusion must shed: with the two dominant branches firing roughly equally,

$$R_n \approx \frac{1}{2} \frac{P_{xs}}{\bar{Q}} \approx 0.9 \times 10^{12} \text{ s}^{-1} \text{ per watt of excess power.}$$

WHERE

R_n Expected neutron emission rate (s^{-1}) if the excess power P_{xs} were vacuum-physics d+d fusion.

$\bar{Q}$ Mean energy per fusion over the two ~50/50 branches, $\approx 3.65 \text{ MeV}$ (Chapter I §5).

Fleischmann and Pons claimed watts of heat and neutrons at of order 10^4 s^{-1} — an admission, right in the founding paper, that the neutrons were about *a billion times* too few for the heat. The authors embraced the discrepancy as evidence of "an hitherto unknown nuclear process"; the physics community read it as evidence that the heat was not nuclear. And then even the 10^4 evaporated. The neutron rate had been inferred from a gamma-ray line: neutrons escaping the cell should be captured by protons in the water bath, emitting photons at 2.224 MeV (the deuteron's binding energy — established physics, measured to sub-keV precision). Richard Petrasso and colleagues at the MIT Plasma Fusion Center dissected the published peak (*Nature* 339, 183, 1989, with an exchange at *Nature* 339, 667): it sat at the wrong energy (near 2.2 MeV but inconsistent with 2.224 MeV once instrument calibration was traced); its width was a factor of about two *narrower* than their NaI detector's resolution permits any real line to be; the Compton edge that must accompany a 2.224 MeV photopeak at 1.99 MeV was absent; and the counting rate was internally inconsistent with the claimed neutron flux by a factor of about forty. Secondary accounts also report that

the peak's position shifted between the announcement video and the published figure; no formal misconduct finding ever adjudicated that allegation, and this book reports it only as reported. What is beyond dispute is the conclusion: the gamma line was an instrumental artifact, and with it fell the only direct nuclear evidence in the original paper.

This is the "dead graduate student" arithmetic of Chapter I doing its forensic work in real time. By spring 1989 the claim had already assumed its lasting logical shape: heat allegedly far above chemistry, nuclear ash far below the heat. Either the calorimetry was wrong, or the nuclear physics of a lattice differs from vacuum d+d in both rate and branching — Huizenga's "miracles," stacked.

§5 The replication wave

The spring and summer of 1989 produced one of the largest spontaneous replication efforts in the history of physics. Hundreds of groups tried; the decisive audits came from laboratories that measured heat *and* nuclear products simultaneously, with stated error budgets.

Caltech. Nathan Lewis, Charles Barnes and colleagues (*Nature* 340, 525, 1989) ran the full battery: calorimetry with stirred, carefully calibrated cells, plus searches for neutrons, gammas, tritium and ^4He . They found nothing above background on every channel — and, constructively, showed that an *unstirred* F&P-style cell develops temperature gradients that a single thermistor reads as phantom excess heat. Their companion analysis of the published calorimetric evidence (Miskelly et al., *Science* 246, 793, 1989) argued the original claims were consistent with mundane errors, recombination among them. At the American Physical Society meeting in Baltimore (May 1–2, 1989), Steven Koonin — whose molecular fusion-rate calculation anchored Chapter I — went further than the data, attributing the episode to "the

incompetence and delusion of Doctors Pons and Fleischmann"; when nine of the session's leading speakers were polled at a press conference, eight said they were at least 95% confident the Utah heat claim could be ruled out. The line drew applause and marks the moment the controversy's tone hardened on both sides.

Harwell. The UK Atomic Energy Authority's laboratory (Williams et al., *Nature* 342, 375, 1989) ran the most systematic search: three independent calorimeter designs, high-efficiency neutron and gamma detection, and a wide range of electrode and electrolyte combinations — with Fleischmann himself serving as a consultant to the laboratory. The result was null across the board, and the paper doubled as a field guide to false positives: neutron-counter electrical noise, cosmic-ray background variation, calibration error in simple calorimeters, and the slow electrolytic enrichment of tritium already present in heavy water.

MIT. The Plasma Fusion Center's team (Albagli et al., *J. Fusion Energy* 9, 133, 1990) combined calorimetry with neutron and gamma counting, helium mass spectrometry, tritium assay and surface analysis, concluding that "within estimated levels of accuracy, no excess power output or any other evidence of fusion products was detected." One caveat belongs on the record: in 1991 the MIT science writer Eugene Mallove resigned, alleging that the group's Phase-II calorimetry curves had been improperly baseline-shifted before publication; MIT stood by the work, the allegation appeared only in advocacy venues, and it has never been substantiated in the peer-reviewed literature. It is reported here as an allegation, because the field's later mythology leans on it heavily.

Inside Pons's own laboratory. The most surgical audit was Salamon et al. (*Nature* 344, 401, 1990): a University of Utah physics team monitored Pons's own operating cells — cells contemporaneously said to be producing excess heat — for five weeks with gamma detection, bounding fusion power for the known reactions at between 10^{-12} and 10^{-6} W — and, with neutron-track detectors over a 67-hour stretch, bounding neutron emission at roughly one per second. For publishing this, the team received a letter from Pons's attorney, C. Gary Triggs, demanding retraction under threat of suit; physicists protested publicly, colleagues offered to help fund the team's

legal defense, and Triggs later apologized in writing. The episode did lasting reputational damage quite apart from the physics.

Set the wave's outcome precisely: it did not prove that no cell ever produced excess heat. It established that the claimed effect did not travel — not to better calorimeters, not to better nuclear diagnostics, not even to instruments carried into the discoverers' own laboratory — and that every nuclear channel, measured independently, sat at or below background, a factor of 10^9 or more beneath the heat. In the frame of this book: the community tightened the error bar, and the Gamow-factor miracle vanished from every instrument that could have seen its ash.

§6 The institutional verdicts: 1989 and 2004

The U.S. Department of Energy convened a panel under its Energy Research Advisory Board (ERAB), co-chaired by nuclear chemist John Huizenga and Nobel laureate Norman Ramsey, which reported in November 1989 (report DOE/S-0073). Its language was measured and has been quoted ever since: the experimental results "do not present convincing evidence that useful sources of energy will result from the phenomena attributed to cold fusion," and the evidence for "a new nuclear process termed cold fusion is not persuasive." The panel recommended against any special program or dedicated centers — while its preamble cautioned that the claims could not yet all be categorically proved or disproved, and the report explicitly kept the ordinary peer-review door open for well-focused experiments. Funding agencies heard the first half; the field's survivors clung to the second.

Fifteen years later the DOE looked again, at the request of researchers who argued the evidence base had matured. The 2004 review ("Report of the Review of Low Energy Nuclear Reactions," Office of Science, December 1, 2004) put the claims before eighteen anonymous reviewers, half via written submissions and half at a one-day

presentation. The findings deserve exact statement, because both camps misquote them. On the evidence for excess power: the reviewers were "split approximately evenly" — a genuinely divided expert panel, which is itself a remarkable datum fifteen years after the field's public burial. On the evidence that the effects are nuclear in origin: two-thirds did not find it conclusive, one reviewer was fully convinced, and the remainder were somewhat convinced; most noted that the effects are not reproducible on demand. The panel's summary judged its conclusions "similar to those found in the 1989 review," and — nearly unanimously — recommended that agencies entertain *individual, well-designed proposals* through normal peer review. No dedicated program was created then, and none existed until the developments of Chapter VIII.

NOTE

Read the 2004 split carefully: it separates two questions this book keeps separate everywhere. "Is there an unexplained thermal anomaly?" drew an even split. "Is it nuclear?" drew two-to-one skepticism. The gap between those two numbers — experts half-persuaded a measurement anomaly exists, largely unpersuaded of its interpretation — is the epistemic state the rest of this book explores, and it is precisely the error-bar question versus the Gamow-factor question.

§7 What survived 1989

Three things outlived the collapse. The first is a hypothesis. Fleischmann and Pons ran months-long electrolysis on cathodes prepared with obsessive electrochemical care, and later work claimed the excess-heat effect switches on only above a loading threshold near $x_{\text{D/Pd}} \approx 0.875\text{--}0.9$ — a regime most 1989 replicators, running weeks rather than months on off-the-shelf palladium, plausibly never reached. That

claim is contested and lab-specific (it comes chiefly from SRI International's program, Chapter V), but it received an ironic endorsement in 2019, when the Google-funded reinvestigation (Berlinguette et al., *Nature* 570, 45, 2019) found that sustaining $x_{D/Pd} \geq 0.875$ is so materially difficult that the original claims were, in their words' effect, never tested at their own operating point. The loading hypothesis does not rescue 1989 — it merely renders the null results less total than they seemed — and Chapter V weighs it on its own evidence.

The second is a discipline. Every calorimetric sin catalogued in 1989–1994 — open-cell recombination, unstirred gradients, calibration drift, the 1.23 V trap, boil-off entrainment — became a design rule: closed cells with catalytic recombiners, stirred or position-insensitive calorimeters, blank cells run in parallel, stated error budgets. The canonical peer-reviewed audit of the original calorimetry, Wilson et al. (*J. Electroanal. Chem.* 332, 1, 1992), found that Fleischmann and Pons's calibration procedure neglected the change in electrochemical input power during heater pulses and concluded they had significantly overestimated the excess — though, in a wrinkle the advocacy literature has leaned on ever since, Wilson's own reanalysis did not reduce every run to zero (the pro-LENR writer Beaudette reads Wilson's tables as leaving one cell with ~40% unexplained excess; that is Beaudette's reading, not Wilson's conclusion). Fleischmann and Pons rebutted in the same journal and conceded nothing. Chapter III makes this exchange — the best short course in calorimetric error ever published — its centerpiece.

The third is a sociological case study that science is still metabolizing. Simultaneous-discovery pressure, patent lawyers scheduling physics announcements, claims propagated by fax and press wire ahead of data, a flagship journal (*Nature*, which never published the original claim) becoming the venue of record for its demolition, and careers ended on both sides — Utah's National Cold Fusion Institute closed in June 1991; Pons and Fleischmann moved to a Toyota-funded laboratory in France that closed in 1998, tens of millions spent, without a marketable result. The standard histories (Huizenga's from inside the ERAB panel; Taubes's from inside the laboratories) read the episode as pathological science; the science-studies literature (Simon's *Undead Science*) reads it as a lesson in how scientific closure actually

happens, and how a community can keep working after it. Both readings are useful; neither changes a single number in this chapter.

AHEAD OF THEORY

Strictly, 1989 was the opposite of a result ahead of theory: established theory (branching ratios, capture gammas, tunneling rates) audited the results and demolished them. What survives as genuinely unresolved is narrower, and it is a matter of record rather than physics: no published audit ever reduced *every* Fleischmann–Pons calorimetric run to an identified error — the mainstream verdict rests on irreproducibility, identified systematics in some runs, and the absent nuclear ash, not on a run-by-run refutation. That residue — an unexplained thermal record with no nuclear signature and no mechanism — is the seed of every claim audited in Chapters III–VIII, and it enters Chapter IX's ledger as the field's founding IOU.

FURTHER READING

Fleischmann, M. & Pons, S. *Electrochemically induced nuclear fusion of deuterium*, Journal of Electroanalytical Chemistry 261, 301–308 (1989); erratum 263, 187–188 (1989) — the original preliminary note and the erratum adding M. Hawkins; DOI 10.1016/0022-0728(89)80006-3.

Petrasso, R. D., Chen, X., Wenzel, K. W., Parker, R. R., Li, C. K. & Fiore, C. *Problems with the γ -ray spectrum in the Fleischmann et al. experiments*, Nature 339, 183–185 (1989) — the dissection of the 2.2 MeV line; exchange with Fleischmann et al. at Nature 339, 667 (1989).

Lewis, N. S., Barnes, C. A., Heben, M. J. et al. *Searches for low-temperature nuclear fusion of deuterium in palladium*, Nature 340, 525–530 (1989) — the Caltech multi-channel null replication; DOI 10.1038/340525a0.

Williams, D. E., Findlay, D. J. S., Craston, D. H. et al. *Upper bounds on 'cold fusion' in electrolytic cells*, Nature 342, 375–384 (1989) — the Harwell systematic search and its catalogue of false-positive mechanisms; DOI 10.1038/342375a0.

Energy Research Advisory Board. *Cold Fusion Research*, U.S. Department of Energy report DOE/S-0073 (November 1989) — the first institutional verdict, co-chaired by J. Huizenga and N. Ramsey.

U.S. Department of Energy, Office of Science. *Report of the Review of Low Energy Nuclear Reactions* (December 1, 2004) — the 18-reviewer re-examination; source of the "split approximately evenly" and "nearly unanimous" findings quoted in §6.

Huizenga, J. R. *Cold Fusion: The Scientific Fiasco of the Century*, University of Rochester Press, 1992 — the inside chronicle from the ERAB co-chair; the "miracles" framing used throughout this book.

Chapter production note: researched, drafted, and adversarially verified by Claude subagents (draft agent: `claude-fable-5`; verification agent: `claude-fable-5`, citations web-checked), orchestrated by `claude-fable-5`. See the [methods appendix](#) for the full research process.

CHAPTER III

The Art of the Calorimeter

Every claim of anomalous heat is really a claim about a temperature sensor, a calibration constant, and an error budget — and phantom watts are easier to make than real ones.

BEFORE WE START

Chapter I built the Gamow factor — the number that says lattice fusion should not happen. Chapter II told the story of 1989, when two electrochemists said their measurements showed it happening anyway. This chapter is about the other number in the book's frame: the error bar. A calorimeter is just an instrument that counts joules going into and out of a box, and in principle nothing could be simpler. In practice, an electrolytic cell is one of the least cooperative boxes imaginable: it bubbles, evaporates, stores chemical energy, changes its own liquid level, and releases heat in places the calibration never visited.

We will meet the three instrument families the field relies on, write down the real bookkeeping equations each one uses, and then do something the 1989 press conference did not: catalogue the specific, quantified ways each design can manufacture "excess heat" out of nothing. With that catalogue in hand we can weigh the strongest surviving heat datasets — and the strongest critiques of them — and end with a checklist for what a believable excess-heat measurement would have to look like. The physics in this chapter is entirely conventional; what is contested is only whether particular experiments beat their own error budgets.

TERMS YOU'LL MEET

calorimeter

An instrument that measures heat flow — here, the total thermal power leaving an electrochemical cell.

isoperibolic calorimeter

A design that holds the surroundings at constant temperature and infers cell power from how far the cell runs hotter than its bath ("isoperibolic" = constant surroundings).

Dewar flask

A vacuum-jacketed vessel (a laboratory thermos). Fleischmann and Pons used silvered Dewar cells so heat would leave mainly by radiation.

heat-transfer coefficient

The constant of proportionality between the cell–bath temperature difference and the power flowing out; watts per kelvin.

calibration constant

The measured value of the heat-transfer coefficient, determined by injecting a known power with a resistive heater. Everything hangs on its stability.

joule heating

Heat produced by current flowing through electrical resistance — the known, mundane part of a cell's power input.

thermoneutral potential

The cell voltage at which electrolysis neither heats nor cools the cell: 1.481 V for light water, about 1.527 V for heavy water. Voltage above this appears as heat; the rest leaves as gas chemistry.

open cell

An electrolysis cell that vents its evolved gases; the chemical energy they carry must be subtracted on paper rather than measured.

closed cell

A sealed cell in which evolved gases are recombined internally, so all input power eventually appears as measurable heat.

recombination

Deuterium and oxygen gas reacting back into heavy water inside the cell, releasing their

	chemical energy as heat where the bookkeeping did not expect it.
catalytic recombiner	A catalyst (typically platinum) placed in a closed cell to force complete recombination in a known location.
faradaic efficiency	The fraction of electrolysis current that actually produces the expected gases; if it is below 100%, open-cell energy accounting silently breaks.
mass-flow calorimeter	A design that captures the cell's heat in a pumped coolant stream and computes power from flow rate, specific heat, and temperature rise — first-principles bookkeeping.
specific heat capacity	The energy needed to warm one gram of a substance by one kelvin; for water, $4.18 \text{ J g}^{-1} \text{ K}^{-1}$.
Seebeck effect	A temperature difference across a junction of two conductors produces a voltage — the physics of thermocouples.
Seebeck envelope calorimeter	A box whose entire wall is a thermoelectric sensor, so its output voltage is proportional to total heat flux regardless of where inside the heat originates.
thermopile	Many thermocouples wired in series to amplify the Seebeck voltage; the wall material of an envelope calorimeter.
excess power (P_{xs})	Measured output power minus all accounted input power. The quantity every claim in this book stands on.
systematic error	A repeatable bias in a measurement — wrong calibration, unmodeled heat path — that no amount of averaging removes.
random error	Noise that fluctuates from reading to reading and shrinks as the square root of the number of measurements averaged.

calibration-constant shift (CCS)	Kirk Shanahan's proposed systematic: if the place where heat is released inside the cell moves, the effective calibration constant moves with it, creating phantom excess.
loading ratio ($x_{\text{D/Pd}}$)	The atomic ratio of deuterium to palladium in the cathode; proponents claim excess heat requires loading above about 0.875.
current density	Electrolysis current per unit cathode area (mA cm^{-2}); a second claimed threshold variable.
entrainment	Liquid droplets carried out of a boiling cell with the vapor; counting them as vapor overstates the energy that left.
heat after death	Fleischmann and Pons's term for cells that allegedly kept producing heat after the current was switched off — reported in the regime with the worst error bars.
blind calibration	Calibrating an instrument by someone who does not know which runs are "live," so expectation cannot steer the constants.
signal-to-noise ratio	The size of the claimed effect divided by the size of the measurement uncertainty; the single number that decides whether a measurement means anything.

§1 The measurement problem: milliwatts riding on watts

Start with the shape of the problem. A typical Fleischmann–Pons-type cell draws several watts of electrical power. The anomalies claimed in the peer-reviewed literature are mostly tens to hundreds of milliwatts — 5–30% of input in the strongest datasets, under 1% in many others. The signal, if it exists, is a small difference between two large numbers, sustained for days. This is the *error bar* half of the book's

unifying frame: where Chapter I asked how experiments might raise the tunneling probability, this chapter asks how tightly the heat measurement can be trusted — because every excess-heat claim lives or dies by exactly that number.

"Excess heat" has a precise meaning: energy out minus energy in, integrated over the run,

$$E_{\text{xs}}(t) = \int_0^t [P_{\text{out}}(t') - P_{\text{in}}(t')] dt',$$

WHERE

E_{xs} Scalar (joules): cumulative excess energy up to time t . A claim is only as good as this integral and its uncertainty.

P_{out} Total thermal power leaving the cell (W), the calorimeter's job to measure.

P_{in} Total accounted input power (W): electrical power minus whatever leaves as chemical energy in vented gases.

t' Integration variable (s); runs last days to weeks, so small steady biases integrate into large fake energies.

The subtlety hides inside P_{in} . In an open cell, the current splits water and the gases leave, carrying chemical energy with them. Only the cell voltage *above* the thermoneutral potential appears as heat:

$$P_{\text{in}} = (E_{\text{cell}} - E_{\text{tn}}) I, \quad E_{\text{tn}} = \frac{\Delta H}{nF} = \begin{cases} 1.481 \text{ V} & \text{H}_2\text{O} \\ 1.527 \text{ V} & \text{D}_2\text{O} \end{cases}$$

WHERE

- E_{cell} Measured cell voltage (V), typically 3–10 V under load.
- I Electrolysis current (A).
- E_{tn} Thermoneutral potential (constant, V): the enthalpy of water formation expressed as a voltage. Fleischmann and Pons used a working value of 1.54 V for D_2O , which folds in vapor carried off at their bath temperature.
- ΔH Enthalpy of formation of the liquid from its gases (constant): 285.8 kJ mol⁻¹ for H_2O , 294.6 kJ mol⁻¹ for D_2O .
- n, F Electrons transferred per molecule ($n = 2$) and the Faraday constant ($F = 96,485 \text{ C mol}^{-1}$).

1.23, 1.48, OR 1.53 VOLTS?

Three different "water-splitting voltages" circulate, and picking the wrong one manufactures heat from thin air. The reversible potential, 1.229 V, is the free-energy (ΔG) value and ignores the entropy the cell exchanges with its surroundings. The thermoneutral values above are the enthalpy (ΔH) values, which is what heat bookkeeping needs. And the heavy-water value differs from light water by 46 mV — use 1.481 V for a D_2O cell and, at one ampere, you have just invented 46 mW of phantom excess, comfortably above many claimed signals. During initial loading the bookkeeping shifts again, because deuterium is being absorbed by the cathode rather than evolved as gas.

Notice what this equation already implies: in an open cell, part of the energy budget is not measured at all — it is *assumed*, on the premise that every coulomb makes gas and every molecule of gas leaves. Both premises can fail, and §4 will price the failure. First, the instruments.

§2 The isoperibolic calorimeter: heat as a temperature difference

The 1989 instrument, and still the most common, is the isoperibolic calorimeter: the cell sits in a constant-temperature bath, runs slightly hotter than the bath, and the temperature difference is read as a power through a calibrated coefficient. At steady state, in its simplest linearized form,

$$P_{\text{out}} = k_c (T_{\text{cell}} - T_{\text{bath}}),$$

WHERE

k_c The heat-transfer coefficient (W K^{-1}) — the *calibration constant*. It is measured, not derived: a resistive heater injects a known power and the temperature response is recorded.

$T_{\text{cell}}, T_{\text{bath}}$ Cell and bath temperatures (K). For $k_c = 0.8 \text{ W K}^{-1}$ and 10 W of input, the cell runs 12.5 K above its bath.

Fleischmann and Pons's actual cells were silvered Dewar flasks, engineered so that heat left mainly by radiation, and their working equation (Fleischmann et al., *J. Electroanal. Chem.* 287, 293 (1990)) is the full unsteady balance with a radiative transfer term:

$$C_p M \frac{d(\Delta\theta)}{dt} = (E_{\text{cell}} - E_{\text{tn}}) I + Q_f + P_{\text{heater}} - k_R [(\theta_b + \Delta\theta)^4 - \theta_b^4]$$

WHERE

$C_p M$	Heat capacity of the cell contents (J K^{-1}) — it <i>falls</i> as electrolyte is consumed, a built-in drift term.
$\Delta\theta, \theta_b$	Cell–bath temperature difference and absolute bath temperature (K); the quartic bracket is the radiative form of the loss term.
Q_f	The putative anomalous ("excess") power (W) — the unknown being solved for.
P_{heater}	Known power injected by the calibration heater (W).
k_R	Radiative heat-transfer coefficient (W K^{-4}): the calibration constant of this design, extracted by fitting the response to heater pulses.
$P_{\text{evap}}, P_{\text{gas}}$	Power carried away by evaporation and by the sensible + latent enthalpy of the evolved gas stream (W). The latent heat of vaporization differs between the isotopes — $\approx 2070 \text{ J g}^{-1}$ for D_2O at its boiling point against $\approx 2257 \text{ J g}^{-1}$ for H_2O — another constant that silently matters.

Read that equation as an audit trail. The left side is thermal inertia; on the right, one term is the answer (Q_f) and every other term is a place for the answer to hide. The calibration constant k_R is determined by nonlinear regression against heater pulses; the heat capacity drifts; evaporation depends on temperature and bubbling; the gas term depends on the open-cell assumptions of §1. The design's virtues are real — it is cheap, sensitive, and fast — but its output is only as good as the claim that k_R measured during a calibration pulse still applies hours later, at a different liquid level, with a different bubble pattern, and with heat possibly being released at a different place inside the cell.

NOTE

Advocates report astonishing precision for this design: Miles and Fleischmann quoted uncertainties as low as $\pm 0.1 \text{ mW}$ for the silvered-Dewar cells. Treat that figure as a claim about the *random* error under ideal conditions, made by the method's proponents; independent laboratories running comparable cells quoted error bars hundreds of times larger (MIT's 1989 phase-II calorimetry, for instance, stated $\pm 40 \text{ mW}$), and the disputes

of §4 and §6 are almost entirely about *systematic* error, which no stated precision addresses.

§3 Following the heat out: mass-flow and Seebeck designs

The cleanest answer to "is your calibration constant real?" is to build an instrument with almost nothing to calibrate. In a mass-flow calorimeter, the cell sits inside a jacket through which coolant is pumped at a measured rate, and essentially all the cell's heat ends up in the stream:

$$P_{\text{out}} = \dot{m} c_p (T_{\text{out}} - T_{\text{in}}) (1 + \varepsilon_{\text{loss}}),$$

WHERE

$\dot{m}$	Coolant mass flow rate (g s^{-1}), measured directly — e.g. by weighing the effluent.
c_p	Specific heat of the coolant ($\text{J g}^{-1} \text{K}^{-1}$; 4.18 for water) — a handbook constant, not a fitted one.
$T_{\text{out}} - T_{\text{in}}$	Temperature rise of the stream across the cell (K).
$\varepsilon_{\text{loss}}$	The small calibrated fraction of heat that escapes without joining the stream — below 2% in the SRI instruments. This residue is the only place a calibration dispute can live.

Every factor is either measured on the spot or a handbook constant; the fitted correction is a few percent of the total rather than 100% of it. This was the choice of

Michael McKubre's group at SRI International, whose labyrinth-sealed flow calorimeters anchor the dataset of §5. The price is complexity and sluggishness — a flow calorimeter integrates over minutes, so it cannot resolve the fast "bursts" that isoperibolic cells report. Whether that is a bug or a diagnostic is a fair question to hold on to.

The third family closes the geometry instead of the plumbing. A Seebeck envelope calorimeter surrounds the entire cell with thermopile walls: every parcel of heat leaving the box, in any direction, from any internal location, must cross the sensor, and the summed voltage reads total heat flux. Position-independence is the point — it is the design most immune to the "the heat moved" failure mode of §4. Edmund Storms, the design's most prolific LENR practitioner, reports long-term stability of ± 60 mW over months and ± 10 mW under favorable conditions (figures from the field's own journal, *J. Condensed Matter Nucl. Sci.*, so read them as practitioner claims rather than audited specifications).

A useful summary: the isoperibolic design measures a temperature and *infers* a power; the mass-flow design measures a power almost directly; the Seebeck design integrates a power over all directions. The field's evidentiary ladder in §5 climbs exactly this ordering.

§4 The failure catalog: how to make phantom watts

Four failure modes account for most of the thirty-five-year calorimetry argument. Each is quantitative, and each has a published exchange attached to it.

Recombination. Open-cell bookkeeping subtracts $E_{tn}I$ on the assumption that all evolved gas leaves. If a fraction f of the D_2 and O_2 instead recombines inside the cell

— on the electrode, on a catalytic speck, in the foam — that fraction's chemical energy reappears as real heat the ledger did not expect:

$$P_{\text{recomb}} = f E_{tn} I \leq 1.527 \text{ V} \times I.$$

WHERE

f Recombination fraction (dimensionless, 0–1); equivalently a faradaic efficiency below one has the same fingerprint.

P_{recomb} Phantom "excess" power (W). At $I = 1$ A, complete recombination fakes ~1.5 W — larger than almost every claimed signal.

This was the thrust of the 1989 Caltech critique (Miskelly, Lewis and colleagues, *Science* 246, 793 (1989)) and of Jones and Hansen's faradaic-efficiency papers; Chapter II's widget let you price it directly. The community's structural fix was the closed cell with catalytic recombiner, standard in serious work after about 1992: with recombination forced and internal, the E_{tn} term drops out of the bookkeeping entirely.

Calibration-constant shift. The subtlest entry is Kirk Shanahan's calibration-constant shift (CCS) hypothesis (*Thermochim. Acta* 387, 95 (2002)). No calorimeter collects heat with perfectly uniform efficiency over its interior; the calibration constant is really a weighted average over *where* the heat is released during calibration. If the heat's location later moves — say, recombination begins at the electrode surface rather than in a recombiner at the top — the true constant shifts by a few percent while the experimenter keeps using the old one. Shanahan argued that a 1–3% unrecognized shift quantitatively reproduces reported excess-power curves in mass-flow data. The arithmetic of the phantom is worth displaying, because it is the heart of this chapter's widget:

$$P_{\text{xs}}^{\text{inf}} = k_a (T_{\text{cell}} - T_{\text{bath}}) - P_{\text{in}} = P_{\text{xs}}^{\text{true}} + \delta (P_{\text{in}} + P_{\text{xs}}^{\text{true}}), \quad \delta = \frac{k_a - k}{k}$$

WHERE

$P_{\text{xs}}^{\text{inf}}, P_{\text{xs}}^{\text{true}}$	The excess power the experimenter infers versus whatever excess actually exists (W).
k_a	The calibration constant the experimenter <i>assumes</i> (W K ⁻¹) — measured once, with the heater.
k_t	The heat-transfer coefficient actually in force during the live run (W K ⁻¹).
δ	The fractional calibration error (dimensionless). Because it multiplies the full input power, $\delta = +1\%$ on a 10 W cell manufactures 100 mW of phantom excess — precisely the scale of many published claims.

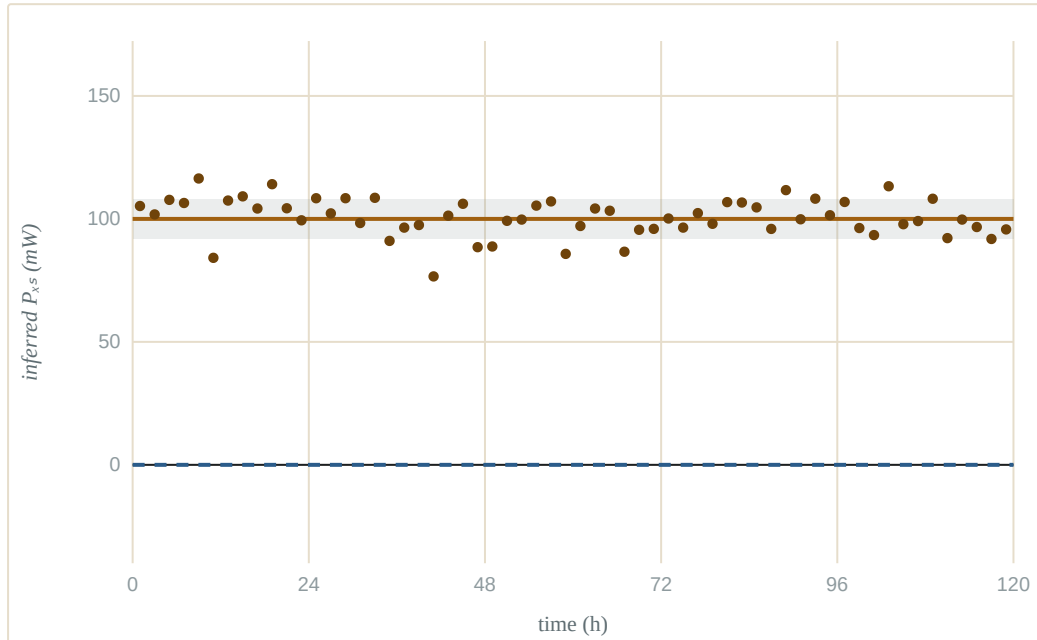
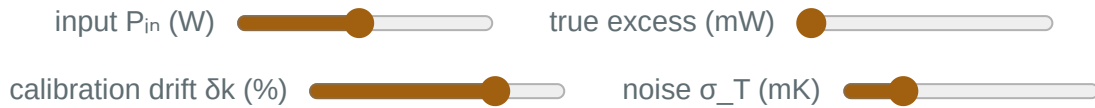
The rebuttal record is genuinely two-sided and should be reported as such. A ten-author response (Marwan et al., *J. Environ. Monit.* 12, 1765 (2010)) answered Shanahan's critique in the same journal with three main points: Seebeck envelope calorimeters integrate heat flux over nearly the full solid angle and are close to position-independent, yet also report excess; measured recombination in the relevant cells is far below the level CCS needs; and CCS offers no reason why the supposed shift should correlate with deuterium loading and appear in D₂O but not H₂O cells. Shanahan maintains the rebuttals misconstrue the mechanism. No neutral party has settled the exchange with a purpose-built experiment, and both sides of it are peer-reviewed — though it is worth noticing that the venue, an environmental-monitoring journal, is an odd home for the field's central metrology dispute.

Heat leaks and cold fingers. Any unmodeled conduction path — a bundle of sensor leads, a support rod, a falling liquid level exposing glass — changes the effective heat-transfer coefficient between calibration and measurement. Wilson and colleagues' reanalysis (§6) located a specific version in the Fleischmann–Pons protocol: the calibration procedure neglected the change in electrochemical input power during the heater pulse itself.

Positive feedback. If the calibration constant depends on temperature (radiation's T^4 guarantees some dependence) and the analysis linearizes it at the calibration point, then anything that warms the cell — including a small real or phantom excess — is scored with the wrong constant, inflating the apparent excess further. Claims of excess that *grows* with temperature need this mundane amplifier excluded first.

INTERACTIVE · THE CALORIMETER TRUTH MACHINE

dial the input · drift the calibration · add noise



$\Delta T = 12.50 \text{ K}$ · true excess = **0 mW** · inferred = **100 mW** (mean of readings 101 mW)
phantom from $\delta k = +100 \text{ mW}$ · single-reading noise $\sigma = 8 \text{ mW}$ · averaging shrinks σ ,
never the phantom

● inferred readings — inferred mean (bias) - - true excess ■ $\pm 1\sigma$ noise band

An idealized isoperibolic cell with $k_t = 0.8 \text{ W K}^{-1}$: the true physics sets $T_{\text{cell}} = T_{\text{bath}} + (P_{\text{in}} + P_{\text{xs}}^{\text{true}})/k_t$, while the experimenter infers $P_{\text{xs}}^{\text{inf}} = k_a(T_{\text{cell}} - T_{\text{bath}}) - P_{\text{in}}$ from noisy temperature readings. Set the true excess to zero and drift the calibration by +1% on a 10 W input: a steady 100 mW phantom appears, indistinguishable point-by-point from a genuine signal of the size many papers report. Note what averaging does and does not fix — the scatter shrinks with integration time, but the phantom is systematic and survives forever.

§5 The datasets that survive best

Judged against the failure catalog — closed cells, first-principles calorimetry, controls, published error analysis — three bodies of excess-heat work are generally ranked strongest, by proponents and by neutral reviewers alike. None has survived hostile replication, for a reason §6 makes precise.

SRI/EPRI (McKubre and colleagues). The flow-calorimetry program at SRI International, funded by the Electric Power Research Institute, is the field's best-instrumented electrolytic dataset: closed cells with catalytic recombiners, mass-flow calorimetry with $\varepsilon_{\text{loss}} < 2\%$, and hydrogen cells run as controls in the same instruments. McKubre et al. reported (in the peer-reviewed core paper, *J. Electroanal. Chem.* 368, 55 (1994), with the bulk of the data in EPRI contract reports TR-104195 and TR-107843) excess power in D/Pd cells but not H/Pd cells, typically 3–10% of input at watt scale, and organized their observations into an empirical parametrization:

$$P_{\text{xs}} \propto (x - x^\circ)^2 (i - i^\circ) \left| \frac{dx}{dt} \right|, \quad x^\circ \approx 0.875, \quad i^\circ \approx 200\text{--}250 \text{ mA cm}^{-2}$$

WHERE

- | | |
|--------------|---|
| x | The loading ratio $x_{\text{D/Pd}}$ (dimensionless): deuterium atoms per palladium atom in the cathode. |
| x° | The claimed loading threshold: below $x \approx 0.875$, no excess; in the EPRI program, every cell that reached $x > 0.95$ reportedly showed excess and none below 0.90 did. |
| i, i° | Cathode current density and its claimed threshold (mA cm^{-2}). |
| $ dx/dt $ | The deuterium flux through the interface — excess correlates with loading <i>changing</i> , not merely being high. |
-

Handle this equation with register discipline: it is SRI's parametrization of SRI's data, published in full only in conference proceedings and contract reports, with a cell-specific proportionality constant that was never made universal. It is not a law of nature; it is the field's most careful description of when its flagship anomaly is said to appear.

The Energetics–ENEA–SRI three-lab campaign. Energetics Technologies (Israel) claimed large excess using "SuperWave" current modulation, with a showcase run (ETI-64) asserting roughly $25\times$ energy gain; the protocol was then run at SRI and at ENEA (Italy's national energy agency) using ENEA-fabricated palladium foils, with significant excess reported in roughly 70% of runs. This is the field's best example of protocol transfer across laboratories — and it must be weighed with its caveats: every venue involved is a conference proceeding or symposium volume, none of it passed independent peer review, and the "replication" was performed by collaborating laboratories sharing personnel and materials, not by adversaries.

Lonchamp's engineering replication. At CEA Grenoble, Georges Lonchamp rebuilt the Fleischmann–Pons ICARUS-type cells to specification — reportedly the only group to execute the original protocol faithfully — and observed modest excess, of order 5–10% during normal electrolysis, with larger apparent excess during boil-off episodes (ICCF-6 proceedings, 1996; conference-grade, and the boil-off regime is precisely where entrainment errors are worst, as Lonchamp himself noted).

AHEAD OF THEORY

Two features of the SRI dataset outrun any explanation, mundane or exotic, and belong in Chapter IX's ledger. First, *isotope specificity*: H/Pd cells in the same calorimeters showed nothing while D/Pd cells showed excess — either the strongest control in the field or evidence of a deuterium-specific systematic, and Shanahan's CCS, being isotope-agnostic, struggles with it. Second, the *loading threshold*: no theory predicts an onset at $x_{\text{D/Pd}} \approx 0.875$, and if the integrated energy densities are taken at face value (of order MJ per cm^3 of palladium, hundreds of eV per atom), they exceed any conceivable chemistry by two orders of magnitude. These are results ahead of theory only if they are results at all — which is exactly what §6 disputes.

§6 The critiques of even the best

The canonical peer-reviewed critique of the founding dataset is Wilson et al. (General Electric), "Analysis of experiments on the calorimetry of LiOD-D₂O electrochemical cells," *J. Electroanal. Chem.* 332, 1 (1992) — published, deliberately, in the same journal as the claims. Their reanalysis of the Fleischmann–Pons 1990 paper found specific systematic errors: the calibration-pulse method neglected the change in electrochemical input power during the heater pulse, the treatment of evaporation and the thermoneutral voltage was faulted, and unaccounted recombination was not excluded. Their conclusion was that Fleischmann and Pons significantly overestimated their excess heat. Two honest footnotes: Fleischmann and Pons published a combative same-issue rebuttal conceding nothing; and Wilson's own corrected accounting did not zero out every cell — the advocate Charles Beaudette later argued that roughly 700 mW of excess survived Wilson's reanalysis in one cell, a partisan reading of Wilson's tables that Wilson himself did not endorse. Neither side conceded, and the exchange remains the single best entry point into open-cell error accounting.

What about SRI? McKubre's papers are notable within the field for carrying explicit error analysis — itemized uncertainty budgets in which, in the best runs, the claimed excess exceeds the stated combined error many times over. That matters because it converts the dispute from "did they measure carefully?" to "is the error model complete?" — which is exactly where Shanahan's CCS aims. The unresolved question is not arithmetic but the completeness of the systematic-error catalog; a 1% effect the model omits defeats a 0.5% error budget no matter how honestly computed.

The modern benchmark is the Google-funded program (Berlinguette et al., *Nature* 570, 45 (2019)) — the first time in decades that well-resourced outsiders built state-

of-the-art calorimetry specifically for these claims. Their published verdict: no evidence of excess heat anywhere they looked, including 420 calorimetric tests of nickel–hydrogen systems with ~50 mW resolution. For the palladium claims the result is subtler and more interesting: the program found that the claimed prerequisite loading, $x_{\text{D}/\text{Pd}} \geq 0.875$, could be reached only rarely and held almost never — exceeded twice in several dozen attempts — and that the indirect loading measures older work relied on can badly mislead. Read carefully, the Nature paper is simultaneously the strongest modern null result *and* an explanation of why nulls are guaranteed: the anomaly, if real, lives in a materials regime that essentially no replication (including, mostly, Google's own) ever occupied. It refutes the claims' reproducibility, not — by its own account — their operating point.

§7 A specification for belief

This chapter's failure catalog can be inverted into a positive specification — what a heat claim would need before the error-bar half of our frame stops being the live explanation:

Closed cell, so recombination is forced and the thermoneutral subtraction disappears from the books. *Mass-flow or Seebeck calorimetry*, so the measurement is first-principles or position-independent rather than hanging on one fitted constant. *Blind calibration*, with live and control runs interleaved and the analyst ignorant of which is which. *Signal at least ten times the total stated error*, systematic terms included — not 1.5σ bumps. *Days-long integration*, because energy, not power, is what chemistry cannot fake at MJ scale. *Isotopic and material controls in the same instrument* — H_2O beside D_2O , platinum beside palladium. And *independent replication with a shared, written protocol*, by a laboratory with no stake in the outcome.

Almost nothing in the literature meets the full specification. The SRI dataset meets more of it than anything else — closed cells, flow calorimetry, hydrogen controls, explicit error budgets — and fails at the final clause, hostile replication, which the Google program suggests may be a materials problem before it is an honesty problem. The specification is also the chapter's summary judgment: calorimetry is a solved measurement problem wrapped around an unsolved reproducibility problem. The instruments exist; what has never existed is an anomaly that travels. Keep both of the book's numbers in view — the Gamow factor says the heat should not be nuclear, and the error bar says much of it may not even be heat. Whatever remains is the subject of the next chapter, which asks whether the claimed nuclear ashes fare any better.

FURTHER READING

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CHAPTER IV

The Claimed Nuclear Evidence

A fire that leaves no ash is not a fire — so for thirty-five years the field has searched its cells for the nuclear ash, and this chapter weighs everything it claims to have found.

BEFORE WE START

Chapter III ended on an uncomfortable truth: calorimetry alone, however careful, can never prove a nuclear origin — it can only measure heat that chemistry seems unable to supply. The decisive evidence must be nuclear: new atoms that were not there before, or radiation caught in the act. Since 1989 the field has advanced five candidate signatures — helium-4 correlated with heat, tritium, transmuted heavy elements, charged-particle tracks in plastic detectors, and low-level neutrons. This chapter takes them one at a time: the strongest peer-reviewed claim, the replication record, the magnitudes, and the standing critique.

One idea does most of the work here, and it comes straight from Chapter I's yardstick: *commensurability*. Every nuclear product carries a known energy, so any measured product rate converts directly into "watts explained," and any claimed excess power converts into a predicted product rate. A claim whose heat and ash match at the right ratio is forensically interesting; a claim whose ash would explain a billionth of its heat is really two unrelated claims. Keep the exchange rate in mind throughout: one watt of deuterium fusion making helium-4 means about 260 billion helium atoms per second, every second, for as long as the heat flows.

TERMS YOU'LL MEET

nuclear ash

The product nuclei a nuclear reaction must leave behind — helium, tritium, or transmuted elements — in amounts fixed by the energy released.

commensurability

The bookkeeping test: does the measured amount of nuclear product match the measured heat at the known energy-per-reaction? The master move of this chapter.

helium-4 (^4He)

The ordinary, stable isotope of helium; the claimed dominant ash of LENR, and also about 5.2 parts per million of the air around every experiment.

mass spectrometry

Weighing atoms and molecules by deflecting their ions; the technique used to count helium atoms in cell gases.

parts per million (ppm)

A concentration unit: one molecule per million. Atmospheric helium sits near 5.2 ppm — the ever-present contamination benchmark.

in-leakage

Atmospheric helium seeping into a cell or sample flask through seals or glass walls, mimicking a nuclear source.

retention fraction

The fraction of helium born in a metal cathode that escapes into the gas where it can be measured; the rest stays trapped in the solid.

anodic stripping

Electrochemically dissolving the outer layers of a cathode to release gases trapped inside it — how SRI hunted for retained helium.

tritium (t)

Radioactive hydrogen-3, half-life 12.3 years; produced by one of the two normal d+d fusion branches, so its presence or absence is diagnostic.

liquid scintillation counting

Detecting tritium's weak beta decay by mixing the sample into a cocktail that flashes light for each decay.

autoradiography

Letting a radioactive sample expose photographic film to map where the activity sits — used at BARC to image tritium in titanium targets.

transmutation	The conversion of one chemical element into another by a nuclear process — the most extraordinary of the LENR claims.
permeation	Gas diffusing through a solid membrane; the Iwamura experiments drive deuterium gas through layered palladium foils.
multilayer	A sandwich of thin films — here palladium and calcium oxide layers tens of nanometers thick — whose structure is claimed to be essential to transmutation.
XPS	X-ray photoelectron spectroscopy: identifies which elements sit in the top few nanometers of a surface by the energies of ejected electrons.
ICP-MS	Inductively coupled plasma mass spectrometry: dissolves a sample into a plasma and counts trace elements atom by atom — sensitive to nanograms and below.
ToF-SIMS	Time-of-flight secondary-ion mass spectrometry: sputters ions off a surface and weighs them, mapping trace elements with high mass resolution.
mass interference	Two different species with nearly the same mass (an atom vs. a molecular ion) landing on the same spectrometer peak — a classic way to "detect" an element that isn't there.
mononuclidic	An element with only one stable isotope (like praseodymium-141), so isotope ratios cannot distinguish a nuclear origin from contamination.
isotopic signature	The ratios of an element's isotopes; natural sources have fixed ratios, so anomalous ratios argue against ordinary contamination.
CR-39	A clear plastic that records the passage of charged particles as latent damage trails, revealed as pits by etching in hot lye.
triple track	Three etch pits splitting from a common point in CR-39 — the signature of a carbon nucleus shattered into three alpha particles by a fast neutron.

background	The signal an instrument reports with no source present — cosmic rays, radon, ambient helium, tritium in ordinary water. Every claim is an excess above this.
upper limit	The largest signal that could have hidden in a null measurement; how physics states "we looked and found nothing" quantitatively.
control cell	An identical cell run with the interesting ingredient swapped out (light water for heavy water); the experiment's alibi against artifacts.
blind analysis	Analyzing samples without knowing which are "live" and which are controls, so expectation cannot steer the result.
statistical significance	How unlikely a signal is to be a background fluctuation, counted in standard deviations (σ); large excesses can still have mundane, non-statistical explanations.
contamination	The mundane arrival of the sought-for substance from the environment — the null hypothesis that stalks every trace-level claim in this chapter.
conference proceedings	Papers published from a meeting without full journal peer review; much of this field's most-quoted evidence lives there, which matters for how hard it can be leaned on.

§1 The forensic standard: commensurability

Chapter I established what deuterium fusion must produce; Chapter III established how excess heat is measured. This chapter joins them. The rule is arithmetic: if an excess power P_{xs} comes from a nuclear reaction releasing energy Q per event, the products must appear at the rate

$$R = \frac{P_{\text{xs}}}{Q} = \frac{1 \text{ W}}{23.85 \text{ MeV} \times 1.602 \times 10^{-13} \text{ J/MeV}} \approx 2.62 \times 10^{11} \text{ atoms}$$

WHERE

R	Scalar, s^{-1} : the rate at which product nuclei must be created if the heat is nuclear.
P_{xs}	Excess power (W), as measured by the calorimetry of Chapter III.
Q	Energy released per reaction (MeV). The number shown uses $Q = 23.85 \text{ MeV}$, the $\text{d}+\text{d} \rightarrow {}^4\text{He}$ Q-value from Chapter I.
$1.602 \times 10^{-13} \text{ J/MeV}$	Unit conversion constant (note: this is the joule value of one MeV; the more familiar 1.602×10^{-19} is one eV).

This single equation is the chapter's master move. It converts any claimed product into "watts explained" and any claimed wattage into a predicted ash rate. One watt attributed to $\text{d}+\text{d} \rightarrow {}^4\text{He}$ demands 2.62×10^{11} helium atoms per second. One watt attributed to the ordinary $\text{d}(\text{d},\text{p})\text{t}$ branch demands 1.5×10^{12} tritons per second — and, per Chapter I, a lethal accompaniment of neutrons. Run the equation backwards on a tritium claim of 10^9 atoms per hour and you get about two-tenths of a microwatt: real forensic evidence, perhaps, but evidence for a process millions of times too feeble to explain the watt-scale heat it is offered alongside. Every section below applies this test, and the scorecard in §8 records the verdicts.

Two honest complications will recur. First, products can hide: helium born inside a palladium cathode may stay dissolved in the metal, so the gas-borne rate is only a retention-corrected fraction of production. Second, products can arrive uninvited: helium is in the air, tritium is in heavy water stocks, and trace elements are on fingerprints and tweezers. The whole art of this chapter's experiments — and of their critics — is the fight over those two corrections.

§2 Helium-4 correlated with heat: the field's best card

If the heat of Chapter III is real and nuclear, and neutrons and gammas are absent (Chapter I's dead-graduate-student argument), then by elimination the field's favored hypothesis is $d+d \rightarrow {}^4\text{He}$ with the energy somehow delivered as heat. That hypothesis makes a sharp prediction: helium in the cell gas at 2.62×10^{11} atoms per second per watt, in heat-producing cells and only in heat-producing cells.

The canonical test is Melvin Miles and co-workers at the Naval Air Warfare Center, China Lake (with Benjamin Bush and J. J. Lagowski at the University of Texas), published in the peer-reviewed *Journal of Electroanalytical Chemistry* in 1993. Effluent gas from Pd/D₂O electrolysis cells was collected in flasks and sent for helium analysis *blind* — the analysts did not know which flasks came from heat-producing runs. Miles reported helium in the gas of heat-producing cells at 10^{11} – 10^{12} atoms per second per watt of excess power — a band that brackets the commensurate rate — while H₂O control cells and no-heat runs showed none. Across three campaigns, his summary count was 18 of 21 heat-producing experiments with helium detected, and 30 of 33 experiments consistent overall (heat with helium, or no heat and no helium).

Two later programs report the same order of magnitude. Michael McKubre's group at SRI International replicated the "Case" gas-loading configuration (palladium on carbon catalyst under D₂ at ~170–200 °C) in metal-sealed vessels and reported helium growing monotonically with integrated excess heat — in some runs rising *above* the ~5.2 ppm atmospheric level, which in-leakage cannot do, since leaks equilibrate toward ambient rather than past it. In SRI's electrolysis run "M4," the raw ratio came out near 31 MeV per helium atom; after anodic stripping and D₂O flushing to recover helium retained in the cathode, SRI reported recovering approximately 104% of the helium expected at 23.8 MeV per atom. At ENEA Frascati, Apicella, Violante and co-workers reported a preliminary correlation of laser-triggered excess energy with helium rising above instrument background, using high-resolution mass spectrometry

to separate ^4He (4.0026 u) from the D_2 molecule (4.028 u) — the mass interference that haunts all helium work. Edmund Storms's 2010 review in *Naturwissenschaften* aggregates roughly a dozen heat–helium data sets and argues the slope clusters near 25 ± 5 MeV per atom.

NOTE — WHERE EACH CLAIM LIVES

Venue matters here more than anywhere else in the book. Miles (1993) and Storms (2010) are journal peer-reviewed. But the *tightest* Q-value agreements — SRI's M4 at "104% of 23.8 MeV," the Case replications, the ENEA correlation — were published in ICCF conference proceedings and internal EPRI/SRI reports, not refereed journals. The field's best quantitative numbers are, by the standards of Chapter I's sources, its least independently audited.

Now the critiques, which are serious. First, *the background problem*: the helium concentrations in most of these measurements sit near or below the ≈ 5.2 ppm helium content of ordinary air, so atmospheric in-leakage — through seals, or by diffusion through the very glass of Miles's collection flasks — is a mundane candidate source. This was the principal objection of the more negative reviewers in the 2004 DOE panel (Chapter II), and it is why the above-ambient Case results are argued about so fiercely. Second, *the retention circularity*: helium born in the outer microns of a palladium cathode partly stays there, so analyses introduce a retention fraction f_{ret} — and when f_{ret} is *fitted* to bring the slope to 23.85 MeV, the agreement is partly assumed rather than measured. Formally, if only a fraction f_{ret} of the helium reaches the gas, the detected rate and the apparent energy-per-atom become

$$R_{\text{det}} = f_{\text{ret}} \frac{P_{\text{xs}}}{Q}, \quad Q_{\text{app}} \equiv \frac{P_{\text{xs}}}{R_{\text{det}}} = \frac{Q}{f_{\text{ret}}}.$$

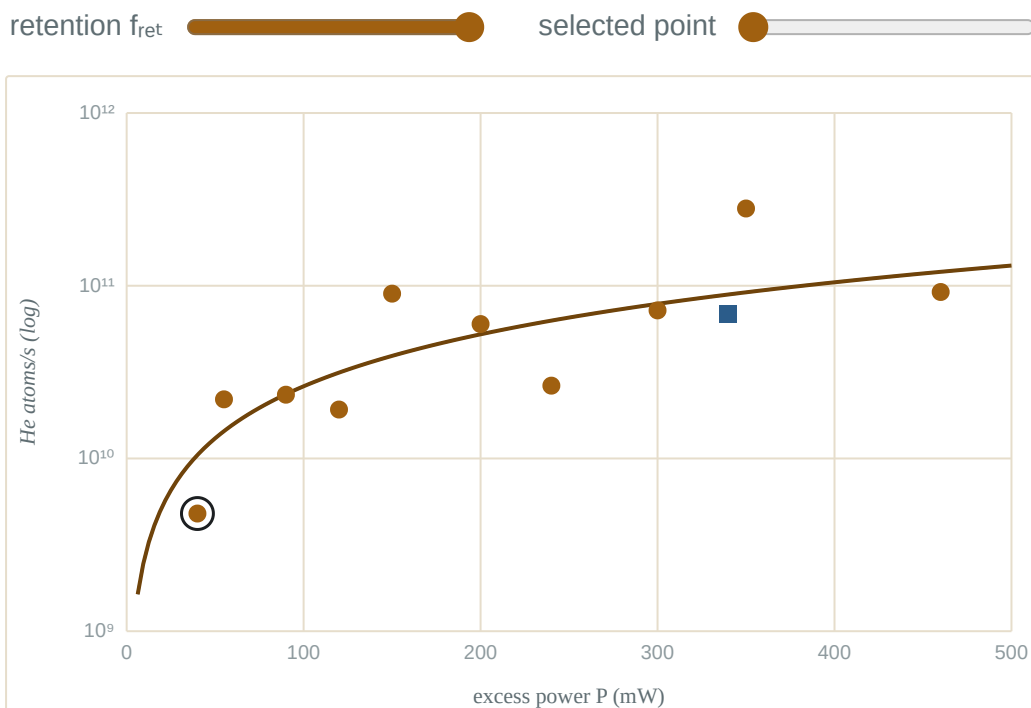
WHERE

- R_{det} Scalar, s^{-1} : the helium arrival rate actually measured in the gas phase.
- f_{ret} Dimensionless fraction (0 to 1): the released — i.e. non-retained — share of the helium produced. Read this way, SRI M4's raw ratio of ≈ 31 MeV per detected atom corresponds to $f_{\text{ret}} \approx 0.77$ before stripping ($23.85/31$ — an inference from the ratio itself, not a number SRI measured independently).
- Q_{app} The *apparent* MeV-per-detected-atom. At $f_{\text{ret}} = 0.5$, a true 23.85 MeV process masquerades as 47.7 MeV per detected atom — which is why raw slopes above 23.85 are read by proponents as retention and by critics as a fudge factor.
-

Third, the calorimetric denominator itself is contested: Kirk Shanahan has argued (in a peer-reviewed exchange in the *Journal of Environmental Monitoring*, 2010, answered by Marwan and ten co-authors in the same volume) that calibration-constant shifts could manufacture the excess power against which the helium is correlated. The widget below lets you play the whole argument yourself: the data band, the commensurability line, and the retention slider that tilts it.

INTERACTIVE · THE HEAT-HELIUM LEDGER

slide retention · select a point



Miles-class (representative): $P = 40 \text{ mW} \cdot R = 4.8 \times 10^9 \text{ atoms/s}$

implied **52.0 MeV** per detected atom; at $f = 1.00$, **52.0 MeV** per produced atom (2.18× the 23.85 MeV benchmark)

line check: at 100 mW the $R = f \cdot P/Q$ line sits at $2.6 \times 10^{10} \text{ atoms/s}$

● Miles-class (representative) ■ SRI M4 raw — $R = f \cdot P/Q$ --- full release ($f = 1$)

Excess power versus helium detection rate on a log axis, against the commensurability line $R_{\text{det}} = f_{\text{ret}} P_{\text{xs}}/Q$ with $Q = 23.85 \text{ MeV}$: at $P_{\text{xs}} = 100 \text{ mW}$ and full release the line sits at $2.62 \times 10^{10} \text{ atoms/s}$, and at $f_{\text{ret}} = 0.5$ each detected atom masquerades as 47.7 MeV. The round points are *representative* Miles-class points drawn from his reported 10^{11} – $10^{12} \text{ atoms s}^{-1} \text{ W}^{-1}$ band (J. Electroanal. Chem. 346, 99 (1993)) — illustrative placements, not digitized data — and the square point carries SRI M4's reported raw ratio of $\approx 31 \text{ MeV}$ per atom (conference-grade source). Select a point to read its implied MeV per detected atom and judge for yourself how much work the retention slider does.

§3 The gamma problem: 24 MeV with nowhere to go

Suppose the correlation is real. Then the field's own best evidence creates its sharpest theoretical crisis. In vacuum, $d+d \rightarrow {}^4\text{He}$ is not merely rare (a $\sim 10^{-7}$ branch, Chapter I) — it is *obligated by kinematics* to emit its energy as a 23.8 MeV gamma ray, because a lone helium nucleus cannot carry away 23.85 MeV of energy while conserving momentum. No LENR experiment reports 23.8 MeV gammas. So the heat–helium hypothesis requires the energy to leave through some unknown channel that delivers megaelectronvolts to the lattice as mere heat.

How tightly is that unknown channel constrained? Peter Hagelstein — MIT electrical engineer and the field's most mathematically serious theorist — answered in a peer-reviewed 2010 paper in *Naturwissenschaften*. Take the heat–helium correlation at ~ 24 MeV per atom as input, and suppose each helium nucleus is born with kinetic energy E_α . A fast alpha ploughing through deuterated palladium knocks deuterons forward, and some of those strike other deuterons hard enough to fuse by the ordinary $d(d,n){}^3\text{He}$ branch — so a fast-alpha scenario predicts *secondary neutrons* in calculable proportion. Measured neutron emission from Fleischmann–Pons-type cells is bounded between roughly 0.008 and 0.8 neutrons per joule of excess energy. Working the secondary-yield calculation backwards:

$$0.008 \leq Y_n \leq 0.8 \text{ n/J} \implies E_\alpha \lesssim 6.2\text{--}20.2 \text{ keV} \ll Q = 23.85 \text{ MeV}$$

WHERE

Y_n Measured upper limits on neutrons emitted per joule of excess energy (n/J), from published searches on active cells.

E_α The birth kinetic energy of the ${}^4\text{He}$ nucleus (keV). The bound is the largest E_α consistent with the neutron limits, given standard knock-on and secondary-fusion cross sections.

Read the inequality slowly: more than 99.9% of the reaction energy must appear in a form that is neither a gamma ray nor kinetic energy of any particle — parceled out, somehow, in quanta no larger than tens of keV. Nothing in accepted nuclear physics or QED does this. The constraint is itself mainstream-published and widely accepted *conditional on the correlation being real*; it is the reason Chapter IX's theories are all, in one way or another, machines for shredding one 24 MeV quantum into a billion phonons.

AHEAD OF THEORY

Ledger item #1 for Chapter IX. The heat–helium correlation, if real, is an experimentally claimed Q-value with no allowed exit channel: two-body kinematics demands a gamma that is not seen, and Hagelstein's constraint (Naturwissenschaften 97, 345 (2010)) caps every particle's energy at $\sim 10^{-3}$ of the reaction energy. No accepted mechanism fragments a nuclear quantum this way. This is the single sharpest gap between the field's best evidence and known physics.

§4 Tritium: real detections, impossible bookkeeping

Tritium should be easy. It is rare in nature, unmistakable in a liquid scintillation counter, and produced by half of ordinary d+d fusions. The tritium record is accordingly the field's strangest: clear positive detections in respectable venues, at levels that explain essentially none of the heat.

The cleanest single paper is Fritz Will, Krystyna Cedzynska and Denton Linton in the *Journal of Electroanalytical Chemistry* (1993): four out of four palladium-wire cathodes of one preparation, electrolyzed in tightly closed D₂SO₄ cells, yielded up to 2.1×10^{11} tritium atoms over seven days against a 4×10^9 -atom background —

with high deuterium loading ($x_{\text{D/Pd}}$ above about 0.8) apparently necessary, and parallel H_2SO_4 control cells showing nothing. Will, a respected electrochemist (formerly of General Electric), also published closed-system assays showing commercial palladium does not arrive pre-loaded with hidden tritium — pre-empting the obvious contamination objection to his own result. Earlier and larger in bulk is the corpus from the Bhabha Atomic Research Centre in India: Iyengar and Srinivasan's overview in *Fusion Technology* (vol. 18, 1990; the journal was peer-reviewed, though notably receptive to cold fusion in that era) collects reports from roughly a dozen independent BARC teams — tritium in electrolytic and gas-loaded palladium and titanium, imaged by autoradiography on titanium targets. At Los Alamos, Thomas Claytor's group reported tritium generation from pulsed deuterium plasma discharges on palladium wires, at rates of order 10^9 – 10^{10} atoms per hour, with D_2 runs up to 25 times matched H_2 controls — though in laboratory reports and conference papers, not refereed journals.

Now the bookkeeping. Run §1's equation backwards: 2.1×10^{11} tritium atoms in seven days is a production rate of 3.5×10^5 atoms per second, which at the d(d,p)t branch's 4.03 MeV corresponds to about 2×10^{-7} watts. The excess-heat claims of Chapters II–III are watts. The tritium, taken at face value, explains roughly one part in 10^7 of the heat — so tritium and heat are not two witnesses to one story; they are two different anomalies that happen to share apparatus. Worse, the BARC teams' headline quantitative result is that neutrons accompany their tritium at a ratio near 10^{-8} , where vacuum d+d physics demands approximately one — an eight-order-of-magnitude branching inversion.

AHEAD OF THEORY

Ledger item #2. Claimed tritium with neutron-to-tritium ratios of 10^{-7} – 10^{-9} (BARC; Claytor) inverts the near-50/50 d+d branching of Chapter I by up to nine orders of magnitude. No mechanism in accepted few-body nuclear physics suppresses the ${}^3\text{He} + \text{n}$ channel relative to t+p at low energy by anything like this factor. If the tritium is nuclear, the nuclear physics in the lattice is not vacuum d+d physics — which is also, of course, the skeptic's simplest reason to doubt that it is nuclear at all.

The critiques here are not hypothetical; they landed. Kevin Wolf of Texas A&M, whose tritium findings were among the most cited, traced his own signals to pre-existing contamination of the cathodes — "my tritium was an impurity" (*Science* 248, 1301 (1990)). In the same issue, Gary Taubes alleged that tritium results from another Texas A&M laboratory were consistent with deliberate spiking (*Science* 248, 1299 (1990)); university inquiries ruled out proven fraud but found the evidence unconvincing — a shadow the episode never fully escaped. Open cells also concentrate the trace tritium naturally present in heavy water as electrolysis preferentially boils off the lighter isotopes. Will's closed cells and assayed materials were designed to defeat exactly these objections, and no published critique has specifically overturned his paper; but neither has any independent laboratory replicated his protocol at journal level, and the BARC program's results faded through the 1990s without a modern, blinded confirmation.

§5 Transmutation: the boldest claim, the thinnest margin

The most extraordinary claim in this book comes from Mitsubishi Heavy Industries. Yasuhiro Iwamura and colleagues reported — in the mainstream, peer-reviewed *Japanese Journal of Applied Physics* (2002) — that when deuterium gas permeates for about a week through a nanostructured foil (40 nm of palladium over alternating CaO and Pd layers on bulk palladium) with a ~100-nanogram layer of cesium on its surface, the cesium steadily *disappears* and praseodymium *appears* in its place, at the 10^{14} – 10^{15} -atom scale, tracked day by day with XPS and confirmed with ToF-SIMS and ICP-MS. Strontium, similarly treated, appeared to become molybdenum. The pattern is arresting:



WHERE

- ^{133}Cs , ^{141}Pr The sole stable isotopes of cesium and praseodymium. The claimed step is +4 protons and +4 neutrons — as if two entire alpha particles were absorbed at once.
- ^{88}Sr , ^{96}Mo The same +8-nucleon jump. The claimed molybdenum was dominated by ^{96}Mo , whose natural abundance is only 16.7% — an isotopic signature no ordinary contamination should carry.
-

The claimed controls behave as a believer would hope: no permeation, no praseodymium; hydrogen instead of deuterium, no praseodymium; no CaO layer, no praseodymium. And uniquely in this field, there is a peer-reviewed partial replication by an unaffiliated industrial laboratory: Hioki and colleagues at Toyota Central R&D Labs (*Jpn. J. Appl. Phys.* 52, 107301 (2013)) implanted the cesium by ion beam — closing the wet-chemistry contamination route — and reported praseodymium increases after deuterium permeation relative to unpermeated controls, though at substantially smaller quantities than Mitsubishi's, not always clear of their own contamination scatter, and framed by the authors themselves with conspicuous caution.

Against this stands the best-resourced replication attempt in the field's history: the U.S. Naval Research Laboratory (David Kidwell, Kenneth Grabowski and colleagues) worked with Mitsubishi's own protocols, failed to reproduce the effect, discovered environmental praseodymium contamination in the Mitsubishi laboratory itself (down to a contaminated pair of tweezers), and argued that the claimed quantities — 10^{14} – 10^{15} atoms is roughly 20–200 *nanograms* — sit comfortably within realistic contamination and mass-interference budgets (a ^{141}Pr peak is impersonated by cesium-oxide and barium-hydride molecular ions in SIMS and ICP-MS). Praseodymium is mononuclidic, so no isotope ratio can acquit it. The strongest counter-evidence would be the anomalous ^{96}Mo -dominated isotopics in the strontium channel, which contamination cannot mimic — but Kidwell disputed that very measurement as a misassignment, and it has never been independently

confirmed. Note the epistemic asymmetry: the positive claims are journal-published, while the NRL critique was presented at ICCF-15 (2009) and circulated in reports but — per the available record — never itself passed through peer review. Both sides of the dispute are thus incompletely audited.

AHEAD OF THEORY

Ledger item #3. Even granting the observations, no theory — mainstream or field-internal — predicts a clean, radiation-free transfer of exactly 8 nucleons whose rate depends on a 2-nanometer CaO layer. Widom–Larsen neutron-capture chains and Takahashi's multi-deuteron condensates (Chapter IX) both predict product distributions different from what Iwamura reports. The claim is ahead of even the field's own theories.

§6 Energetic particles: pits in plastic

From 2007 the SPAWAR group in San Diego (Pamela Mosier-Boss, Stanislaw Szpak, Frank Gordon, Lawrence Forsley) pursued a clever variant: instead of loading bulk palladium for weeks, *co-deposit* palladium and deuterium simultaneously from a $\text{PdCl}_2/\text{LiCl}/\text{D}_2\text{O}$ bath onto a thin cathode pressed against a CR-39 track detector — chemistry builds the loaded lattice atom by atom, and the plastic keeps a permanent, integrating record of any charged particle that strikes it. In the peer-reviewed *European Physical Journal — Applied Physics* (40, 293 (2007)), they reported copious etch pits on chips from D_2O co-deposition runs — morphologically consistent, they argued, with alpha particles of roughly 1 MeV and below — while H_2O runs, no-current soaks, and CuCl_2 substitutions produced few or none. (A citation warning for readers chasing sources: this paper is often mis-cited as "Eur. Phys. J. A 40, 293 (2009)" — a different journal entirely; the correct venue is EPJ

Applied Physics, 2007.) In *Naturwissenschaften* (96, 135 (2009)) they added the exotic centerpiece: a handful of *triple tracks* — three lobes radiating from a common vertex, the textbook signature of a carbon nucleus in the plastic shattered into three alphas, a reaction requiring neutrons above roughly 9.6 MeV. Deuterium–deuterium fusion makes 2.45 MeV neutrons; 9.6 MeV implicates something like the 14.1 MeV neutrons of d+t fusion — which would require a tritium-producing step first, stacking one unaccepted mechanism atop another.

The replication story is instructive because it splits cleanly in two. The *observation* replicates: in the 2007 "Galileo Project" organized by writer Steven Krivit, multiple laboratories — including the avowedly skeptical Earthtech in Austin — reproduced the pitting readily. The *interpretation* does not. Ludwik Kowalski, in a peer-reviewed comment (*Eur. Phys. J. Appl. Phys.* 44, 287 (2008); SPAWAR reply immediately following), showed the dominant pits' diameter-versus-depth systematics are inconsistent with alpha particles or any lighter nuclear projectile. Earthtech's protected-detector tests found CR-39 in contact with electrolyte suffers gross chemical attack — etch rates some fifty-fold above normal — and that chips shielded from electrolyte contact showed no tracks above background (reported on the web, not peer-reviewed). Candidate mundane sources in the published exchange include chemical attack by the electrolyte, micro-explosions of recombining D₂/O₂ bubbles, and corona damage from the cell's electric field. The triple tracks are too rare — of order a few per detector — to settle statistically, and even taken at face value they imply neutron rates around a few per hour: some ten orders of magnitude below heat-commensurate rates. Whatever made them, it cannot be the main event.

§7 Neutrons: the clearest negative in the field

Neutrons are where the field's forensics are least ambiguous — against it. The one *Nature*-level positive is Steven Jones and colleagues (BYU, 1989; the "other" cold

fusion announcement of Chapter II): a neutron spectrometer signal near 2.45 MeV — the $d(d,n)^3\text{He}$ fingerprint — from Ti/Pd electrolysis, implying a fusion rate of order 10^{-23} per deuteron pair per second. Note the scale: that is a picowatt-class effect, about 10^{13} times *smaller* than Fleischmann–Pons heat requires. Jones's claim, modest as it was, did not survive: Moshe Gai's low-background counting experiment (Yale/BNL, *Nature* 340, 29 (1989)) set limits well below the BYU signal, and Michael Salamon's team spent five weeks with detectors beside Pons's *own operating cells* (*Nature* 344, 401 (1990)), bounding neutron emission at roughly one per second and fusion power at 10^{-12} – 10^{-6} watts — during periods the cells were allegedly active. The dramatic 1989–1990 reports of neutron *bursts* from cycled titanium–deuterium gas systems (BYU–Los Alamos) were later withdrawn when the signals were traced to high-voltage breakdown artifacts in the ^3He counter tubes. And the best-instrumented modern search, the Google-funded program (Berlinguette et al., *Nature* 570, 45 (2019); Chapter VIII), fielded state-of-the-art neutron diagnostics on highly loaded Pd systems and found nothing anomalous in the parameter space it could reach.

Why do sporadic ultra-low-level neutron claims neither save nor damn the heat claims? Because of §1's arithmetic, run in both directions. They cannot *save* the heat: a few neutrons per hour, or even per second, converts to picowatts–nanowatts of d+d fusion, off by ten to twelve orders of magnitude from claimed watts. But they cannot *damn* it either, on the field's own terms: the aneutronic-helium hypothesis of §2–§3 predicts almost no neutrons, so neutron silence is consistent with it — which is precisely what makes that hypothesis so hard to falsify with a neutron counter, and why the heat–helium correlation, not the neutron ledger, is where the field's case must stand or fall. What the neutron record *does* establish, beyond reasonable dispute, is Chapter I's conclusion: whatever happens in these cells, it is not ordinary two-body d+d fusion at watt scale.

§8 The scorecard

Here is the chapter in one table. "Commensurate?" asks §1's question: does the product rate match the claimed heat at the known Q-value? Verdicts are this book's, argued in the sections above.

Product	Strongest citation	Replication	Commensurate?	Standing crit
^4He + heat	Miles, J. Electroanal. Chem. 346, 99 (1993)	SRI, ENEA — conference-grade only	Claimed yes (~ 24 MeV/atom, with retention corrections)	Near-ambient levels; in-leakage fitted retention contested calorimetry
Tritium	Will, J. Electroanal. Chem. 360, 161 (1993); BARC, Fusion Technol. 18 (1990)	Multi-lab 1989–95; none modern, journal-level	No — $\sim 10^{-7}$ of claimed heat; n/t inverted $\sim 10^8\times$	Demonstrated contamination (Wolf); spiking allegation (Tau open-cell enrichment)
Transmutation	Iwamura, Jpn. J. Appl. Phys. 41, 4642 (2002)	Partial (Hioki/Toyota 2013); NRL failed	Not tested against heat; quantities are nanogram-scale	Contamination masses; mass interference; N found lab Pr contamination
CR-39 tracks	Mosier-Boss, Eur. Phys. J. Appl. Phys. 40, 293 (2007); Naturwiss. 96, 135 (2009)	Pitting replicates (incl. skeptics); interpretation does not	No — implied fluxes $\sim 10^{10}$ below heat-commensurate	Chemical/mechanical damage (Kowa Earthtech prototype chip nulls)
Neutrons	Jones, Nature 338, 737 (1989)	Contradicted (Gai; Salamon in Pons's lab); bursts withdrawn	No — deficit of 8–12 orders vs any heat claim	Instrumental artifacts; limits below the claim

The honest bottom line, in the register discipline of this book. *Established*: no nuclear product has ever been observed commensurate with claimed excess heat through ordinary d+d channels, and neutron emission at heat-commensurate levels is excluded by multiple peer-reviewed null measurements — including in Pons's own laboratory.

Contested: the heat–helium correlation is the strongest strand — peer-reviewed at its core, with a claimed slope bracketing 23.85 MeV per atom — but its tightest numbers live in conference proceedings, its signals flirt with atmospheric background, and its retention correction is partly circular; tritium and transmutation have journal-published positives standing against demonstrated contamination episodes and failed high-quality replications; none of it reaches proof standard. And several strands, *if real*, are not merely unexplained but ahead of any theory, mainstream or otherwise — the 24 MeV with no exit channel, the inverted branching ratio, the 8-nucleon jumps. Chapter IX will collect those IOUs. The *Gamow factor and the error bar* frame it as ever: nothing in this chapter enlarges the tunneling probability; everything in it is a fight over whether the measurement — of atoms this time, not watts — can be trusted at parts-per-million stakes. First, though, Chapters V–VII visit the one variable every experimenter agrees matters: how much deuterium you can force into the metal, and what it does when it's there.

FURTHER READING

Miles, M. H., Hollins, R. A., Bush, B. F., Lagowski, J. J. & Miles, R. E. *Correlation of excess power and helium production during D₂O and H₂O electrolysis using palladium cathodes*, Journal of Electroanalytical Chemistry 346, 99–117 (1993) — the canonical heat-helium paper; blind helium analyses, 18/21 correlation.

Hagelstein, P. L. *Constraints on energetic particles in the Fleischmann-Pons experiment*, Naturwissenschaften 97, 345–352 (2010) — the peer-reviewed bound forcing any helium birth energy below ~20 keV; §3's equation, DOI 10.1007/s00114-009-0644-4.

Will, F. G., Cedzynska, K. & Linton, D. C. *Reproducible tritium generation in electrochemical cells employing palladium cathodes with high deuterium loading*, Journal of Electroanalytical Chemistry 360, 161–176 (1993) — the cleanest closed-cell tritium claim, with materials assays against contamination.

Iwamura, Y., Sakano, M. & Itoh, T. *Elemental analysis of Pd complexes: effects of D₂ gas permeation*, Japanese Journal of Applied Physics 41, 4642–4650 (2002) — the Cs→Pr and Sr→Mo transmutation claims, DOI 10.1143/JJAP.41.4642.

Mosier-Boss, P. A., Szpak, S., Gordon, F. E. & Forsley, L. P. G. *Triple tracks in CR-39 as the result of Pd-D co-deposition: evidence of energetic neutrons*, Naturwissenschaften 96, 135–142 (2009) — the triple-track claim; read with Kowalski's comment, Eur. Phys. J. Appl. Phys. 44, 287 (2008). DOI 10.1007/s00114-008-0449-x.

Jones, S. E. et al. *Observation of cold nuclear fusion in condensed matter*, Nature 338, 737–740 (1989) — the low-level neutron claim; read with Salamon et al., Nature 344, 401–405 (1990), the in-situ null in Pons's laboratory. DOI 10.1038/338737a0.

Storms, E. *Status of cold fusion (2010)*, Naturwissenschaften 97, 861–881 (2010) — a proponent's peer-reviewed aggregation of the heat-helium data sets; useful for the claims map, to be read with this chapter's critiques alongside. DOI 10.1007/s00114-010-0711-x.

Chapter production note: researched, drafted, and adversarially verified by Claude subagents (draft agent: `claude-fable-5`; verification agent: `claude-fable-5`, citations web-checked), orchestrated by `claude-fable-5`. See the [methods appendix](#) for the full research process.

CHAPTER V

Loading the Lattice: Palladium as a Materials Problem

Strip away the nuclear controversy and an experimental question remains that any metallurgist can love: how much deuterium can you force into a piece of palladium, how do you know it stayed, and why do some pieces take it while their twins refuse?

BEFORE WE START

This chapter puts the nuclear question aside almost entirely. Palladium soaked in deuterium is a real and remarkable material — a dense metallic hydride that has been studied since 1866 — whether or not anything nuclear ever happens inside it. We will meet its two solid phases and the stress of converting between them, the electrochemical trick that turns a volt of cell overpotential into the equivalent of enormous gas pressure, and the surprisingly delicate art of measuring how much deuterium is actually inside a cathode while the experiment runs.

The reason this materials science belongs in a book about contested claims is a single empirical assertion, made most precisely by Michael McKubre's group at SRI International: that excess heat appears *only* when the deuterium-to-palladium ratio is pushed above roughly 0.875 — a regime most 1989 replication attempts never reached, and, as a well-funded 2015–2019 program discovered, one that is genuinely hard to reach on purpose. If you believe the claim, it converts a scandal into a specification. If you doubt it, it is an unfalsifiable escape hatch. Either way, it made loading the field's central materials number, and this chapter examines how that number is produced, measured, and argued over.

TERMS YOU'LL MEET

palladium (Pd)

A silvery noble metal of the platinum group, famous since 1866 for absorbing hundreds of times its own volume of hydrogen gas.

face-centered cubic (fcc)

Palladium's crystal structure: atoms at the corners and face centers of a repeating cube.

interstitial site

A gap between the atoms of a crystal lattice where a small foreign atom can sit.

octahedral site

The interstitial gap at the center of six surrounding metal atoms in an fcc crystal — where hydrogen sits in palladium. There is exactly one per Pd atom.

metal hydride

A metal with hydrogen dissolved into (or bonded within) its lattice; PdD is an *interstitial* hydride, still metallic and conducting.

loading ratio (x)

The atomic ratio $x = x_{\text{D/Pd}}$: deuterium atoms per palladium atom. The chapter's central variable.

α phase

The dilute solid solution: a little D dissolved in nearly pure Pd, x of order a few percent at room temperature.

β phase

The concentrated hydride phase, $x \gtrsim 0.6$ at room temperature, with a visibly expanded lattice.

miscibility gap

A range of compositions a system cannot adopt as one uniform phase; between α and β , palladium splits into coexisting regions of both.

phase diagram

The map of which phase is stable at each composition, temperature, and pressure.

critical point

The temperature and pressure above which two coexisting phases merge into one; for Pd–H, near 290 °C and ~20 bar.

lattice parameter

The edge length of the repeating crystal cube, measured in ångströms; it swells as hydrogen

	loads.
Sieverts' law	For dilute dissolution, absorbed atomic hydrogen scales as the <i>square root</i> of gas pressure — because H_2 must split into two atoms to enter.
fugacity	The "effective pressure" of a gas in thermodynamic bookkeeping: the pressure an ideal gas would need to match the real gas's chemical driving force.
overpotential	The extra voltage, beyond the equilibrium value, applied at an electrode to drive its reaction at a finite rate.
current density (i)	Electric current per unit electrode area, here in mA per cm^2 ; the knob that sets how hard electrolysis pushes.
Faraday constant (F)	The charge of a mole of electrons, 96,485 coulombs — the conversion rate between current and reaction rate.
surface poison	A trace impurity that adsorbs on the cathode and alters its surface chemistry — blocking either deuterium entry or deuterium escape.
resistance-ratio method (R/R_0)	Inferring loading from the electrical resistance of the cathode, normalized to its hydrogen-free value — cheap, continuous, and treacherously two-valued.
four-terminal measurement	Passing current through two leads and sensing voltage with two others, so contact resistance doesn't contaminate the reading.
operando X-ray diffraction	Measuring the crystal's lattice parameter by X-ray diffraction <i>while the experiment runs</i> , giving a direct, calibration-free composition gauge.
hydrogen embrittlement	The cracking and weakening of metals infiltrated by hydrogen — the same lattice strain that makes high loading self-defeating.
dislocation	A line defect in a crystal along which planes of atoms are mis-stacked; loading stress multiplies

	them.
grain	A single-crystal region within an ordinary polycrystalline metal; its boundaries are fast paths and trap sites for hydrogen.
crystallographic texture	A statistical preference of grains for particular orientations, set by rolling and annealing history.
annealing	Heating a metal to heal defects and reset its microstructure — one of the levers in cathode-preparation recipes.
co-deposition	Electroplating palladium from solution <i>while</i> deuterium evolves, so the metal is born loaded instead of being filled afterwards.
dendrite	The branching, high-surface-area metal deposit that co-deposition produces — nothing like a polished bulk cathode.

§1 A metal that drinks deuterium

Thomas Graham noticed in 1866 that palladium absorbs prodigious volumes of hydrogen; it remains the textbook interstitial hydride. Deuterium enters the metal as single atoms and sits in the octahedral sites of palladium's fcc lattice — one such site per Pd atom, so complete filling corresponds to one D per Pd. The composition is tracked by a single number:

$$x \equiv x_{\text{D/Pd}} = \frac{N_{\text{D}}}{N_{\text{Pd}}}, \quad 0 \leq x \lesssim 1.$$

WHERE

x The loading (dimensionless): the atomic ratio of dissolved deuterium to palladium, averaged over the cathode.

N_D, N_{Pd} Counts of deuterium and palladium atoms in the sample.

The Pd–H(D) phase diagram — assessed comprehensively by Manchester, San-Martín and Pitre (1994), with the thermodynamics reviewed by Flanagan and Oates (1991) — has two solid phases below a critical point near 290 °C and about 20 bar. The α phase is a dilute solution, a few percent D at room temperature. The β phase is the true hydride, beginning around $x \approx 0.6$ at room temperature. Between them lies a miscibility gap: a loading cathode does not pass smoothly from empty to full but coexists as α and β regions, with a phase front sweeping through the metal. In the dilute limit the uptake follows Sieverts' law, $x \propto \sqrt{P}$ — a first hint that the currency of loading is the chemical potential of *atoms*, not the pressure of molecules.

Loading physically swells the crystal: the lattice parameter grows from 3.89 Å in pure Pd to about 4.03 Å in the β phase — roughly 3.5% in length, 10% in volume. That expansion has two consequences that run through the whole chapter. First, density: pure Pd packs 6.8×10^{22} atoms per cm³, so a cathode loaded to $x \approx 0.9$ holds roughly $5\text{--}6 \times 10^{22}$ deuterium atoms per cm³ — comparable to liquid deuterium itself ($\sim 5 \times 10^{22}$ atoms/cm³). Electrochemistry at room temperature quietly achieves a deuterium density that otherwise requires cryogenics. Second, stress: a 10% volume mismatch between coexisting α and β regions is enormous by metallurgical standards, and §5 shows it is the main reason high loading is so hard to keep.

NOTE

Everything in this section is established materials science, replicated for over a century and summarized in mainstream reviews. The controversy in this chapter begins only when loading numbers are *correlated with heat* (§4). Keep the two registers separate: the hydride is real; the threshold is a claim.

§2 Volts as pressure: electrochemical loading

Why electrolyze heavy water instead of simply pressurizing the metal with D₂ gas? Because a cathode is a pressure amplifier. During electrolysis, D₂O is reduced at the palladium surface, depositing adsorbed D atoms that either pair up and bubble away as D₂ or dissolve into the lattice. Driving the electrode harder — applying a cathodic overpotential η_c — raises the electrochemical potential of those adsorbed atoms. If the surface stayed in equilibrium with D₂ gas, the gas would need an effective pressure — a fugacity — larger by an exponential factor:

$$\frac{f_{\text{D}_2}}{f^\circ} = \exp\left[\frac{2F\eta_c}{RT}\right], \quad \log_{10} \frac{f_{\text{D}_2}}{f^\circ} = \frac{2F\eta_c}{RT \ln 10} \approx 33.8 \frac{\eta_c}{1 \text{ V}} \quad (T = 298 \text{ K})$$

WHERE

f_{D_2} The equivalent fugacity of D₂ (scalar, pressure units): the gas pressure that would exert the same chemical driving force on the metal.

f° The reference fugacity, 1 atm.

F Faraday constant, 96,485 C mol⁻¹ — a mole of electron charge.

η_c Cathodic overpotential (V), taken positive here; the factor 2 counts the two electrons transferred per D₂ molecule.

R Gas constant, 8.314 J mol⁻¹ K⁻¹.

T Absolute temperature (K).

Taken literally, the numbers are outlandish: one volt of overpotential is "worth" 10^{34} atmospheres. The honest reading is that this is the *ideal, reversible ceiling* — the Nernstian bound assuming the adsorbed layer equilibrates with the gas. Real cathodes never approach it: the same surface kinetics that evolve D_2 bubbles vent the driving force, and careful measurements in the hydrogen-charging literature find *actual* equivalent fugacities of order thousands of atmospheres, not 10^{34} (for steel cathodes, Liu, Atrens and co-workers measured up to $\sim 6.5 \times 10^3$ atm — Corrosion Science 87, 239, 2014). Still, thousands of atmospheres from a bench-top power supply is the whole game: it is why an electrolytic cell, and not a gas bottle, was Fleischmann and Pons's instrument, and why loading in practice is a *steady state* — a tug-of-war between electrochemical injection and leakage back out through the same surface — rather than a filling of a tank.

THE SURFACE IS THE VALVE

Because injection and escape both happen at the electrode surface, the achievable loading is controlled less by voltage than by surface chemistry: whatever slows the recombination of adsorbed D atoms into D_2 keeps the "pressure" high. This is why trace electrolyte impurities — silicates dissolved from the glass cell, boron, aluminium, even the lithium of LiOD — figure so prominently in loading recipes, and why two chemically "identical" cells can equilibrate at very different x . The valve, not the pump, sets the level.

§3 Watching x in real time: the R/R_0 curve

You cannot weigh a cathode while it electrolyzes. The field's workhorse gauge is electrical: dissolved deuterium scatters conduction electrons, so the cathode's resistance rises as it loads. Measured with a four-terminal AC bridge and normalized

to the unloaded value, the ratio R/R_0 traces a well-established but awkward curve: it climbs to a maximum of about 2.0 near $x \approx 0.75$ for D/Pd, then *falls* with further loading — through roughly 1.7 near $x \approx 0.93$ and on down toward ~ 1.5 as $x \rightarrow 1$, per the calibration curve McKubre and Tanzella fitted to SRI's data (restated most completely in the field's house journal: JCMNS 29, 129, 2019). Physically, resistivity peaks where the part-filled interstitial sublattice is maximally disordered; as the sites approach complete filling, order returns and scattering drops.

The consequence is that the gauge is *two-valued*: a reading of $R/R_0 = 1.7$ is consistent with a mediocre loading of $x \approx 0.5$ or an excellent one of $x \approx 0.93$. To know which branch you are on, you must track the whole history — watch the ratio rise, crest, and descend — and trust that nothing (a crack, a temperature drift, a partial de-load) silently walked the sample back over the peak. Every quantitative loading claim of the 1990s runs through this inference. McKubre's group, who did more than anyone to calibrate and standardize the method, were explicit that it measures the *average* loading of the bulk, and published calibrations only alongside stated caveats; Zhang and co-workers devoted a seventeen-page paper to the method's failure modes (temperature coefficients, phase coexistence, surface damage, geometry changes from cracking; *J. Electroanal. Chem.* 528, 1, 2002).

The sharpest modern audit came from the Google-funded program (§6): Benck and co-workers, cross-checking resistance against operando X-ray diffraction — which reads composition directly off the swelling lattice parameter — found that at high loading the resistance proxy can seriously mislead, precisely in the regime where the contested claims live. A field whose central empirical law has a two-valued, artifact-prone instrument on its x -axis has a measurement problem before it has a physics problem — the "error bar" half of this book's refrain, transplanted from calorimetry to composition.

§4 The threshold claim: heat begins at 0.875

Now the contested part. Through the 1990s, McKubre's SRI group ran mass-flow calorimetry (Chapter III) on dozens of Pd cathodes whose loading, current, and interfacial deuterium flux were logged continuously. Out of that program came the field's most-quoted empirical regularity: excess power appeared only in heavily loaded cells, and its magnitude was parametrized as

$$P_{\text{xs}} = k (x - x^\circ)^2 (i - i^\circ) \left| \frac{dx}{dt} \right|, \quad x^\circ \approx 0.875, \quad i^\circ \approx 200\text{--}250 \text{ mA cm}^{-2}$$

WHERE

P_{xs}	Excess power (W): heat out beyond all accounted input, per Chapter III's bookkeeping.
x°	The claimed loading threshold (dimensionless): below it, no excess at any current. Onsets quoted between 0.84 and 0.88.
i, i°	Electrolytic current density and its claimed threshold (mA cm^{-2}).
dx/dt	The rate of change of loading — a deuterium <i>flux</i> through the interface; steady full cells reportedly made less heat than cells being actively worked.
k	A cell-specific proportionality constant. It was never given as a universal number — which is exactly what separates a parametrization from a law.

Read the fine print in the attribution. The flow-calorimetry methods and the D/Pd-versus-H/Pd comparison were peer-reviewed (McKubre et al., *J. Electroanal. Chem.* 368, 55, 1994); the correlation itself was published in conference proceedings (ICCF-3, 1993) and EPRI contract reports, and restated years later in the field's house journal (JCMNS 4, 32, 2011). Within the EPRI program the claim is stark: every cell that reached $x > 0.95$ showed excess heat, and no cell below 0.90 did. As a sociological explanation it is elegant — 1989's replicators, working with off-the-shelf palladium

and no loading diagnostics, would mostly have sat near the ambient equilibrium $x \approx 0.7$, comfortably below the claimed threshold, and found nothing.

The standing critiques are equally clear. The correlation comes from one laboratory's own data, parametrized post hoc; the constant k varies cell to cell; the x -axis is the two-valued resistance gauge of §3; and no hostile laboratory ever ran the protocol at $x \geq 0.95$ to check it — because, as §6 shows, reaching that regime is itself a research problem. Shanahan's calibration-shift critique of the calorimetry (Chapter III) applies here undiminished. So the threshold functions in the literature less as a tested law than as a boundary condition proponents place on replication: a claim that is simultaneously the field's best excuse and its most concrete, falsifiable materials prediction.

AHEAD OF THEORY

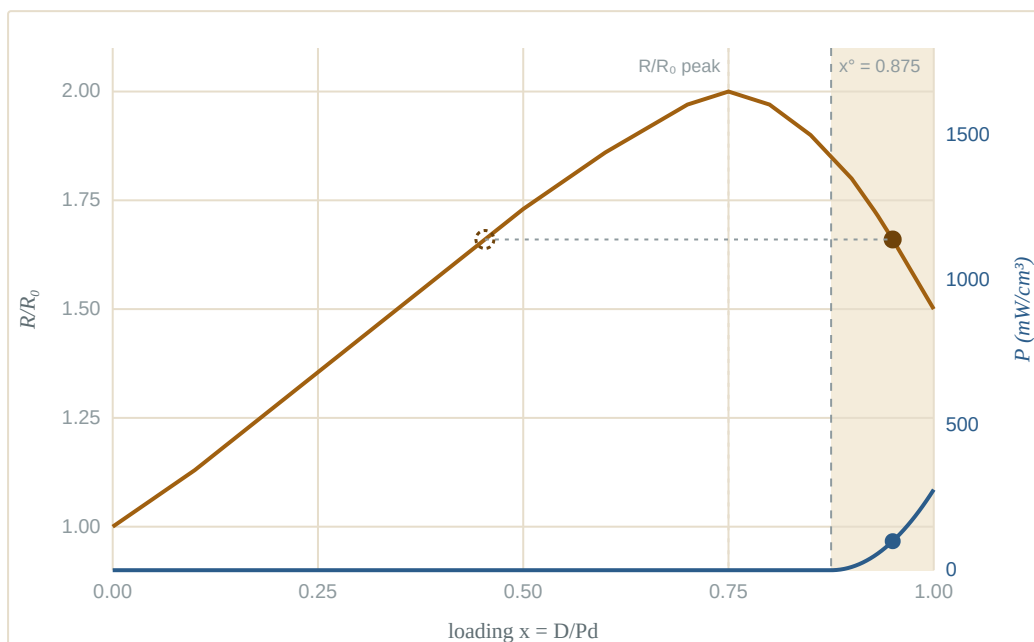
No theory — mainstream or fringe — *predicts* 0.875. The claimed threshold does not coincide with any known feature of the Pd–D phase diagram (the β phase is complete well below it; the ordered near-stoichiometric phases sit above it), and the quadratic dependence and flux term are purely descriptive. If the correlation is real, it is a lone empirical regularity awaiting a mechanism; if it is an artifact, it is a remarkably specific one. Chapter IX's ledger of results-ahead-of-theory carries it with that double flag.

INTERACTIVE · THE LOADING THRESHOLD

slide loading · slide current

target loading x

current density i (mA/cm²)



$x = 0.950$ · $R/R_0 = 1.66$ (same reading as $x = 0.45$) · $i = 250$ mA/cm²
predicted $P_{xs} \approx 100$ mW/cm³ of cathode

— R/R_0 calibration (left axis) — predicted P_{xs} at this i (right axis) | $x^\circ = 0.875$ threshold

The resistance gauge and the threshold claim in one picture. The amber curve is a representative R/R_0 calibration for D/Pd — anchored at the established peak of ≈ 2.0 near $x \approx 0.75$, descending through ≈ 1.7 at $x \approx 0.93$ toward ≈ 1.5 at $x = 1$ (following the polynomial calibration of McKubre & Tanzella, JCMNS 29, 129, 2019; exact falling-branch values vary between laboratories) — with the ghost marker showing the *other* loading that gives the same reading. The slate curve is SRI's empirical correlation $P_{xs} = k(x - 0.875)^2(i - i^\circ)$ with $i^\circ = 200$ mA cm⁻², the flux factor folded into k , and k normalized illustratively so that $x = 0.95$, $i = 250$ mA cm⁻² gives 100 mW per cm³ of cathode (the constant is cell-specific and was never published as universal — McKubre et al., ICCF-3 1993; JCMNS 4, 32, 2011). Slide the loading below 0.875 and the prediction collapses to zero at any current: the believer's one-picture story of why 1989's replications failed.

§5 Why nine-tenths is hard

If high loading were easy, the threshold claim would have been settled in a season. It is not easy, for reasons that are pure materials science. The $\alpha \rightarrow \beta$ conversion's 10% volume mismatch means a loading cathode is a solid trying to expand inside itself: the β front puts the interior under stresses that nucleate dislocations, multiply defects, and open microcracks — the same physics as hydrogen embrittlement. Every crack is a new internal surface where dissolved D can recombine to D_2 and escape; a cracked cathode is a leaky tank that no overpotential can fill. High loading is therefore self-limiting: push too fast and the act of loading destroys the capacity to load.

The protocols that emerged read like recipes because they are recipes: load gently at low current density to move the α/β front slowly; only then raise the current toward the claimed i° ; manage temperature (solubility falls as temperature rises); add trace species deliberately (aluminium, boron, silicates) to condition the surface valve of §2; avoid poisons that catalyze recombination. And beneath the protocol lies the *materials lottery*, the field's open secret: nominally identical cathodes behave differently, and only some batches "work." The most systematic response was Vittorio Violante's program at ENEA Frascati, which spent two decades correlating loading success and (claimed) excess-heat occurrence with measurable metallurgy — rolling and annealing schedules, grain size, hardness, and crystallographic texture (the fraction of grains near $\langle 100 \rangle$ orientation) — and claimed, in conference proceedings and reports, reproducible $x > 0.95$ in foils manufactured to spec, with ENEA-fabricated foils used in the SRI and Energetics replication campaigns of Chapter III. The metallurgy itself is solid and was adopted by other laboratories; the correlation with heat remains proceedings-grade — it has never been published in a mainstream refereed journal.

THE LOTTERY AS EVIDENCE

Both camps claim the batch-to-batch capriciousness as support. To proponents it shows the effect depends on uncontrolled microstructural variables — exactly what you expect if some rare defect geometry or impurity site is the active ingredient. To skeptics it is the classic signature of an artifact hunted through an uncontrolled parameter space: with enough cathodes and no pre-registered criterion, some batch will always "work." The materials science that would distinguish these — full characterization before, during, and after a blinded, adversarial run — is precisely what §8 specifies and what has never been done.

§6 The Google program buys a better gauge

The most consequential modern test of this chapter's claims came from the Google-funded, four-laboratory program (2015–2019) whose public summary appeared in *Nature* (Berlinguette et al., 570, 45, 2019). One of its three thrusts aimed squarely at the loading question: build cells that reach and *hold* $x_{\text{D/Pd}} \geq 0.875$, instrument them properly, and see. Two results matter here. First, the program's materials paper (Benck et al., *Chemistry of Materials* 31, 4234, 2019) reached a maximum loading of $x = 0.96 \pm 0.02$ — measured by operando X-ray diffraction rather than resistance, using aqueous, polymer, and ceramic solid electrolytes — and demonstrated along the way that the resistance-ratio proxy misleads at high x . Second, and more sobering: across several dozen attempts, the program exceeded the 0.875 benchmark *only twice*, transiently; sustaining the claimed active regime under calorimetry proved out of reach even for a well-funded team that wanted to get there. No excess heat was observed in any properly instrumented run.

The epistemics deserve to be stated with care, because the paper itself states them with care. The program's null is devastating to any reading of the literature in which

high loading is *easy* and heat is *common* — it explains thirty years of failed replications in one stroke. But it is explicitly *not* a refutation of the SRI correlation, because the correlation's own operating point was barely ever reached: a threshold claim is untested, not falsified, by experiments that stay below the threshold. (Proponents add that the record 0.96 was achieved with protium, not deuterium, and not in a heat-measuring cell.) The authors' verdict — no anomalies found, parameter space genuinely under-explored, better tools now exist — is the rare sentence in this field that both camps quote approvingly. What the program unambiguously delivered is the gauge this chapter has been missing: loading numbers you do not have to take on faith.

§7 Skipping the queue: co-deposition

There is a claimed shortcut around the whole loading problem. In co-deposition, introduced by Stanislaw Szpak and Pamela Mosier-Boss at the Navy's SPAWAR laboratory in San Diego (Szpak, Mosier-Boss and Smith, *J. Electroanal. Chem.* 302, 255, 1991), no bulk cathode is loaded at all. Palladium is electroplated from PdCl_2 dissolved in D_2O onto an inert substrate *while* deuterium evolves, so each freshly deposited atomic layer forms already saturated: effectively instant high loading, in minutes instead of weeks, without the α/β stress history of a bulk foil. The deposit is a black, branching, dendritic sponge of enormous surface area — as far from Violante's engineered foils as palladium gets. Proponents report that co-deposition cells show their claimed effects promptly and, by the standards of this field, reproducibly; the approach supplied the CR-39 track-detector evidence — including the disputed "triple tracks" attributed to energetic neutrons — that Chapter IV examined in detail.

The critiques mirror the design. The plated film is thin and its mass tiny, so claimed heat signals are small and the calorimetry correspondingly delicate; the violent surface chemistry of simultaneous plating and gas evolution is exactly the

environment in which chemical damage to a plastic track detector is most plausible (Kowalski's published critique of the CR-39 interpretation, and the wider exchange, are covered in Chapter IV); and the signature claims have not been reproduced outside the network of collaborating laboratories. Co-deposition's real significance for *this* chapter is conceptual: it concedes the premise that bulk loading is the bottleneck, and bets the field's chips on the near-surface, high-flux, defect-rich regime instead — the same corner of parameter space where §4's flux term $|dx/dt|$ already pointed.

§8 A materials specification for a definitive experiment

This chapter's payoff is that the controversy now has a bill of materials. Whatever one believes about the heat, the loading claim is concrete enough to specify the experiment that would test it, component by component: cathodes metallurgically characterized before the run (texture, hardness, grain size, impurity assay — Violante's checklist); loading driven by a protocol capable of $x \geq 0.95$ and *measured by operando diffraction*, not by the two-valued resistance gauge alone (Benck's contribution); current density held above the claimed i° with deliberate flux excursions (McKubre's correlation); heat measured by a calorimeter of Chapter III's best class, with Chapter IV's nuclear diagnostics running throughout; and the whole thing executed with pre-registered criteria by a team with no stake in the outcome. Every element exists; they have never been assembled in one adversarial experiment. That assembly is Chapter X's subject.

Notice, finally, where this chapter sits in the book's frame — *the Gamow factor and the error bar*. Loading is the field's claimed lever on the first number: proponents' theories (Chapter IX) need the dense, confined, screened deuterium of the β phase, and the threshold claim asserts the tunneling-relevant conditions switch on only near full occupancy. Materials characterization is the second number: until x itself is measured with an instrument that cannot lie about which branch it is on, the field's

central empirical law has error bars on its *x*-axis as well as its *y*. The next chapter turns to the one place where a metal lattice has been *proven* — in accelerator beams, by mainstream groups, with mainstream referees — to change a nuclear reaction rate: electron screening.

FURTHER READING

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Szpak, S., Mosier-Boss, P. A. & Smith, J. J. *On the behavior of Pd deposited in the presence of evolving deuterium*, Journal of Electroanalytical Chemistry 302, 255–260 (1991) — the original co-deposition paper (§7).

Chapter production note: researched, drafted, and adversarially verified by Claude subagents (draft agent: `claude-fable-5`; verification agent: `claude-fable-5`), orchestrated by `claude-fable-5`. See the [methods appendix](#) for the full research process.

CHAPTER VI

Electron Screening: The Accepted Anomaly

Fire deuterons into a slab of palladium and the fusion rate comes out wrong — too high, by a factor the best available theory cannot explain — and this time the referees, the replications, and the mainstream journals all agree the effect is real.

BEFORE WE START

So far the anomalies in this book have been contested: heat that some calorimeters see and others don't, ashes that appear in one lab and vanish in the next. This chapter is different. Since 2001, at least four independent accelerator groups — in Berlin and Szczecin, at Tohoku in Japan, in Bochum, and at Berkeley — have fired low-energy deuteron beams into deuterium-loaded metals and measured the fusion yield. In every metal tested, the yield at the lowest energies is larger than bare-nucleus physics predicts. The standard way to book the excess is a "screening energy": the metal's electrons cancel part of the Coulomb repulsion, as if every collision got a small energy discount. The discount is real, replicated, and published in mainstream journals. It is also, quantitatively, a scandal: measured values run two to eight times larger than any published calculation, and the four groups disagree with each other by factors of two to three.

This makes electron screening the book's pivot. It is an existence proof that the lattice is not a spectator — condensed-matter physics measurably reaches into a nuclear reaction rate. But the chapter's second job is arithmetic honesty: we will extrapolate the measured discount down to room temperature, carefully, with the Chapter I machinery, and watch it fall tens of orders of magnitude short of

rescuing cold fusion. A real anomaly and an insufficient one — both things are true at once, and the whole field lives in the space between them.

TERMS YOU'LL MEET

electron screening

The partial cancellation of the repulsion between two nuclei by the negative charge of electrons around and between them.

screening energy (U_e)

The single number (in eV) that summarizes screening: the effective energy boost a colliding pair receives, so that $\sigma_{\text{scr}}(E) = \sigma_{\text{bare}}(E + U_e)$.

screening length

The distance over which surrounding charge neutralizes a test charge; inside it, the two nuclei feel nearly the full repulsion minus a constant offset.

adiabatic limit

The maximum screening a bound electron can supply if it adjusts infinitely quickly to the approaching nuclei — about 27–28 eV for deuterons on deuterium.

united-atom limit

The thought experiment behind the adiabatic limit: as two nuclei merge, their electron cloud becomes that of the combined ("united") atom, whose deeper binding energy is donated to the collision.

gas target

A cell of D_2 gas used as the target; the clean benchmark, with no lattice and only molecular electrons.

beam–target experiment

An accelerator setup: a deuteron beam of known energy strikes a deuterated target and detectors count the charged products or neutrons.

lab vs center-of-mass energy

For a beam deuteron hitting a resting deuteron, only half the beam energy is available to the relative motion: $E_{\text{cm}} = E_{\text{lab}}/2$.

thick-target yield

The reaction count from a beam stopping entirely inside the target — an integral over the slowing-down path, which must be unfolded to get a cross section.

stopping power

The rate at which a projectile loses energy in matter; poorly known below a few keV, and a

metal deuteride

major systematic in screening measurements.

A metal (Pd, Ti, Zr, Ta...) with deuterium dissolved in its lattice, written e.g. PdD_x; the target material in all these experiments.

stoichiometry

The actual deuterium-to-metal ratio in the beam spot — hard to know, changed by the beam itself, and the leading suspect for inter-group disagreement.

enhancement factor (f)

The ratio of the measured (screened) cross section to the bare-nucleus one at the same energy; the observable everything else is fitted to.

conduction electrons

Electrons in a metal that are not tied to any one atom and move quasi-freely; the candidate source of the extra screening.

Fermi energy

The energy of the most energetic conduction electrons even at absolute zero — several eV in metals, corresponding to temperatures of tens of thousands of kelvin.

degenerate electron gas

An electron population whose behavior is set by quantum statistics (the Pauli principle), not temperature; conduction electrons at room temperature are deeply degenerate.

Thomas–Fermi screening

The standard quantum-statistical description of how a degenerate electron gas screens a charge, with an essentially temperature-independent screening length.

jellium

The simplest metal model: ions smeared into a uniform positive background with an electron gas on top; used for first-principles screening estimates.

dielectric function

The response function $\epsilon(q, \omega)$ describing how a material's charges rearrange around a disturbance; the machinery behind the best screening calculations.

Debye screening

The classical (Boltzmann) screening law for hot dilute plasmas; invoked controversially by the

plasma discharge

Bochum group to explain metal screening.

A low-voltage gas discharge used to bombard a target with ions of a few hundred eV to a few keV — a way to reach energies below what beams can deliver cleanly.

internal pair conversion

A nuclear de-excitation that emits an electron–positron pair instead of a gamma ray; the claimed decay mode of a proposed ${}^4\text{He}$ threshold resonance.

threshold resonance

A nuclear state sitting almost exactly at the energy where a reaction channel opens; if one exists in ${}^4\text{He}$ at the d+d threshold, low-energy fusion would be modified.

ion track

The femtoseconds-long trail of dense excitation a slowing ion leaves in a solid; one group proposes fusion occurs thermally inside these transient hot zones.

carbon buildup

The slow growth of a carbon film on any beam-struck surface from residual vacuum-chamber hydrocarbons; it shifts the effective beam energy and mimics physics.

sputtering

Erosion of the target surface by the beam itself, which continually changes the very deuterium density one is trying to hold fixed.

§1 A wall with a discount: the screened potential

Chapter I treated two colliding deuterons as naked point charges. Real deuterons in a target are dressed: bound electrons, molecular electrons, or a metal's conduction sea surround them, and that negative charge cancels part of the repulsion. The geometry is simple. Electrons rearrange over a characteristic screening length — ångströms for atoms, fractions of an ångström for a dense electron gas. The nuclear action happens at femtometers, thousands of times deeper in. So over the entire region where

tunneling is decided, the electron cloud is effectively a distant, static shell, and inside a shell of charge the potential is lowered by a *constant*. The barrier keeps its shape and loses altitude:

$$V_{\text{scr}}(r) \approx \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 r} - U_e, \quad r \ll R_{\text{scr}},$$

WHERE

$V_{\text{scr}}(r)$ Scalar field of separation r : the potential energy of the pair with electrons included.

U_e The screening energy (constant, eV): the depth of the electron-induced offset. Tens of eV for atoms; the subject of this chapter for metals.

R_{scr} The screening length (constant, of order 0.01–0.1 nm in a metal) — vastly larger than the femtometer scales where tunneling is decided.

A constant shift of the potential is exactly equivalent to a constant boost of the collision energy. That single observation, made precise by Assenbaum, Langanke and Rolfs in 1987, is the entire formalism of this chapter: the screened cross section at energy E is the bare cross section at $E + U_e$,

$$\sigma_{\text{scr}}(E) = \sigma_{\text{bare}}(E + U_e), \quad f(E) = \frac{\sigma_{\text{scr}}(E)}{\sigma_{\text{bare}}(E)} \approx \exp\left[\pi\eta(E) \frac{U_e}{E}\right]$$

WHERE

- E Center-of-mass collision energy (keV). For a deuteron beam on resting deuterons, $E = E_{\text{lab}}/2$ — the classic factor-of-two trap.
- $f(E)$ The enhancement factor (dimensionless): how many times the screened cross section exceeds the bare one. It grows explosively as E falls toward U_e .
- $\eta(E)$ The Sommerfeld parameter from Chapter I, with $2\pi\eta = \sqrt{E_G/E}$ and the d+d Gamow energy $E_G = 986 \text{ keV}$.
-

WHY $\pi\eta$, NOT $2\pi\eta$

The exponent looks like half a Gamow factor, and it is. The tunneling exponent is $2\pi\eta(E) = \sqrt{E_G/E}$; shift E by a small U_e and expand to first order: since $\sqrt{E_G/E}$ falls like $E^{-1/2}$, its change is $-\frac{1}{2}\sqrt{E_G/E} (U_e/E) = -\pi\eta U_e/E$. The factor $\frac{1}{2}$ from differentiating the square root converts $2\pi\eta$ into $\pi\eta$. Transcribing $2\pi\eta$ into the enhancement formula — a common error in the secondary literature — overstates every enhancement quadratically.

Note what the formula promises. At high energy, screening is invisible: a 50 keV collision barely notices a discount of a few hundred eV. But $f(E)$ has $1/E$ inside an exponential riding on $\eta \propto E^{-1/2}$, so as the energy drops the enhancement blows up. Screening is a low-energy phenomenon by construction — which is precisely why it belongs in this book, and why measuring it means fighting an accelerator down to energies where beams barely work at all.

§2 The honest benchmark: gas targets and the adiabatic limit

How big should U_e be? For an isolated atom or molecule there is a clean upper bound, the adiabatic limit. Picture the collision slowly: as the two deuterium nuclei approach, their shared electron cloud deforms continuously into the cloud of the united atom — a helium-like system whose electrons are more tightly bound. The binding-energy difference is released into the relative motion of the nuclei: the electrons, sliding downhill into deeper orbits, pay the collision an energy subsidy. For deuterons on atomic or molecular deuterium that bookkeeping tops out near $U_e \approx 27\text{--}28\text{ eV}$ — a couple of Rydbergs, the natural scale of atomic physics.

And in gas targets, that is exactly what is measured: $U_e = 25 \pm 5\text{ eV}$ for the d+d reaction in D_2 gas, established by the Rolfs group's accelerator program in the 1990s. Theory and experiment agree. This matters more than it may look: it certifies the method. The whole apparatus — thick-target yields, stopping-power corrections, the extraction of U_e from a fit of $f(E)$ against the bare cross section — was validated on a system where the answer was known. The screening story does not begin with an anomaly; it begins with a confirmation.

NOTE

The same physics runs the other way in stellar interiors, where plasma screening enhances solar fusion rates and must be *divided out* of laboratory data to get the bare S-factor astrophysicists need. Screening entered nuclear physics as a nuisance parameter for the Sun — the LUNA underground program measured it to correct solar reaction rates — years before metals made it a puzzle in its own right.

§3 The metal surprise

In 2001, Konrad Czerski and collaborators in Berlin put deuterated *metals* — Al, Zr, Ta — under a 5–60 keV deuteron beam and published the yields in *Europhysics Letters* (a mainstream venue). The extracted screening energies came out around an order of magnitude above the gas-target value: for Ta, roughly 300 eV. The following year Jiro Kasagi's group at Tohoku University reported in the *Journal of the Physical Society of Japan* a systematic scan across target materials at beam energies down to 2.5 keV, with screening energies of $U_e = 310 \pm 20 \pm 50$ eV for Pd, $200 \pm 15 \pm 40$ eV for Fe, and — strangest of all — $600 \pm 20 \pm 75$ eV for palladium oxide, where the d+d yield at 2.5 keV (lab) ran about fifty times the bare expectation. Then in 2004 Francesco Raiola, Claus Rolfs and the Bochum group published a survey of 58 deuterated materials in *The European Physical Journal A*: most metals showed large screening — Pd near 800 eV on their analysis — while insulators, semiconductors and a few metals like Au showed gas-like values of tens of eV. A periodic table of screening, with a sharp metal/non-metal divide.

What does theory allow? The adiabatic limit says ~ 28 eV from bound electrons. Adding everything a metal legitimately has — conduction electrons treated as a Thomas–Fermi degenerate gas, the ion lattice's polarization, cohesion effects — the best available calculations land near 100–300 eV at the very most; Czerski's own improved dielectric-function model, cohesion effects included, gives about 130 eV for Pd. The measured values sit a factor of two to eight above *any* published calculation. That is the anomaly. It has now stood for a quarter century, in refereed journals, replicated in kind by independent groups on three continents.

Replicated in kind — but not in magnitude, and the disagreement is itself part of the puzzle. For the same metal, palladium: Kasagi (Tohoku, 2002) reports ≈ 310 eV; Raiola (Bochum, 2004) reports ≈ 800 eV; Huke, Czerski and colleagues, in the field's most careful error analysis (*Physical Review C* 78, 015803, 2008), report 296 ± 15 eV and argue that much of the spread comes from what the beam does to the target: sputtering and beam-induced diffusion change the deuterium density in the first nanometers of the surface — exactly the layer where the lowest-energy reactions happen — so a fit that assumes uniform stoichiometry books a density artifact as

screening. Add poorly known stopping powers below a few keV and carbon buildup on the beam spot, and a factor-two systematic is easy to assemble. The Huke analysis is the standing caveat over every number in this section; it reduces the anomaly, but even its own corrected value, **296 eV**, is still more than double the best theory.

AHEAD OF THEORY

This one is uncontested — the cleanest entry in the whole ledger of Chapter IX. Measured d+d screening energies in deuterated metals (roughly 300–800 eV depending on group and method) exceed every published first-principles calculation (~30 eV adiabatic; ~100–300 eV with all solid-state effects) by factors of two to eight, and the measurements disagree among themselves by factors of two to three. Every party to the dispute — Bochum, Berlin/Szczecin, Tohoku, Berkeley — agrees an unexplained enhancement exists; no accepted model reproduces its magnitude or its material pattern (large in Pd, PdO, Ta; small in Au, insulators, semiconductors). This is the "electron screening puzzle" of low-energy nuclear physics: mainstream, peer-reviewed, and open since 2001.

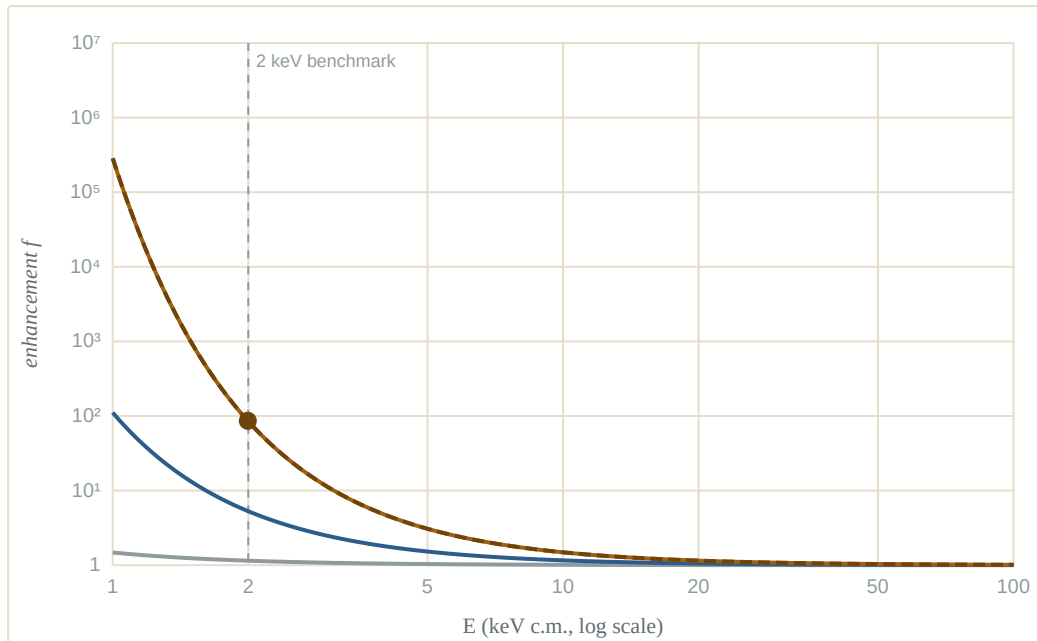
INTERACTIVE · THE SCREENING DISCOUNT

slide U_e · move the energy marker

screening U_e (eV)



marker energy E



$E = 2.00 \text{ keV}$ · $\pi\eta = 11.11$ · $f \approx 86\times$ (first-order) · $22\times$ (exact shifted)

$U_e = 800 \text{ eV} \rightarrow$ orders of magnitude gained at 2 keV: **1.93**

— 25 eV (D_2 gas) — 300 eV (theory max) — 800 eV (Pd, measured) — your U_e

The enhancement factor $f(E) = \exp[\pi\eta U_e/E]$ with $\pi\eta = \frac{1}{2}\sqrt{986 \text{ keV}/E}$, plotted for the three canonical scales of U_e : 25 eV (D_2 gas, matching theory), 300 eV (the most theory can plausibly supply in a metal), and 800 eV (Bochum's measured Pd value). At 10 keV the curves barely separate; at 2 keV the 800 eV discount is worth a factor of ~ 85 — nearly two orders of magnitude the Gamow factor never had to pay. The readout also shows the exact shifted-energy form $\frac{E}{E+U_e} e^{2\pi\eta(E)-2\pi\eta(E+U_e)}$, which is smaller: the first-order formula overshoots once E stops being large compared to U_e .

§4 A contested explanation: Debye's ghost in a Fermi sea

The Bochum group proposed a disarmingly classical explanation. Treat a metal's quasi-free conduction electrons as a hot dilute plasma, and they gather around a deuteron in a Debye cloud whose radius shrinks — and whose screening energy therefore grows — as temperature falls:

$$R_D = \sqrt{\frac{\epsilon_0 k_B T}{e^2 n_{\text{eff}}}}, \quad U_e^{\text{Debye}} = \frac{e^2}{4\pi\epsilon_0 R_D} \propto \sqrt{\frac{n_{\text{eff}}}{T}}.$$

WHERE

R_D The Debye radius (scalar, length): the size of the classical screening cloud.

$k_B T$ The thermal energy (constant at fixed temperature; ≈ 0.025 eV at room temperature).

n_{eff} The fitted density of "quasi-free" electrons participating in the cloud (scalar, m^{-3}) — the model's one adjustable knob.

The model fits the periodic-table pattern tolerably with of order one quasi-free electron per metal atom, and it makes a sharp prediction: $U_e \propto 1/\sqrt{T}$. Remarkably, Raiola et al. reported the prediction confirmed — in a 2006 follow-up (*Eur. Phys. J. A* 27, s01, 79) the screening in deuterated Pt fell as the target was heated from 20 °C to 340 °C, tracking the Debye curve; the authors called it the first observation of a temperature-dependent nuclear cross section.

The critique is not subtle, and it is now the majority view. Conduction electrons at room temperature are not a classical gas — they are a deeply degenerate Fermi system, with Fermi energies of several eV, thousands of times the thermal energy. Their screening is Thomas–Fermi screening, set by the Fermi energy and essentially blind to temperature; applying Boltzmann statistics to them is a category error, whatever the fit quality. Huke et al. (PRC 2008) put it bluntly: the Debye–Hückel picture is "clearly excluded" on basic solid-state grounds, and Kasagi's group leveled

the same objection independently. What of the measured temperature trend? A metal deuteride heated by hundreds of degrees changes its deuterium solubility, diffusivity and surface chemistry all at once — the very systematics flagged in §3 track temperature too. The temperature dependence is a contested single-group result carrying an inadmissible theory; the enhancement it was meant to explain, however, does not go away when the model does. The anomaly survives its own worst explanation.

§5 What screening buys at room temperature — done honestly

Now the calculation this chapter owes the rest of the book. Suppose we take the most generous measured number at face value — $U_e = 800 \text{ eV}$ in Pd — and ask what it does for two deuterons at thermal energy, $E \approx 0.025 \text{ eV}$. The first-order formula $f(E) = e^{\pi\eta U_e/E}$ is useless here: it assumes $U_e \ll E$ and diverges nonsensically in the opposite regime. The honest move is the exact shifted form, $\sigma_{\text{scr}}(E) = \sigma_{\text{bare}}(E + U_e)$: at $E \approx 0$, the collision effectively happens at U_e itself, so the tunneling exponent is the Gamow factor evaluated at 0.8 keV,

$$2\pi\eta(U_e) = \sqrt{\frac{E_G}{U_e}} = \sqrt{\frac{986 \text{ keV}}{0.8 \text{ keV}}} \approx 35.1, \quad e^{-35.1} \approx 5.7 \times 10^{-16}.$$

Compare Chapter I: the bare thermal Gamow factor was $e^{-6280} \sim 10^{-2728}$.

Screening at 800 eV has collapsed an exponent of 6,280 down to 35 — it has closed *most of the exponent*. This is why screening haunts every theoretical conversation in

this field: no other measured, replicated effect comes within shouting distance of that kind of leverage on the Gamow factor.

But a penetration probability is not a rate, and here the extrapolation must be done with adult supervision. If 10^{-16} per barrier assault were the right number, any plausible attempt frequency — molecular vibrations run at $\sim 10^{14}$ per second — would predict vigorous, easily detectable fusion in every slab of deuterated palladium on Earth. It is not observed. The arithmetic is telling us the transplant is illegitimate, and Huke et al. and the recent Metzler review both say why: a beam-derived U_e describes the *dynamic* response of the electron gas to a fast keV projectile sweeping through it. Two thermal deuterons parked at neighboring interstitial sites, $\gtrsim 0.17$ nm apart, are a different problem — a molecular-type bound system, whose fusion rate is governed by the Koonin–Nauenberg calculation of Chapter I and rigorously capped by the Leggett–Baym bound (*Phys. Rev. Lett.* 63, 191, 1989): no equilibrium electronic arrangement in condensed matter can raise the tunneling rate to anywhere near observable levels.

So the honest ladder, in fusions per deuteron pair per second, reads:

$\lambda_{\text{fus}} \approx 3 \times 10^{-64}$ for the bare D_2 molecule (Koonin–Nauenberg) $\rightarrow$ at most $\sim 10^{-45}$ with static screening from consistent theory ($U_e \approx 100\text{--}150$ eV equivalent), the ceiling enforced by Leggett–Baym $\rightarrow$ versus the $\sim 10^{-11}$ needed for a watt per cubic centimeter of PdD. Measured against the ~ 53 missing orders of magnitude, legitimate screening buys perhaps 10–20 of them. It closes a third of the exponent — call it half if you insist on force-feeding the most anomalous beam values into an equilibrium calculation where they don't belong — and none of the remainder. The anomaly is real; the rescue is not.

NOTE

Keep the two registers of this chapter separate. That metals screen keV-beam fusion far more than theory predicts: established, replicated, unexplained. That screening of any measured magnitude produces watt-level fusion at 300 K: excluded by the same

formalism that certifies the anomaly, by some thirty orders of magnitude. A surprising discount on the barrier is not a hole in the wall.

§6 The sub-keV frontier

Everything hinges on the lowest energies, where $f(E)$ is largest — and where experiments are hardest. Below a few keV a deuteron in a solid stops within nanometers, so the signal comes entirely from a surface layer whose deuterium content the beam itself is churning; stopping powers there are extrapolations; absolute yields are tiny; and every hour of beam time grows carbon on the spot. The modern campaigns attack the regime with new tricks, and report new anomalies.

At Berkeley Lab, Schenkel and colleagues (*J. Appl. Phys.* 126, 203302, 2019) sidestepped the accelerator entirely with a pulsed plasma discharge, driving deuterons of less than 2 keV onto Pd wires. Neutron yields exceeded the bare-nucleus expectation more than a hundredfold, describable by $U_e = 1000 \pm 250$ eV — the largest value ever reported, with the authors' own caveats prominently attached: the ion energy distribution in a discharge is broad, and the surface deuterium density during plasma loading is poorly known. The same group's ARPA-E-funded follow-up moved to calibrated beams to pin those systematics down; a December 2025 preprint from the group (Karahadian et al., arXiv:2512.06212 — a preprint, not yet peer-reviewed) reports sub-keV yields on *electrochemically loaded* Pd and Ti some eighteen orders of magnitude above the bare extrapolation.

The most consequential recent thread is Czerski's. His Szczecin group pushed thick-target d+d yields in ZrD₂ across seven decades of signal, down to a center-of-mass energy of 675 eV — comparable to U_e itself — and reported in *Physical Review X* (15, 041004, 2025) that below about 5 keV the data show signatures of a new reaction

channel: a proposed 0^+ threshold resonance in ^4He decaying by internal electron–positron pair emission, following the group's own earlier letters — the resonance proposal (PRC 106, L011601, 2022) and first indications of electron emission (PRC 109, L021601, 2024). The group's zirconium-target analyses fit screening energies near 340 eV. This is register two, stated carefully: PRX is a top-tier venue and the claim passed hard review, but it is a single group's result, no independent confirmation exists, and no such resonance appears in the evaluated $A = 4$ level compilations. If it is real, the most-studied fusion reaction in physics has had a hidden low-energy channel for eighty years. A separate manuscript from the same group (arXiv:2409.02112, journal status unconfirmed) goes further, interpreting a sub-2.5-keV yield plateau as *thermal* fusion inside transiently molten ion tracks — a preprint-stage claim that should be read as a hypothesis, not a result.

§7 The pivot: what an accepted anomaly proves

Recall the book's two numbers: the Gamow factor and the error bar. Every chapter so far has been about the error bar — whether calorimetry and nuclear diagnostics can be trusted. This chapter is the book's purest Gamow-factor chapter, and its lesson is double-edged.

Edge one: the lattice is not vacuum. A metal measurably, reproducibly modifies a nuclear reaction rate, by more than the best theory can explain, and mainstream nuclear physics has carried that open anomaly on its books for twenty-five years without resolving it. Anyone who says condensed matter cannot touch nuclear physics is wrong by direct experiment. Edge two: the touch is a caress, not a demolition. The same formalism that certifies a factor of 50 at 2.5 keV certifies tens of missing orders of magnitude at 0.025 eV. Screening moves the exponent further than anything else in this book — and still not remotely far enough.

There is, however, a third way to read the numbers, and it sets up the next chapter. The enhancement is enormous precisely where deuterons carry a few keV. So do not ask the lattice to fuse thermal deuterons; *inject hot ones* and let the lattice's anomalously large screening — plus its billion-fold density advantage over any plasma — do what it demonstrably does. That is exactly the design logic NASA published in 2020: an external driver makes keV deuterons inside a deuteride, where screened fusion proceeds at measurably enhanced rates. Chapter VII takes up that story, alongside the Google program's reassessment of the field's error bars.

FURTHER READING

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Czerski, K., Huke, A., Biller, A., Heide, P., Hoefft, M. & Ruprecht, G. *Enhancement of the electron screening effect for d+d fusion reactions in metallic environments*, Europhysics Letters 54, 449–455 (2001) — the first metal-target measurements (Al, Zr, Ta, 5–60 keV), DOI 10.1209/epl/i2001-00265-7.

Kasagi, J., Yuki, H., Baba, T., Noda, T., Ohtsuki, T. & Lipson, A. G. *Strongly enhanced DD fusion reaction in metals observed for keV D⁺ bombardment*, Journal of the Physical Society of Japan 71, 2881–2885 (2002) — the material scan of §3, including PdO at 600 eV and the $\times 50$ yield at 2.5 keV, DOI 10.1143/JPSJ.71.2881.

Raiola, F. et al. *Enhanced electron screening in d(d,p)t for deuterated metals*, European Physical Journal A 19, 283–287 (2004) — the 58-material survey and the Debye-model proposal, DOI 10.1140/epja/i2003-10125-0; temperature follow-up: *Enhanced d(d,p)t fusion reaction in metals*, Eur. Phys. J. A 27, s01, 79–82 (2006), DOI 10.1140/epja/i2006-08-011-0.

Huke, A., Czerski, K., Heide, P., Ruprecht, G., Targosz, N. & Żebrowski, W. *Enhancement of deuteron-fusion reactions in metals and experimental implications*, Physical Review C 78, 015803 (2008) — the definitive systematic-

error analysis; Pd at 296 ± 15 eV vs ~ 130 eV theory; the Debye model excluded; arXiv:0805.4538.

Schenkel, T. et al. *Investigation of light ion fusion reactions with plasma discharges*, Journal of Applied Physics 126, 203302 (2019) — sub-2-keV plasma-discharge yields on Pd consistent with $U_e = 1000 \pm 250$ eV, with the authors' systematics caveats; DOI 10.1063/1.5109445, arXiv:1905.03400.

Dubey, R., Czerski, K. et al. *Experimental signatures of a new channel of the deuteron-deuteron reaction at very low energy*, Physical Review X 15, 041004 (2025) — yields to $E_{\text{cm}} = 675$ eV and the claimed ^4He threshold-resonance channel of §6; arXiv:2408.07567. Single-group; awaiting independent replication.

Chapter production note: researched, drafted, and adversarially verified by Claude subagents (draft agent: `claude-fable-5`; verification agent: `claude-fable-5`), orchestrated by `claude-fable-5`. See the [methods appendix](#) for the full research process.

CHAPTER VII

The Reassessment: Google and NASA

Thirty years after the press conference, the question came back to Nature and Physical Review C — asked properly this time, and answered in two very different ways.

BEFORE WE START

By 2015 the field described in Chapters II–VI had spent a quarter-century outside the mainstream: its claims lived in conference proceedings and specialty journals, and no top-tier venue would touch the subject. This chapter is about the two programs that changed that — not by validating cold fusion, but by re-opening the question in the places where physics is normally settled. Google quietly funded about thirty researchers across five institutions to re-test the three strongest historical claims with modern instruments, and published the result — a careful null — in *Nature*. NASA's Glenn Research Center, meanwhile, published in *Physical Review C* a demonstration that real d–d fusion can be made to happen *inside* a deuterated metal, provided you supply the energy from outside with MeV photons.

Our two running numbers organize the pair neatly. The Google program was an assault on the *error bar*: build calorimeters, loading probes, and tritium detectors good enough that an anomaly could not hide — and find that none does, down to tens of milliwatts. The NASA program was an assault on the *Gamow factor*: put keV-scale kinetic energy into deuterons that are already sitting in a lattice at enormous density, and let the metal's electrons help from there. One tightened the measurement; the other raised the tunneling probability — with hot deuterons, which is exactly why it is not cold fusion. Both left the field with

something it had lacked since 1989: peer-reviewed coordinates for what is actually known.

TERMS YOU'LL MEET

Nature Perspective

A commissioned, peer-reviewed review-and-opinion article in *Nature* — a venue statement that a topic is worth mainstream attention, distinct from a research Article.

null result

An experiment that finds no effect at its stated sensitivity. Informative in proportion to how good the sensitivity is.

loading ratio ($x_{\text{D/Pd}}$)

The atomic ratio of dissolved deuterium to palladium; proponents' claimed threshold for excess heat is $x_{\text{D/Pd}} \gtrsim 0.875$ (Chapter V).

operando measurement

Measuring a material's state (here, its hydrogen content) continuously *while* the experiment runs, rather than inferring it before or after.

X-ray diffraction (XRD)

Reading a crystal's lattice spacing from the angles at which it scatters X-rays; the Pd lattice swells measurably as hydrogen loads, so XRD gives a direct loading gauge.

resistance-ratio proxy

Inferring loading from the electrical resistance of the Pd electrode — the field's traditional gauge, shown by the Google teams to mislead badly at high loading.

coefficient of performance (COP)

Heat out divided by power in; COP = 1 means ordinary chemistry and physics, COP > 1 would mean excess heat.

glow discharge

A low-pressure plasma sustained by a voltage between electrodes — the regime of neon signs, and of the Google program's pulsed-plasma experiments.

beam–target fusion

Fusion produced by accelerating ions into a stationary target; conventional, well-calibrated, and the workhorse for measuring screening.

screening potential (U_e)	The effective energy boost that surrounding electrons give a fusing pair (Chapters I and VI); measured in eV, and anomalously large in metals.
bremsstrahlung	"Braking radiation": the continuous X-ray/gamma spectrum emitted when fast electrons decelerate in a heavy target. Its maximum photon energy equals the electron beam energy (the <i>endpoint</i>).
linac	A linear accelerator; NASA used a medical-type electron linac to make its bremsstrahlung photons.
photodisintegration	Breaking a nucleus apart with a photon; for the deuteron, $\gamma + \text{d} \rightarrow \text{n} + \text{p}$ once the photon exceeds the deuteron's binding energy.
deuteron binding energy (B_d)	The 2.2246 MeV holding a deuteron's proton and neutron together — the threshold a photon must beat to split it.
photoneutron	A neutron released by photodisintegration.
knock-on collision	An elastic collision in which a fast particle transfers part of its kinetic energy to a struck nucleus, setting it in motion inside the material.
elastic scattering	A collision that conserves total kinetic energy — the billiard-ball case, governed purely by masses and angles.
neutron moderation	The stepwise slowing of neutrons by elastic collisions with light nuclei; the same kinematics NASA runs in reverse to heat deuterons.
lattice confinement fusion (LCF)	NASA's name for fusion of energetic deuterons occurring inside a deuterated metal, where the lattice supplies fuel density and electron screening while an external driver supplies the energy.

erbium deuteride (ErD₃)

A metal hydride that packs deuterium to roughly 10^{23} atoms per cm³ — about a billion times the fuel density of a tokamak plasma. Titanium deuteride (TiD₂) is similar.

Oppenheimer–Phillips reaction

A deuteron grazing a heavy nucleus can hand over just its neutron, its proton never surmounting the Coulomb barrier — a "stripping" reaction that needs less energy than full fusion.

scintillator

A material that flashes light when radiation deposits energy in it; the flash shape and size encode the particle's type and energy.

pulse-shape discrimination

Telling neutrons from gamma rays by the differing time-profile of their scintillation flashes — hard to do perfectly inside an intense gamma field.

spectrum unfolding

Reconstructing the true energy spectrum of incoming particles from a detector's indirect response — a model-dependent inversion, and a classic home for systematic error.

overpotential

The extra voltage applied to an electrode beyond the thermodynamic minimum; it acts like pressure, forcing more deuterium into the metal.

breakeven

The point where a fusion system releases as much energy as it consumes. Neither program came anywhere near it, and neither claimed to.

ARPA-E

The U.S. Department of Energy's high-risk research agency, whose 2022–2023 LENR funding program is the subject of Chapter VIII.

§1 Why Google funded it

The Google program began, by its organizers' own account, with a question of due diligence rather than belief. Matt Trevithick and Ross Koningstein of Google asked whether the 1989 claims had ever been tested with instruments good enough to settle them — and concluded, after canvassing the literature and the field's own veterans, that they had not. Even Michael McKubre, the field's most careful electrochemist (Chapter III), conceded in 2018 that "to my knowledge there is no written replicable procedure or protocol." So from 2015 to late 2019, Google funded roughly USD 10 million of work by about thirty researchers across the University of British Columbia, MIT, the University of Maryland, Lawrence Berkeley National Laboratory, and Google itself — with an unusual mandate: publish whatever you find, positive or null, in peer-reviewed venues.

The public disclosure came all at once, in a 2019 *Nature* Perspective: Berlinguette, Chiang, Munday, Schenkel, Fork, Koningstein and Trevithick, "Revisiting the cold case of cold fusion." Its headline sentence is the most-quoted null in the field's modern history: the team "found no evidence of anomalous effects claimed by proponents of cold fusion that cannot otherwise be explained prosaically." But the same paper argues, with equal care, that the parameter space where the claims live — very highly loaded metal hydrides, calorimetry under extreme conditions, nuclear reactions at eV-to-keV energies inside lattices — is genuinely under-explored, and that the instruments built to explore it have value regardless of the answer. Both halves of that sentence matter; the rest of this chapter unpacks them.

NOTE

A Perspective is not a research Article, and the program's primary data live in about a dozen separate peer-reviewed papers cited below. The *Nature* venue mattered less for the data than for the permission it granted: after May 2019 it was once again respectable to say, in a mainstream journal, that this parameter space deserved instruments.

§2 Three thrusts, three answers

The program organized itself around the three strongest historical claims, and its findings are best kept separate, because they are of three different kinds.

Thrust A: highly loaded Pd–D electrochemistry. The Fleischmann–Pons claim, refined by McKubre's correlation (Chapter V), holds that excess heat requires sustained loading $x_{\text{D/Pd}} \gtrsim 0.875$. The Google teams found that merely *reaching* that regime — and knowing you have reached it — is brutally hard. Over several dozen electrochemical runs the 0.875 benchmark was exceeded only twice, fleetingly. In a companion *Chemistry of Materials* study, Benck et al. pushed palladium to a loading of $x = 0.96 \pm 0.02$ (with protium, not deuterium) and, crucially, showed by operando X-ray diffraction that the field's traditional resistance-ratio proxy for loading can badly mislead at high x — meaning that decades of claimed loadings above 0.9 carry an unquantified calibration risk. In properly instrumented runs, calorimetry at tens-of-milliwatt sensitivity found no excess heat. The honest verdict, stated candidly in the Perspective itself, is not "refuted" but "not yet tested at scale": the claimed regime was too hard to hold long enough to run the heat measurement proponents ask for.

Thrust B: pulsed plasma discharges. To re-test the 1990s Los Alamos claims of tritium from pulsed deuterium discharges onto palladium wires, Thomas Schenkel's group at Berkeley Lab ran benchtop glow-discharge experiments for days to weeks at a time: peak ion current densities of 0.1 to several A/cm² at keV-scale cathode voltages. The LENR-style result was null — no tritium above background, and no 2.22 MeV neutron-capture gammas beyond what ordinary beam–target d–d fusion predicts. But the conventional physics produced the program's most interesting number: at center-of-mass energies $E_{\text{cm}} \lesssim 2 \text{ keV}$, the measured d–d neutron yields ran more than $100\times$ above the bare-nucleus extrapolation, consistent with a screening potential of $U_e = 1000 \pm 250 \text{ eV}$ in the deuterated palladium target

(Schenkel et al., *J. Appl. Phys.* 126, 203302, 2019). Recall from Chapter VI the operative formula:

$$f(E) = \frac{\sigma_{\text{screened}}(E)}{\sigma_{\text{bare}}(E)} \approx \exp\left[\pi\eta(E) \frac{U_e}{E}\right], \quad \pi\eta(E) = \frac{1}{2}\sqrt{\frac{E_G}{E}},$$

WHERE

- $f(E)$ Enhancement factor (dimensionless): how many times the screened cross section exceeds the bare one at center-of-mass energy E . Valid for $U_e \ll E$ (Assenbaum–Langanke–Rolfs, Chapter I §4).
- U_e Screening potential (eV) — the effective energy boost from the target's electrons. Convert to keV before dividing by E ; mixed units silently inflate f .
- $\eta(E)$ Sommerfeld parameter (dimensionless, Chapter I); for d+d, $E_G = 986 \text{ keV}$, so $\pi\eta = \frac{1}{2}\sqrt{986/E[\text{keV}]}$.

Because f grows explosively as E falls toward U_e , sub-keV yield anomalies are diagnostic — and systematics-prone. The Schenkel paper is careful on this point: the sub-keV baseline depends on stopping powers and on the deuterium depth profile at the target surface, and the authors present $U_e \approx 1000 \text{ eV}$ as "a possible explanation," not a measurement of new physics.

AHEAD OF THEORY

A screening potential of $1000 \pm 250 \text{ eV}$ in palladium sits far above what adiabatic-limit many-body theory readily supplies (roughly a few hundred eV at best), and it corroborates the earlier Raiola et al. beam measurements of $U_e \approx 800 \text{ eV}$ in Pd (Chapter VI). The *existence* of anomalously large screening in metals is multiply replicated across independent groups; its *magnitude* disagrees between groups by factors of 2–3, one 2023 palladium target showed no enhancement at all, and thirty-plus years after the screening formula was written down there is still no accepted first-principles calculation reproducing hundreds of eV in PdD_x. This — not excess heat —

is the solidest anomaly the Google program touched, and Chapter IX's ledger will collect it.

Thrust C: nanostructured metal hydrides at high temperature. To confront the Ni–H "reactor" claims of the 2010s (Chapter IV's cautionary tail), the UBC team built a gram-scale calorimeter that operates at up to 1232 °C under 33 bar of hydrogen with roughly 50 mW power resolution (MacLeod et al., *Rev. Sci. Instrum.* 88, 084101, 2017) — and ran 420 samples of hydrided metal powders through it. The measured coefficient of performance was consistently 1: no anomalous heat beyond chemistry, in by far the largest controlled survey of such materials ever published. Proponents can and do argue that some specific fuel recipe was missed; no critique has identified which.

§3 What the program concluded — and the tools it left

Summed up, the Google program's conclusion has two clauses. First: across the three re-examined claims, no evidence of anomalous effects — no excess heat at 50 mW resolution in 420 powder samples, no tritium in weeks of plasma discharges, no reliably sustained loading above the claimed threshold, and therefore no confirmation of the 1989-style anomalies anywhere the instruments could see. Second: the interesting corner of parameter space — $x_{\text{D/Pd}}$ held above 0.875 *under* precision calorimetry — was reached only twice, transiently, so its emptiness is asserted by extrapolation, not observation. The program's authors state both clauses plainly; readers who quote only one are editorializing.

The lasting products were instruments and methods, published as ordinary materials science: the MacLeod high-temperature, high-pressure calorimeter; Benck's operando-XRD protocol for hydride stoichiometry, which replaced a folklore loading

gauge with a crystallographic one; and a run of palladium-hydride surface science from the same collaboration — Johnson et al. in *Nature Materials* on how hydrogen uptake nucleates at lattice-strained nanocrystal vertices, and Jansonius et al. in *Angewandte Chemie* on how elastic strain tunes hydrogen absorption. None of these is about cold fusion; all of them make any future loading claim easier to check. That is what tightening the error bar looks like in practice.

The program also had a coda, published after six more years of quiet work by the same UBC–Berkeley axis. In 2025, Chen, Maiwald, Schauer and colleagues in Berlinguette's group reported in *Nature* a benchtop device — the "Thunderbird" reactor — that bombards a palladium target with keV deuterons while electrochemically loading it with deuterium in situ. Switching the electrochemistry on raised the beam-driven d–d fusion neutron rate by $15 \pm 2\%$ (three campaigns: 135.5, 142.9 and 138.6 neutrons per second beam-only, rising to 156.7, 159.2 and 164.3 with loading), with about one volt of overpotential doing the fuel-densifying work of roughly 800 atmospheres of D₂ pressure. Read the claim precisely: this is peer-reviewed, mainstream, and *not* cold fusion — the fusion is ordinary hot beam–target d–d, the electrochemistry is a knob that packs more fuel near the surface, there is no net energy, and as of mid-2026 no independent replication has yet been published. But it is the cleanest demonstration on record that the electrochemical and nuclear regimes touch at all: a voltage measurably moved a fusion rate. Whether the 15% is entirely fuel density or partly a change in screening is not fully settled in the paper — a small, well-posed open question of exactly the kind the 2019 Perspective asked for.

§4 NASA's lattice confinement fusion: the mechanism chain

The second reassessment came from an unexpected quarter. In April 2020, a NASA Glenn Research Center team published back-to-back papers in *Physical Review C* — Pines et al. (theory) and Steinetz et al. (experiment) — reporting d–d fusion inside deuterated metals, driven not by an ion beam but by photons. They named the scheme lattice confinement fusion: the metal's job is to hold deuterium at around 10^{23} atoms per cm^3 — on the order of a billion times the ion density of a tokamak plasma — and to supply electron screening; the driver's job is to make a few of those deuterons hot.

The mechanism is a four-step cascade, and each step is textbook physics. First, a linac's electron beam makes bremsstrahlung photons with an endpoint of about 2.9 MeV. Second, any photon above the deuteron's binding energy splits a lattice deuteron:

$$\gamma + \text{d} \rightarrow \text{n} + \text{p}, \quad E_\gamma \geq B_d = 2.2246 \text{ MeV}, \quad T_n \approx T_p \approx \frac{E_\gamma - B_d}{2},$$

WHERE

- B_d Deuteron binding energy (constant): 2.2246 MeV — the photodisintegration threshold. Not to be confused with the 2.9 MeV beam endpoint.
- E_γ Photon energy (MeV), anywhere in the bremsstrahlung continuum up to the endpoint.
- T_n, T_p Kinetic energies of the released neutron and proton. Because their masses are nearly equal and the photon carries little momentum, the surplus $E_\gamma - B_d$ is shared almost equally.
-

An endpoint photon thus yields a photoneutron of about $(2.9 - 2.2246)/2 \approx 0.34 \text{ MeV}$; because the bremsstrahlung spectrum is continuous and weighted toward threshold, NASA's spectrum-averaged photoneutron energy was about 145

keV, with the population below 400 keV. Third — and this is the step that makes neutral particles the ideal middlemen — those photoneutrons elastically scatter off other deuterons. A charged projectile wastes its energy on the lattice's electrons; a neutron gives its energy directly to a nucleus, transferring in a single head-on collision the maximum fraction

$$\frac{T_d^{\max}}{T_n} = \frac{4 m_n m_d}{(m_n + m_d)^2} = \frac{8}{9} \approx 0.889,$$

WHERE

$T_d^{\max}$ Maximum kinetic energy handed to the struck deuteron (keV) — the head-on case; the average over scattering angles is half this, $4/9 \approx 0.44$.

m_n, m_d Neutron and deuteron masses; the mass ratio 1 : 2 fixes the $8/9$. This is standard neutron-moderation kinematics, run in reverse: instead of cooling neutrons with light nuclei, NASA heats nuclei with neutrons.

NASA's average knock-on deuteron carried about 64 keV — consistent with $\frac{4}{9} \times 145 \text{ keV}$. Fourth, these hot deuterons collide with the cold deuterium ocean around them, and now Chapter I's cross-section machinery applies with a for-once-comfortable energy in it. Taking the illustrative endpoint case $E \approx 337 \text{ keV}$: the Gamow exponent is $2\pi\eta = \sqrt{986/337} \approx 1.71$, the penetration factor is $e^{-1.71} \approx 0.18$, and

$$\sigma(E) = \frac{S(E)}{E} e^{-2\pi\eta} \approx \frac{56 \text{ keV} \cdot \text{b}}{337 \text{ keV}} \times 0.18 \approx 0.030 \text{ b} = 30 \text{ mb}.$$

WHERE

- $S(E)$ The astrophysical S-factor (Chapter I §4): the measured, slowly varying nuclear part of the cross section; $\approx 56 \text{ keV} \cdot \text{b}$ for d+d near these energies (Bosch–Hale parametrization).
- b The barn, 10^{-24} cm^2 — nuclear physics' unit of cross-sectional area; $1 \text{ mb} = 10^{-3} \text{ b}$.
-

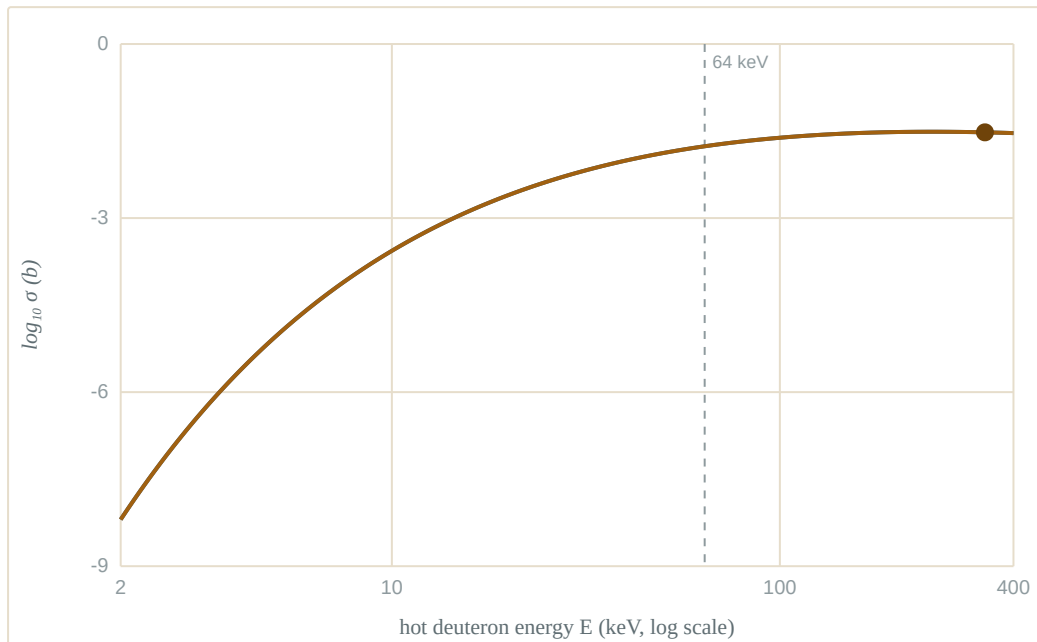
Thirty millibarns is an unremarkable, healthy hot-fusion cross section — and it is available here only because someone paid for hundreds of keV. At the thermal energies Chapter II's electrochemists had to work with, the same formula returns a penetration factor of order 10^{-2728} (Chapter I §3); NASA's cascade does not cheat that exponent, it feeds it. The observed signature was the one Chapter I promised: the 2.45 MeV neutron line of $\text{d}(\text{d}, \text{n})^3\text{He}$, standing on the low-energy photoneutron continuum (the mirror branch's 3.02 MeV protons and 1.01 MeV tritons deposit their energy inside the metal). That line is the fingerprint of real fusion happening inside ErD_3 and TiD_2 — driven, always, by the accelerator next door.

INTERACTIVE · THE LATTICE-FUSION CASCADE

tune the photon · add
screening

photon E_γ (MeV)

screening U_e (eV)



$E_\gamma = \mathbf{2.900 \text{ MeV}} \rightarrow E_d = (E_\gamma - 2.2246 \text{ MeV})/2 = \mathbf{337.7 \text{ keV}} \cdot 2\pi\eta = 1.71$
 $\sigma_{\text{bare}} = \mathbf{30.0 \text{ mb}} \cdot U_e = 0 \text{ eV} \rightarrow f = \mathbf{1.00\times} \cdot \sigma_{\text{screened}} = 30.0 \text{ mb}$

— bare $\sigma(E)$ — screened $f \cdot \sigma(E)$ | NASA mean hot d (64 keV)

The NASA cascade as arithmetic: a photon of energy E_γ splits a deuteron ($B_d = 2.2246 \text{ MeV}$), the surplus is shared, and the resulting hot deuteron of energy $E_d \approx (E_\gamma - B_d)/2$ fuses with cross section $\sigma = \frac{S(E)}{E_d} e^{-\sqrt{E_G/E_d}}$ ($E_G = 986 \text{ keV}$, $S(E) = 56 \text{ keV} \cdot \text{b}$), boosted by screening as $f = e^{\pi\eta U_e/E_d}$. At the 2.9 MeV endpoint, $E_d \approx 338 \text{ keV}$ and $\sigma \approx 30 \text{ mb}$ — and even $U_e = 1000 \text{ eV}$ changes nothing, because screening only matters when E falls toward the keV scale: drag the photon toward threshold and watch f wake up. The chain here is deliberately idealized — it skips the knock-on step's $\leq 8/9$ transfer and the bremsstrahlung average that put NASA's real means at 145 keV (neutrons) and 64 keV (deuterons), and the first-order screening formula strains as E_d approaches U_e .

§5 What NASA showed — and what it didn't

Take the established part first. Fusion really occurred inside a room-temperature metal lattice: the 2.45 MeV line under photon drive is conventional nuclear physics, which is precisely why *Physical Review C* published it. The theory paper's contribution is about the lattice's assist: Pines et al. argue that screening in a deuterated metal — bound and conduction electrons, plus a radiation-induced plasma of knocked-about charges — should be computed from the nonlinear Vlasov problem rather than linearized Debye theory, and that when the projectile energy falls toward U_e the familiar exponential enhancement formula breaks down and the Gamow integral must be done directly for the screened potential. Those are calculational claims layered on the multiply replicated experimental fact (Chapter VI) that metals screen far more strongly — hundreds of eV, against 25 ± 5 eV for D_2 gas — than adiabatic theory explains; no published calculation has independently confirmed or refuted the Vlasov treatment.

Now the contested part. Steinetz et al. also report neutron groups near 4 MeV (in TiD_2) and 5 MeV (in ErD_3) — above anything the 2.45 MeV fusion line plus scattering can produce. They offer two candidate explanations: "boosted" fusion, in which an already-hot deuteron fuses in flight so the neutron inherits extra energy, or Oppenheimer–Phillips stripping on the lattice's heavy nuclei, screened enough to proceed at these energies. The claim rests on spectrum unfolding from organic scintillators — EJ-309 liquid and stilbene — working by pulse-shape discrimination inside an intense bremsstrahlung gamma field, which is a systematics-heavy way to make a spectral discovery. The papers demonstrate neither mechanism, and no independent group has reproduced the feature.

AHEAD OF THEORY

The 4–5 MeV neutron groups are a peer-reviewed observation with two candidate explanations and zero independent replications as of mid-2026. The nearest supporting evidence comes from a programmatically allied group: Gillespie et al. (Texas

Tech/MURR, *Nucl. Eng. Technol.*, 2026) found tritium production in TiD_2 irradiated in a *fast* neutron spectrum running $2.9\text{--}5.1\times$ above conventional-physics simulation, while a mostly thermal spectrum agreed with prediction within 25% — a pattern consistent with the knock-on fusion picture, from collaborators rather than independents. Both results go to Chapter IX's ledger as claims awaiting outside replication.

And the framing NASA itself insists on: this is not cold fusion, and the team says so without hedging — the project's senior experimental physicist, Lawrence Forsley, put it as "What we did was not cold fusion." The fusing deuterons carry tens to hundreds of keV delivered by a megavolt-scale machine; the system is nowhere near breakeven, since the linac consumes vastly more power than the fusion returns; and NASA pitches the work as a route to compact neutron sources and, speculatively, fusion–fission hybrids for space power. What the lattice contributed was fuel density and screening — a warm assist to a hot process. In this book's terms: NASA raised the Gamow factor honestly, by paying for it in beam energy, and demonstrated that the lattice makes the price somewhat lower. Later program briefings and a commercial licensee have claimed far more — fifty-fold neutron-generator gains, 2025 commercialization, hybrid demonstrations — but none of that has peer-reviewed support as of this writing, and we set it aside accordingly.

§6 Why these two programs matter

Measured by their own data, both programs are modest: one found nothing anomalous, the other found conventional fusion in an unconventional container. Measured by what they changed, they are the hinge of the field's modern history — for three reasons.

Venue legitimacy. Between 1990 and 2019, the field's positive claims lived almost entirely in conference proceedings and specialty journals (Chapter IV). "Revisiting

the cold case" put the subject back in *Nature*; Steinetz and Pines put deuterated-metal nuclear reactions in *Physical Review C*; Chen et al. put an electrolytic cell wired to a fusion diagnostic back in *Nature* in 2025. None of these validates the 1989 claims — all of them establish that the underlying questions can pass ordinary peer review when asked with ordinary rigor. Young researchers, and program managers, noticed.

Tool legacies. The Google program left 50 mW calorimetry at 1232 °C, operando loading metrology, and a quantitative low-energy screening dataset; NASA left a photoneutron drive scheme and a worked example of instrumenting a lattice for nuclear diagnostics. The chronic weaknesses of the historical record — loading you couldn't verify, heat you couldn't calibrate, ash you couldn't count — now each have a published, purchasable answer. Whatever one believes about anomalies, the error bars available to the next generation are categorically better.

Recruitment. The programs seeded the funding wave that Chapter VIII follows. Trevithick moved to the venture fund DCVC and presented at an ARPA-E workshop on LENR in October 2021; in September 2022 ARPA-E announced an LENR Exploratory Topic that explicitly cited the Google-sponsored capabilities published in *Nature*; in February 2023 it awarded about USD 10 million across eight projects — including USD 1.5 million to Schenkel's Berkeley group to pin down the screening systematics his 2019 paper flagged, and USD 2 million to MIT to search for neutron emission from laser-stimulated metal hydrides. The through-line is exact: the null program's instruments, people, and open questions became the next program's statement of work. Google tightened the error bar; NASA raised the Gamow factor; between them they rebuilt the field's licence to find out.

FURTHER READING

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Chen, K.-Y., Maiwald, J., Schauer, P. A. et al. *Electrochemical loading enhances deuterium fusion rates in a metal target*, Nature 644, 640–645 (2025) — the Thunderbird reactor's $15 \pm 2\%$ electrochemical enhancement of beam-driven d–d fusion; DOI 10.1038/s41586-025-09042-7.

Chapter production note: researched, drafted, and adversarially verified by Claude subagents (draft agent: `claude-fable-5`; verification agent: `claude-fable-5`), orchestrated by `claude-fable-5`. See the [methods appendix](#) for the full research process.

CHAPTER VIII

New Programs: Japan, Europe, ARPA-E

A generation after 1989 the field's center of gravity sits in Sendai, Szczecin, and a Washington funding office — and every claim is now audited in electronvolts per hydrogen atom.

BEFORE WE START

The previous chapter ended with two institutional verdicts: Google's consortium found no excess heat, and NASA found real — but beam-driven — nuclear reactions in deuterated metals. This chapter surveys what the field itself has been doing since about 2015. Three programs dominate: a Japanese industrial–academic collaboration claiming watt-scale heat from nano-structured metal powders and films, a pair of European Union projects that split cleanly into a positive-claiming one and a negative-reporting one, and a 2023 re-entry by the U.S. government's most adventurous energy agency, ARPA-E, followed in 2026 by DARPA.

To keep our footing we will use one ruler for everything: the energy released per hydrogen atom involved. Chemistry — any chemistry — tops out near one or two electronvolts per hydrogen atom. Nuclear reactions deliver millions. Every claim in this chapter quotes a number on that axis, and every number comes with a venue (a top-tier journal, a mid-tier journal, a field-internal proceedings) and a replication status. Holding all three — the number, the venue, the replication record — at once is the discipline this chapter practices.

TERMS YOU'LL MEET

gas loading	Absorbing hydrogen or deuterium into a metal directly from pressurized gas, usually at elevated temperature — the successor technique to 1989's electrolysis.
nano-composite	A material made of metal nanoparticles (here Pd·Ni or Cu·Ni) embedded in a ceramic support such as zirconia, giving enormous hydrogen-absorbing surface area.
multilayer thin film	A stack of alternating metal layers each a few nanometers thick, deposited by sputtering; the Tohoku experiments use six Cu/Ni bilayers on bulk nickel.
sputtering	A vacuum deposition technique that builds thin films atom by atom by knocking material off a target with ions.
NEDO	Japan's New Energy and Industrial Technology Development Organization, a government agency that funded the 2015–2017 collaborative metal-hydrogen heat project.
specific energy (eV/H)	Total released energy divided by the number of hydrogen atoms absorbed or transferred — this chapter's yardstick for "beyond chemistry."
enthalpy	The heat absorbed or released by a process at constant pressure; hydride-formation enthalpies set the chemical baseline here.
metal hydride	A metal with hydrogen dissolved in its lattice; forming one releases roughly 0.2 eV per hydrogen atom — ordinary, well-tabulated chemistry.
Avogadro's number (N_A)	6.022×10^{23} particles per mole — the bridge from grams of hydrogen to a count of hydrogen atoms.
COP	Coefficient of performance: output power (or energy) divided by input. COP > 1 claimed from a sealed reactor is the headline number to audit.
differential calorimetry	Measuring excess heat as the difference between an active run and a blank (inert or hydrogen-free) run

under nominally identical conditions.

blank run

The control experiment: same apparatus, same temperature program, inactive sample. The claimed signal is active minus blank.

emissivity

How efficiently a surface radiates heat. In a vacuum system at 600–900 °C, heat leaves mostly as radiation, so a drift in sample emissivity mimics excess power.

air-flow calorimetry

Inferring power from the temperature rise and flow rate of air passing over a reactor — simple, but sensitive to flow-profile errors.

JCMNS

The *Journal of Condensed Matter Nuclear Science*, the field's own journal; editorially reviewed by sympathetic peers, not equivalent to mainstream peer review.

venue quality

Shorthand for how adversarial a paper's review was: *Nature*/PRX, then solid specialist journals (JJAP, IJHE), then proceedings and JCMNS.

preprint

A manuscript posted (typically on arXiv) before peer review; citable, but carrying no referee endorsement yet.

independent replication

Reproduction of a result by a group outside the claiming collaboration, with its own apparatus and its own incentives.

Horizon 2020

The EU's 2014–2020 research framework programme, which funded both CleanHME and HERMES.

CleanHME

"Clean Energy from Hydrogen-Metal Systems": the EU project (2020–2025, coordinated at Szczecin) that pursued both reactor heat and accelerator screening physics.

HERMES

The parallel EU project (2020–2024, coordinated at Turku) that attempted rigorous Pd–D replications and reported essentially negative results.

thick-target yield

The reaction count from a beam stopping entirely inside a target — the raw observable of accelerator

screening energy (U_e)

screening experiments.

The effective collision-energy boost from electrons around the fusing nuclei (Chapter VI); measured in metals at hundreds of eV.

ARPA-E

The U.S. Advanced Research Projects Agency—Energy, which funds high-risk energy research; it funded an eight-team LENR portfolio in February 2023.

reference experiment

A standardized, openly documented experiment that any lab can run to compare results — the program-design remedy for a field of one-off claims.

null result

A measurement consistent with zero effect. Publishing them is how fields self-correct; their absence from the literature is itself a data point.

BAA

Broad Agency Announcement — the solicitation document by which DARPA invites proposals for a new program.

ICCF

The International Conference on Cold Fusion (now "Condensed Matter Nuclear Science"), the field's annual meeting since 1990.

§1 The yardstick: electronvolts per hydrogen atom

Chapter I gave us the Gamow factor; Chapter III gave us the error bar. This chapter's claims are all calorimetric, so they live and die by the error-bar side of the frame — and the cleanest way to compare them is to normalize every claimed energy release to the number of hydrogen atoms that could have participated. Define the specific energy per hydrogen atom:

$$E_{/H} = \frac{E_{xs}}{N_H}, \quad N_H = \frac{m_H}{M_H} N_A,$$

WHERE

- $E_{/H}$ Scalar, eV per hydrogen atom: the claimed energy yield normalized to the hydrogen inventory.
- E_{xs} Integrated excess energy of the run (J or kJ) — the calorimeter's output, active minus blank.
- N_H Number of hydrogen (or deuterium) atoms absorbed or transferred during the run (dimensionless count).
- m_H Mass of that hydrogen (g), typically inferred from pressure drop or sample weight change.
- M_H Molar mass of hydrogen (constant, 1.008 g/mol; 2.014 for deuterium).
- N_A Avogadro's number (constant, $6.022 \times 10^{23} \text{ mol}^{-1}$).
-

Chemistry sets the floor rungs of the ladder. Dissolving hydrogen into palladium to form the β -phase hydride releases about **40 kJ** per mole of H_2 — roughly **0.2 eV** per hydrogen atom. Burning hydrogen outright, the most energetic common hydrogen chemistry, releases **286 kJ** per mole of H_2 , or about **1.5 eV** per hydrogen atom. No rearrangement of electrons gets much past that: *a few eV per hydrogen atom is the ceiling of chemistry*. At the other end, d+d fusion to ^4He releases **23.85 MeV** per two deuterons — about $1.2 \times 10^7 \text{ eV}$ per deuteron. Between the ceiling and the fusion line stretch nearly seven empty orders of magnitude where no known process lives, and it is precisely there — tens of eV/H to hundreds of keV/H — that this chapter's claims sit.

A worked example, which the widget below reproduces: an experiment reports $E_{xs} = 100 \text{ kJ}$ of integrated excess heat while absorbing 10 mg of hydrogen. Then $N_H = (0.010/1.008) \times 6.022 \times 10^{23} = 6.0 \times 10^{21}$ atoms and

$$E_{\text{H}} = \frac{10^5 \text{ J}}{1.602 \times 10^{-19} \text{ J/eV} \times 6.0 \times 10^{21}} \approx 104 \text{ eV/H},$$

about seventy times anything chemistry can supply from that hydrogen. If both numerator and denominator are right, something non-chemical happened. The auditor's eye goes straight to the two inputs: the numerator is a calorimeter reading with all of Chapter III's failure modes, and the denominator is *model-laden* — "hydrogen transferred" is inferred from pressure drops and absorption models, and choosing a smaller N_{H} inflates E_{H} arithmetically. Keep both dependencies in view for every claim that follows.

§2 Japan I: the NEDO collaborative phase (2015–2017)

The most sustained institutional excess-heat effort of the past decade is Japanese. From October 2015 to October 2017, a NEDO-funded project on "metal-hydrogen energy" joined Technova, Nissan Motor, and Kobe, Tohoku, Nagoya, and Kyushu universities to cross-check a single claim: that nano-composite powders — palladium–nickel or copper–nickel nanoparticles in zirconia, in batches of order a hundred grams — release anomalous heat when exposed to hydrogen or deuterium gas at 200–300 °C.

The capstone paper, Kitamura, Takahashi et al. in the *International Journal of Hydrogen Energy* (2018), reported excess power of 3–24 W lasting up to several weeks, with repeatable runs near 11–12 W (about 1.3–1.5 W per gram of nickel). Integrated over a run and divided by the hydrogen taken up, the reported specific energy came to tens of eV/H in typical runs — the paper's own best case reaches 90 MJ per mole of hydrogen, roughly 0.9 keV per hydrogen atom — and, in the most aggressive phase-by-phase accountings circulated by the collaboration, hundreds

of keV per deuterium atom. Single-element nanoparticle controls and blank runs showed nothing. The venue matters and cuts both ways: IJHE is a mainstream, peer-reviewed Elsevier journal — one of the few times a multi-watt excess-heat claim has passed ordinary review — but it is a hydrogen-energy journal, not a nuclear-physics one, and its referees judge calorimetry, not nuclear plausibility.

Three audit points temper the headline. First, replication was *internal*: the same apparatus lineage circulated among the six consortium partners, whose senior authors (Takahashi, Kitamura, Iwamura, Itoh, Kasagi) co-publish across nearly every paper in this literature; no laboratory outside the network has published a reproduction. Second, the measurement is differential calorimetry at elevated temperature, where changes in gas thermal conductivity and catalytic surface chemistry between active and blank runs can masquerade as excess power; no adversarial third party has been given the chance to exclude these. Third, the eV/H bookkeeping inherits the denominator problem of §1 — "hydrogen transferred" is a modeled quantity, which is exactly why the same dataset supports quotes from ~10 eV/H to ~270 keV/D.

AHEAD OF THEORY

The NEDO experiments reported comparable excess heat with *light* hydrogen and with deuterium. Any explanation built on d+d fusion fails outright for the pure-H runs, and p + Ni reactions face an even taller Coulomb barrier. By ordinary nuclear reasoning the absence of a strong isotope effect is evidence *against* a nuclear origin; the field instead carries it as an unexplained signature. Chapter IX's ledger takes it up.

§3 Japan II: Clean Planet, Tohoku, and the nano-multilayer line

The successor program moved from powders to precision. At Tohoku University's Condensed Matter Nuclear Reaction Division — funded in part by the startup Clean Planet Inc. — Iwamura, Itoh, and colleagues sputter nano-multilayers (typically six bilayers of ~4 nm Cu on ~20 nm Ni, on bulk nickel), load them with hydrogen gas near 250 °C, then heat them rapidly to 600–900 °C in vacuum and watch the calorimetry. Their flagship paper — Iwamura et al., *Japanese Journal of Applied Physics* 63, 037001 (2024) — reports steady excess power of roughly 4–5 W on inputs of tens of watts (up to ~17% of input), deliberate "heat bursts" triggered by perturbing the input power, and a specific energy exceeding 10 keV per absorbed hydrogen atom — with no neutrons or gamma rays above background. JJAP is a mainstream IOP/JSAP journal; this is among the strongest venues any modern excess-heat claim has reached, and that fact deserves to be stated as plainly as the caveats that follow.

The caveats are structural. In a vacuum system at 600–900 °C, heat leaves the sample almost entirely as radiation, so the excess-power inference rests on calibration curves that assume the sample's emissivity stays put. The group's own materials companion (Hioki et al., *Frontiers in Materials*, 2024) shows by in-situ synchrotron X-ray diffraction that the Cu/Ni multilayer alloys into a single layer by 900 °C under hydrogen — the "active" nanostructure is destroyed at operating temperature, and the surface that radiates at the end of a run is not the surface that was calibrated at the start. An emissivity drift is exactly the artifact this calorimetry cannot distinguish from excess heat, and no independent replication had been published as of mid-2026. Clean Planet's commercial position — patents, investment, and boiler-prototype announcements with claimed energy densities in the MJ-per-gram-of-hydrogen range — belongs in a separate mental file from the JJAP paper: company claims are not peer-reviewed, and the book's scoreboard counts only the journal record.

Two companion claims from the same group push further and rest on thinner ice. A photon-radiation calorimetry study (Kasagi et al., arXiv preprint, later in JCMNS) claims $\geq 410 \pm 108$ keV per hydrogen atom — three orders beyond even the

JJAP figure — and a 2024 arXiv preprint reports helium-3 detected in heat-producing samples, the predicted ash of Takahashi's multi-deuteron condensation model. Both are at arXiv/JCMNS level, unreplicated, and should be weighted accordingly.

AHEAD OF THEORY

Take the JJAP number at face value for a moment: $> 10 \text{ keV}$ per hydrogen atom, with no radiation. Any nuclear process releasing keV-to-MeV energy per event must dispose of that energy, and eighty years of nuclear measurement say it comes out as fast particles and photons. A mechanism that channels *all* of it into lattice heat, leaving nothing for the detectors, has no accepted formulation. The claims sit precisely in the regime — heat far beyond chemistry, radiation at zero — where conventional physics says "artifact." That is either the discovery or the diagnosis; Chapter IX's ledger holds the tension.

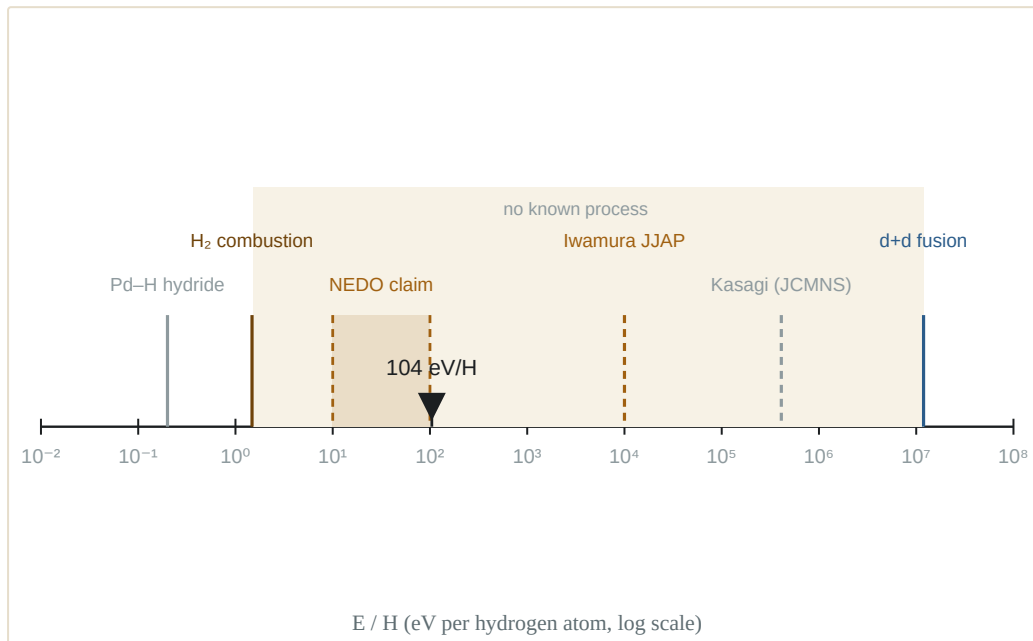
INTERACTIVE · THE BEYOND-CHEMISTRY AUDITOR

set energy · set hydrogen ·
audit

excess energy E (kJ)

absorbed H (mg)

duration (h)



$$E_{xs} = 100 \text{ kJ} \cdot m_H = 10.0 \text{ mg} \rightarrow N_H = 5.97 \times 10^{21} \cdot E/H = 104 \text{ eV}$$

chemistry **cannot** explain this — beyond the 1.5 eV/H ceiling by **$\times 69.6$** · over 170 h the mean power is 163.6 mW

— chemistry | contested claims — d+d fusion ▼ your experiment

The yardstick of §1, live: choose an integrated excess energy E_{xs} and a hydrogen inventory (mass of absorbed H), and the auditor computes $E/H = E_{xs}/N_H$ and places it on the log ladder between hydride chemistry (~ 0.2 eV/H), hydrogen combustion (~ 1.5 eV/H), the NEDO claims (~ 10 – 100 eV/H), the Iwamura JJAP claim (>10 keV/H), and d+d fusion (~ 12 MeV per deuteron). The duration slider converts the same energy into an average power, the number the calorimeter actually has to resolve. Note how a claim can be pushed up the ladder either by more heat *or* by assuming less hydrogen — the denominator is doing quiet work.

§4 Mizuno's R20: the claim that would not propagate

The boldest quantitative claim of the era came from Tadahiko Mizuno in Sapporo: reactors ("R19", "R20") in which palladium is hand-burnished onto nickel mesh, filled with low-pressure deuterium at a few hundred °C. With Jed Rothwell he reported 250 W of excess on 50 W of input, and 2–3 kW on 300 W — a COP of 5–10 — in the *Journal of Condensed Matter Nuclear Science* 29, 21–33 (2019), measured by air-flow calorimetry. The protocol was published openly, the materials cheap, and Rothwell promoted replication energetically. It was, in effect, the field's best-advertised recipe.

The replication record is therefore diagnostic. One attempt (H. Zhang, 2019, documented on community forums) saw ~4 W rising to 9.6 W that faded within hours — about which Mizuno himself expressed reservations; a Hokkaido University of Science group reported a small excess that never reached formal publication; numerous documented attempts found nothing. No independent confirmation exists in any journal, and Mizuno never re-established the kilowatt-class results after his laboratory was disrupted in 2020. The critiques are conventional: single-point air-flow calorimetry is vulnerable to flow-profile and recirculation errors of just the claimed magnitude, and kilowatt-scale power densities should have permitted trivially convincing demonstrations (self-sustained boiling) that were never shown.

The lesson generalizes beyond Mizuno. A genuinely working recipe with COP 5, published openly to a motivated worldwide community, propagates — that is what recipes do. This one did not. When an accessible, extraordinary protocol fails to propagate, the failure itself is evidence about the original measurement, and it should update us on less accessible claims too.

§5 Europe: CleanHME, HERMES, and the ENEA metallurgy line

The European Union ran, in effect, a controlled experiment on the field itself: two Horizon 2020 projects, launched together in 2020, with different postures. CleanHME (grant 951974, €5.6M, sixteen partners, coordinated by Konrad Czerski at the University of Szczecin) pursued hydrogen–metal energy directly; HERMES (grant 952184, ~€4.0M, coordinated by Pekka Peljo at Turku, with partners including TU München and Imperial College) set out to reproduce Pd–D excess-heat conditions with rigorous methods.

HERMES's outcome is quickly told and heavily weighted: it reported essentially negative results, explicitly confirming the Google consortium's conclusion (Chapter VII) that cold fusion remains unproven while the extreme-loading corner of parameter space stays hard to reach. The headline conclusion lives in EU project reporting rather than a journal, but the methodological outputs — loading protocols, operando characterization — were published normally. As with Google, a resourced, competent, motivated team looked and found nothing.

CleanHME's legacy splits in two. Its reactor calorimetry — project communications from the January 2025 final event claim multi-laboratory excess heat with "promising COP" but short-lived excursions — has produced *no peer-reviewed publication as of mid-2026*, so by this book's rules those numbers do not enter the scoreboard. Its accelerator physics, by contrast, is the project's durable achievement and reached remarkable venues: a proposed 0^+ threshold resonance in ^4He near the d+d threshold (Czerski, *Phys. Rev. C* 106, L011601 (2022)), follow-up evidence in PRC (2024), and finally Dubey, Czerski et al., *Physical Review X* 15, 041004 (2025), reporting d+d thick-target yields measured down to about 675 eV center-of-mass — the lowest ever — and signatures that below a few keV the reaction proceeds dominantly through the resonance, decaying by internal electron–positron pair emission. Recall the screening formalism of Chapters I and VI, which is the language of these measurements:

$$f(E) = \frac{\sigma_{\text{screened}}(E)}{\sigma_{\text{bare}}(E)} \approx \exp\left[\pi\eta(E) \frac{U_e}{E}\right], \quad 2\pi\eta = 31.29 Z_1 Z_2 \sqrt{\frac{\mu}{E}},$$

WHERE

$f(E)$ Screening enhancement factor (dimensionless): how much faster the reaction runs than the bare-nucleus expectation at center-of-mass energy E . Valid for $U_e \ll E$; note the single $\pi\eta$, not $2\pi\eta$ (Assenbaum–Langanke–Rolfs).

U_e Screening energy (eV): ~25 eV in gas targets, ~100 eV from defect-free-metal theory, and of order 300–350 eV in the Szczecin group's deuterated-zirconium measurements.

η Sommerfeld parameter (dimensionless); the 31.29 prefactor holds for E in keV (center-of-mass) and reduced mass μ in amu ($\mu = 1.007$ for d+d, so $2\pi\eta \approx 31.4/\sqrt{E_{\text{keV}}}$).

PRX is a top-tier, highly selective venue, and the claim is proportionally extraordinary: a new *dominant* channel in the most-studied fusion reaction in physics, below a few keV, missed for eighty years. One group stands behind it; independent confirmation is the entire question. It is also, notably, the only published mechanism candidate that could in principle connect the accelerator anomalies to heat-without-neutrons claims — though no quantitative bridge from picobarn yields to watts exists, and the authors do not claim one.

Italy's ENEA Frascati laboratory (Vittorio Violante) rounds out the European picture with two decades of palladium metallurgy: rolling and annealing protocols, grain-orientation and surface-texture studies, aimed at reproducibly reaching loadings of $x_{\text{D/Pd}} > 0.9$ — the operational threshold of Chapter V. The loading metallurgy is solid materials science, adopted by SRI and by the Google program; the associated excess-heat and helium correlations live almost entirely in conference proceedings and JCMNS, and the Google program, using ENEA-grade materials knowledge, found no excess heat in over 420 calorimetric tests. The metallurgy replicated; the anomaly did not follow it.

§6 ARPA-E's re-entry — and DARPA's escalation

On 17 February 2023, ARPA-E announced roughly USD 10M across eight projects — the first dedicated U.S. federal LENR funding since the 1989 and 2004 DOE reviews declined to recommend a program. Program Director Scott Hsu had pre-announced the topic at ICCF-24 (the "Solid-State Energy Summit," Mountain View, July 2022); the program's stated philosophy was deliberately agnostic and methods-first: hypothesis-driven experiments, control of variables, modern nuclear diagnostics, standardized and shareable protocols, results — positive or null — publishable in top-tier journals.

The portfolio: Texas Tech (Robert Duncan, USD 1.15M) as a cohort-wide materials-characterization and nuclear-product-detection resource; Lawrence Berkeley National Laboratory (Thomas Schenkel, USD 1.5M, with Jeremy Munday's UC Davis group) interrogating metal hydrides with sub-keV ion beams; MIT (~USD 2M) building a reproducible gas-loading test platform — a team associated in secondary reporting with Hagelstein, Chiang, and Metzler, though the award documents name no PIs; Stanford (USD 1.5M) on nuclear-product detection from deuterated nanoparticles under phonon stimulation; the Energetics Technology Center (USD 1.5M, "CATHODE") electro-depositing Pd/Li from heavy water directly onto a scintillator; Amphionic (USD 296K) on nanoscale inter-surface potential wells; and two University of Michigan awards (USD 1.108M and a second) on D₂ gas cycling through Pd nanocrystals with laser stimulation and microwatt calorimetry, and on Pd-aramid-nanofiber composites. Hsu subsequently left ARPA-E for a DOE fusion-coordination role, with the portfolio passing to Ahmed Diallo.

What has the cohort produced by mid-2026? The honest answer, stated plainly: *no peer-reviewed excess-heat confirmation, no published reference-experiment protocol or detection-standards document, and no published null-result paper either* — the

publish-negative-results promise is so far unrealized in the open literature. The flagship experimental output is a preprint: Karahadian et al. (arXiv:2512.06212, December 2025), from the LBNL/UC Davis team, reporting that in electrochemically loaded Pd and Ti hydrides the D–D fusion yield stops falling exponentially below about 2.5 keV and plateaus — at the lowest energies some 10^{18} times above the bare-nucleus extrapolation. A preprint is a claim awaiting referees; but its qualitative phenomenology converges with the independent, peer-reviewed Szczecin PRX result of §5, from a different laboratory on different funding. The cohort's strongest *refereed* output to date is theory-adjacent: Metzler, Hunt, Hagelstein, and Galvanetto's topical review (*New Journal of Physics* 26, 101202 (2024)) cataloguing known solid-state mechanisms — site proximity, screening, transient lattice dynamics — and arguing each is worth orders of magnitude, while the required total remains more than forty.

AHEAD OF THEORY

Two groups — Szczecin (PRX 15, 041004, peer-reviewed) and LBNL/UC Davis (arXiv:2512.06212, preprint) — now report a *plateau* in sub-keV d+d yields. Static screening cannot produce a plateau: $\sigma_{\text{bare}}(E + U_e)$ merely shifts the exponential cliff, so any single U_e still predicts a falling yield. A flat yield at falling beam energy means either new physics (thermalized-deuteron fusion in ion tracks, a threshold resonance) or an unmodeled systematic (beam tails, surface deuterium). Nothing in accepted theory currently accounts for it — the sharpest experiment–theory gap of the 2020s, logged for Chapter IX.

The institutional story then escalated: in January 2026 DARPA issued a Broad Agency Announcement for MARRS (HR001126S0007; proposers day 20 January, proposals due 12 March, anticipated start around September 2026), explicitly citing the post-2023 evidence of large solid-state rate amplification. The program runs two 18-month phases with reviews at months 12 and 16, and sets a quantitative goal: raise solid-state fusion rates from the current state of the art, of order 0.1 reactions per second per gram, to 10^6 . It is worth converting that *success criterion* into power to calibrate expectations:

$$P = R\bar{Q} = 10^6 \text{ s}^{-1} \text{ g}^{-1} \times 3.65 \text{ MeV} \times 1.602 \times 10^{-13} \text{ J/MeV} \approx 0.6 \mu\text{W}$$

WHERE

P Thermal power per gram of material if the MARRS goal rate is reached (scalar, W/g).

R Fusion reaction rate per gram ($\text{s}^{-1} \text{ g}^{-1}$): MARRS's month-34 goal is 10^6 , up from a stated baseline of ~ 0.1 .

$\bar{Q}$ Mean energy per d+d fusion for the two dominant branches, $\approx 3.65 \text{ MeV}$ (Chapter I).

Total success — a ten-million-fold amplification — yields *microwatts* per gram: a nuclear diagnostics milestone, still a factor of a million or more from an energy technology. That is not a criticism of the program; it is the program's own sober framing, and the right calibration for headlines about "DARPA funding cold fusion." The 2023–2026 federal re-entry has been a bet on *measurement*, and its returns so far are measured in screening physics, not watts.

§7 The mid-2026 scoreboard

Lay the decade's claims on one table, with venue and replication status attached, and the pattern is stark.

Replicated and mainstream: enhanced electron screening in metal deuterides under keV beams (Tohoku 2002, Bochum, Szczecin — Chapter VI), now extended by the sub-keV yield measurements, with the plateau reported independently by Szczecin (PRX, peer-reviewed) and LBNL/UC Davis (preprint). Also replicated and

mainstream: the *negative* results — Google (*Nature*, 2019), HERMES (2024) — from every resourced external program that has audited the electrolytic excess-heat claims.

Peer-reviewed, single-group, unreplicated: Kitamura's nano-composite heat (IJHE, 2018 — internal consortium replication only); Iwamura's multilayer heat (JJAP, 2024); the Czerski group's e^+e^- channel (PRX, 2025). Publication in a serious venue means competent referees found no fatal flaw — not that the claim is true. For the calorimetric entries, the mundane explanations (emissivity drift, gas-conduction shifts, model-laden hydrogen accounting) have never been excluded by an adversarial third party, because no hostile, resourced replication of either Japanese protocol has ever been attempted. That — not another positive paper from the same network — is the concrete open item.

Field-internal or commercial: Mizuno's kilowatts (JCMNS; replication effectively failed), the 410 keV/H photon calorimetry and He-3 detections (arXiv/JCMNS), CleanHME's reactor COP (project communications), Clean Planet's boiler prototypes and every other company announcement. These are claims, not results, and the decade's most reliable regularity is the asymmetry between them and the journal record: company numbers run kilowatts and COPs; refereed numbers run watts and percent-level effects; replicated numbers, so far, run picobarns.

What to watch from here: whether arXiv:2512.06212 survives review and whether anyone outside Szczecin and Berkeley reproduces the plateau; whether any ARPA-E team publishes a reference protocol or a null; whether MARRS's month-12 review reports amplification on schedule; and whether anyone, anywhere, subjects the Iwamura protocol to a hostile replication. Those are empirical questions with dates attached — Chapter X takes up what the answers would mean. But the claims already on the table outrun every accepted mechanism by orders of magnitude, and before we can weigh futures we must face the explanatory gap directly. That is Chapter IX.

FURTHER READING

Kitamura, A., Takahashi, A., Takahashi, K., Seto, R., Hatano, T., Iwamura, Y., Itoh, T., Kasagi, J. et al. *Excess heat evolution from nanocomposite samples under*

exposure to hydrogen isotope gases, International Journal of Hydrogen Energy 43 (33), 16187–16200 (2018) — the NEDO collaborative phase's capstone claim; DOI 10.1016/j.ijhydene.2018.06.187.

Iwamura, Y., Itoh, T., Yamauchi, S. & Takahashi, T. *Anomalous heat generation that cannot be explained by known chemical reactions produced by nano-structured multilayer metal composites and hydrogen gas*, Japanese Journal of Applied Physics 63, 037001 (2024) — the Tohoku/Clean Planet flagship; DOI 10.35848/1347-4065/ad2622.

Mizuno, T. & Rothwell, J. *Excess heat from palladium deposited on nickel*, Journal of Condensed Matter Nuclear Science 29, 21–33 (2019) — the R19/R20 claims; field-internal journal, cited here for the record audited in §4.

Dubey, R., Czerski, K., Das H., G., Kowalska, A., Targosz-Ślęczka, N., Kaczmarski, M. & Valat, M. *Experimental Signatures of a New Channel of the Deuteron-Deuteron Reaction at Very Low Energy*, Physical Review X 15, 041004 (2025) — d+d yields to ~675 eV c.m. and the e⁺e⁻-pair channel claim; arXiv:2408.07567.

Metzler, F., Hunt, C., Hagelstein, P. L. & Galvanetto, N. *Known mechanisms that increase nuclear fusion rates in the solid state*, New Journal of Physics 26, 101202 (2024) — the ARPA-E-era mechanism census; DOI 10.1088/1367-2630/ad091c.

Karahadian, M. E., Colborne, M., Persaud, A., Schenkel, T. & Munday, J. N. *Enhanced nuclear fusion in the sub-keV energy regime*, arXiv:2512.06212 (2025) — the LBNL/UC Davis sub-keV plateau; preprint, unrefereed as of mid-2026.

ARPA-E. *U.S. Department of Energy Announces USD 10 Million in Funding to Projects Studying Low-Energy Nuclear Reactions*, press release and project descriptions, arpa-e.energy.gov (17 February 2023) — the program's goals and the eight-team portfolio of §6.

Chapter production note: researched and drafted by a Claude subagent (draft agent: `claude-fable-5`); every Further-reading citation web-verified against primary sources by the orchestrating lead, running as `claude-opus-4-8` after a mid-project model change (see the methods appendix). See the **methods appendix** for the full research process.

CHAPTER IX

Theories and the Explanatory Gap

Fifty-two orders of magnitude is not a gap that a clever idea jumps on inspiration — so this chapter audits every serious attempt, with the receipts, and ends with the IOUs.

BEFORE WE START

Eight chapters of experiments have left us with a strange inventory: a handful of anomalies that refuse to die (excess-heat reports, the heat–helium correlation, tritium episodes, transmutation patterns, anomalously strong screening), and a mountain of solid physics saying none of it should happen. This chapter turns to the theorists. Since 1989, dozens of mechanisms have been proposed to explain how nuclei might react inside ordinary metals. We audit the serious ones — the proven muon benchmark, static electron screening, phonon-coherence schemes, the Widom–Larsen neutron theory, and the exotica — and we hold each to the same two requirements established in Chapter I: raise the fusion rate by roughly fifty orders of magnitude, and change the nuclear ashes. Each audit ends with a number: how far short, and at which step.

The tone here matters. It would be easy to sneer — most of these theories fail immediately and visibly. It would be equally easy to hand-wave — some of their authors are serious physicists working on a genuinely unsolved problem, and one experimental anomaly in this area (the size of measured screening energies) is real, peer-reviewed, and unexplained. So we do neither. We state what each theory claims, what it would need to be true, what published critiques found, and what remains. The chapter closes with this book's formal ledger: every experimental result from Chapters III–VIII that currently runs ahead of any accepted theory, each flagged with the one measurement that would settle it.

TERMS YOU'LL MEET

lepton

The particle family that includes the electron, the muon, and neutrinos; leptons feel electromagnetism and the weak force but not the strong nuclear force.

muon

An unstable heavy cousin of the electron: same charge, 206.77 times the mass, living 2.2 microseconds on average.

muonic molecule

A hydrogen molecular ion in which a muon replaces the electron; the muon's mass shrinks the molecule about 200-fold.

alpha-sticking

The fate that ends a muon's catalytic career: after a fusion, the muon occasionally binds to the outgoing helium nucleus and is carried away.

cycling rate

How many complete catalysis loops (capture, molecule formation, fusion, release) a muon can execute per second.

effective mass

The apparent mass an electron acquires inside a crystal, set by the band structure; a bookkeeping mass for how it accelerates, not new inertia for particle physics.

quasiparticle

A collective excitation of many particles that behaves like a single particle — a "dressed" electron, a phonon, a plasmon.

screening energy (U_e)

The effective reduction of the Coulomb barrier by nearby electrons, expressed as an energy boost to the colliding pair (Chapters I and VI).

jellium

The idealized metal of many-body theory: a smeared-out uniform positive background filled with an electron gas.

degenerate electron gas

Electrons packed so densely that the Pauli exclusion principle, not temperature, dictates their motion — the actual state of conduction electrons in a metal.

Thomas–Fermi screening	The standard quantum treatment of how a degenerate electron gas shields a charge, with a characteristic screening length of tens of picometers in metals.
Debye screening	The classical (hot, dilute plasma) screening formula; applying it to cold metal electrons is a category error that nonetheless fit some data (§3).
WKB action	The integral G in the semiclassical tunneling formula e^{-G} ; the single number that encodes how opaque a barrier is.
classical turning point	The separation at which the pair's energy equals the barrier height — the outer edge of the classically forbidden region.
attempt frequency	The prefactor ν_0 multiplying the tunneling exponential: roughly how often per second the pair "assaults" the barrier.
zero-point motion	The irreducible quantum jitter a confined particle keeps even at absolute zero.
phonon	The quantum of lattice vibration; energies of milli-electronvolts, the only currency a crystal can pay energy out in.
coherence	Many quantum systems oscillating in phase, so their amplitudes add before squaring — the source of every "collective enhancement" claim in this chapter.
Dicke superradiance	The textbook example of coherent enhancement: N phase-locked emitters radiate collectively, far faster than N independent ones.
spin–boson model	A workhorse model of quantum optics: two-level systems coupled to an oscillator bath; Hagestein's framework is a "lossy" variant.
downconversion (fractionation)	Splitting one large energy quantum into many small ones — here, one 24 MeV nuclear

quantum into $\sim 10^9$ phonons; the load-bearing miracle of coherence theories.

exit channel

The route by which a reaction's energy leaves — which products, with what energies. Chapter IV's evidence problem is an exit-channel problem.

weak interaction

The force behind beta decay; the only interaction that converts protons into neutrons and back.

inverse beta decay

The reaction $e^- + p \rightarrow n + \nu_e$: an electron and a proton merging into a neutron plus a neutrino — possible only if the electron brings enough energy.

ultra-low-momentum neutron

Widom–Larsen's proposed product: a neutron born almost at rest, whose quantum wavelength is huge and whose capture probability is correspondingly enormous.

surface plasmon

A collective sloshing of surface electrons on a metal; Widom–Larsen's proposed source of enormous local electric fields.

mass renormalization

A field-induced shift in a particle's effective mass; the contested first step of the Widom–Larsen chain.

deep Dirac level

A claimed ultra-tight electron orbit, picometers from the nucleus, extracted from solutions of the Dirac equation that standard quantum mechanics rejects as unphysical.

hydrino

A claimed below-ground-state hydrogen atom (Randell Mills); incompatible with quantum electrodynamics and refuted in the mainstream literature.

§1 The bar a theory must clear

Chapter I set the yardstick and every chapter since has leaned on it, so let us state it one last time as an entrance exam. A deuteron pair bound in a D_2 molecule fuses at the Koonin–Nauenberg rate of about 3×10^{-64} per second — the zero point of everything. A watt of nuclear power per cubic centimeter of palladium deuteride, the scale Fleischmann and Pons claimed, requires:

$$\lambda_{\text{need}} = \frac{P_{\text{xs}}/Q}{N_{\text{pairs}}} \approx \frac{2.6 \times 10^{11} \text{ s}^{-1} \text{ cm}^{-3}}{3.4 \times 10^{22} \text{ cm}^{-3}} \approx 8 \times 10^{-12} \text{ s}^{-1} \text{ per pair},$$

WHERE

- λ_{need} Scalar, s^{-1} : the per-pair fusion rate needed to explain the claimed power.
- P_{xs} Claimed excess power, here 1 W per cm^3 .
- Q Energy per reaction; taking the most generous channel, $\text{d}+\text{d} \rightarrow {}^4\text{He}$ at $23.85 \text{ MeV} = 3.82 \times 10^{-12} \text{ J}$.
- N_{pairs} Deuteron pairs per cm^3 in PdD: $n_{\text{D}} \approx 6.8 \times 10^{22} \text{ cm}^{-3}$, so $\approx 3.4 \times 10^{22}$ pairs.
-

From $10^{-63.5}$ to $10^{-11.1}$: **requirement one is a rate enhancement of roughly fifty-two orders of magnitude.** Requirement two is the ashes. Chapter IV established that the observed products — if real — are not those of vacuum $\text{d}+\text{d}$ fusion: helium-4 without gamma rays, tritium without neutrons. The sharpest form of this requirement is a peer-reviewed constraint derived by Peter Hagelstein (Naturwissenschaften 97, 345 (2010)): if the excess heat really is $\text{d}+\text{d} \rightarrow {}^4\text{He}$ at $\sim 24 \text{ MeV}$ per helium, then the measured neutron ceilings (0.008–0.8 neutrons per joule) force the newborn helium nucleus to carry almost none of that energy:

$$E_{\alpha} \lesssim 6.2\text{--}20.2 \text{ keV} \lll Q = 23.85 \text{ MeV}.$$

WHERE

E_{α} Kinetic energy of the ^4He at birth. A fast alpha in a deuterated lattice knocks deuterons forward, and those produce detectable secondary d+d neutrons; the neutron limits therefore bound E_{α} .

More than 99.9% of the reaction energy must leave through some channel that is neither a gamma ray nor an energetic particle. **Requirement two is thus not just changed branching ratios; it is a new exit channel** — a way to pay out a 24 MeV nuclear quantum in keV-or-smaller installments. And there is a third requirement, less often stated: **a mechanism must not predict effects in the many places we have looked and found nothing**. Accelerator measurements pin the d+d cross section and its screening down to sub-keV energies at the few-percent level (Chapter VI); decades of metallurgy, electrochemistry, and geochemistry have processed megatons of hydrogen-loaded metal without unexplained heat, helium, or radioactivity; and the one particle known to catalyze cold fusion — the muon, §2 — behaves in metals exactly as standard physics says. A theory that makes palladium special must explain why palladium refineries are not.

THE LEGGETT-BAYM BOUND, AND THE LOOPHOLE EVERY SURVIVOR USES

In 1989 Anthony Leggett and Gordon Baym proved an exact result (Phys. Rev. Lett. 63, 191 (1989)): in a many-body system *in thermal equilibrium*, the tunneling rate between two nuclei is bounded by what the exact interatomic potential allows, and no electronic environment in equilibrium condensed matter can raise cold d+d fusion to observable levels — the bound sits tens of orders of magnitude below λ_{need} . The theorem has never been challenged in the mainstream literature. Its fine print is therefore load-bearing: every surviving LENR theory locates its miracle in *non-*

equilibrium physics — driven surfaces, transient defects, coherent excitations — precisely because equilibrium is provably closed.

That is the exam: $\times 10^{52}$ on the rate, a new exit channel for the ashes, and no collateral damage to the rest of physics. In the frame this book has used throughout — the Gamow factor and the error bar — a theory's job is to move the Gamow factor; the error bar it must respect. Now the candidates.

§2 The proven catalyst: what the muon actually buys

Cold fusion — genuinely cold, replicated worldwide, quantitatively understood — has existed in laboratories since 1957. The catalyst is the muon. Drop negative muons into cold deuterium and fusion proceeds briskly at cryogenic temperatures. The mechanism is pure geometry. A muonic hydrogen molecule is an ordinary one with the electron's role played by a particle 207 times heavier, and every Coulombic length scales inversely with the orbiting mass:

$$a_{\mu} = a_0 \frac{m_e}{m_{\mu}} = \frac{52.9 \text{ pm}}{206.77} \approx 256 \text{ fm}.$$

WHERE

- a_0 The Bohr radius, 52.9 pm (constant): the size scale of ordinary atoms.
- m_μ The muon mass, $206.77 m_e$ (constant). Using the free mass; the reduced mass in a muonic molecule is slightly smaller.
- a_μ The muonic Bohr radius: the deuterons in a $dd\mu$ molecule sit ~ 500 fm apart instead of 74,000 fm.
-

Squeezing the tunneling gap two-hundred-fold collapses the WKB action, and the in-molecule fusion rate rockets from 10^{-64} s^{-1} to of order $10^8\text{--}10^9 \text{ s}^{-1}$ for $dd\mu$ and $\sim 10^{12} \text{ s}^{-1}$ for $dt\mu$ (Ponomarev's canonical review, Contemporary Physics 31, 219 (1990)). Read that again: *a factor of ~ 200 in one length converts the age-of-the-universe into a nanosecond — about seventy orders of magnitude*. Muon catalysis is the existence proof that "chemistry-scale" rearrangement of the lepton cloud can close the entire Gamow gap. It is also, not coincidentally, the template every "heavy electron" theory in this chapter tries to imitate.

But the muon also teaches the field's hardest economics lesson. A muon lives $2.2 \mu\text{s}$, and each catalytic loop takes time, so the number of fusions one muon delivers obeys a simple budget:

$$\frac{1}{X_\mu} \approx \omega_s + \frac{\lambda_0}{\phi \lambda_c}, \quad \lambda_0 = \frac{1}{\tau_\mu} = 4.55 \times 10^5 \text{ s}^{-1}.$$

WHERE

- X_μ Fusions per muon (dimensionless) — the catalyst's total career.
- ω_s Effective alpha-sticking probability, measured $\approx 0.45\text{--}0.6\%$ for d–t cycles after in-flight stripping is credited.
- λ_c Density-normalized cycling rate: up to $\approx 1.45 \times 10^8 \text{ s}^{-1}$ in dense d–t mixtures.
- ϕ Target density in units of liquid-hydrogen density ($4.25 \times 10^{22} \text{ atoms/cm}^3$).
- τ_μ The muon lifetime, 2.197 μs (constant): the clock that ends every career.
-

At high density the lifetime term shrinks and sticking rules: $X_\mu \approx 1/\omega_s \approx 150$, which is what experiment found — Steven Jones (the same Jones of Chapter IV) reported ~ 150 d–t fusions per muon at LAMPF (Nature 321, 127 (1986)), still roughly the record. One hundred fifty fusions at 17.6 MeV each is ~ 2.6 GeV of output; making one muon at an accelerator costs ~ 5 GeV. Nature permits cold fusion and then, with measured numbers, denies it a profit. For this chapter the muon's real gift is calibration: it fixes the exchange rate between lepton mass and fusion rate. All electron-screening effects in metals combined behave like a lepton of perhaps $1.01\text{--}1.1 m_e$ at the relevant separations. The distance from there to $207 m_e$ — from a few eV of extra binding to a few keV — is the explanatory gap, made flesh. Hold that thought through the widget in §3.

§3 Static screening: real physics, audited to the electronvolt

The most respectable mechanism is the one Chapter VI measured: electrons crowd around and between two approaching deuterons and cancel part of their repulsion, worth a screening energy U_e . Everyone agrees screening exists; the audit is about

magnitudes, and it has three lines. *What consistent theory delivers:* treating palladium's conduction electrons honestly — as a degenerate quantum gas, via Thomas–Fermi, jellium, or dielectric-function methods — gives $U_e \approx 100\text{--}150\text{ eV}$; the Berlin–Szczecin group's careful dielectric calculation gives 134 eV for Pd (Huke et al., Phys. Rev. C 78, 015803 (2008)). *What beams measure:* more. The same paper's re-measurement finds $296 \pm 15\text{ eV}$ in Pd; Kasagi's group at Tohoku found $\approx 310\text{ eV}$ in Pd and $\approx 600\text{ eV}$ in PdO (J. Phys. Soc. Jpn. 71, 2881 (2002)); Raiola and the Bochum collaboration reported $\approx 800\text{ eV}$ in Pd (Eur. Phys. J. A 19, 283 (2004)); a Berkeley plasma-discharge experiment fit $1000 \pm 250\text{ eV}$ (Schenkel et al., J. Appl. Phys. 126, 203302 (2019)). The groups disagree with one another by factors of 2–3 — far outside quoted errors — and the one model that "fit" the material trend, classical Debye screening of quasi-free electrons, is physically inadmissible: conduction electrons at room temperature are degenerate, and Boltzmann statistics simply do not apply to them (the standing critique in Huke et al. and in later reviews).

AHEAD OF THEORY

Sit with the strangeness of this: an *uncontested, replicated-in-kind accelerator measurement* — screening energies in metals two to four times larger than the best available calculation, with a periodic-table pattern (large in Pd, Ta, PdO; absent in Cu, Ag, Au) no model reproduces — has stood unexplained since 2001. It is the only entry in this chapter's ledger (§7) that is mainstream physics by every sociological measure. It licenses nothing about excess heat; it does prove our quantitative command of electrons-near-fusing-nuclei is imperfect at the factor-of-a-few level.

What would be needed: this is where the widget below earns its place, because the arithmetic is exponential and intuition fails. Model the pair with the chapter's one formula — the semiclassical tunneling rate through a screened Coulomb barrier:

$$\lambda_{\text{fus}} = C \nu_0 \exp \left[-\frac{2}{\hbar} \int_{r_n}^{r_t} \sqrt{2\mu(V(r) - E)} \, dr \right], \quad V(r) = \frac{1.44 \text{ eV} \cdot \text{r}}{r}$$

WHERE

ν_0, C	Attempt frequency (10^{16} s^{-1} here) and a dimensionless calibration constant ($C \approx 3.5 \times 10^2$) absorbing the nuclear physics the exponent omits; fixed once by matching the D_2 anchor.
μ	Reduced mass of the deuteron pair, $\mu c^2 = 937.81 \text{ MeV}$ (constant).
r_n, r_t	Nuclear channel radius (4 fm) and the classical turning point where $V = E$ — the ends of the forbidden region.
$V(r), E$	The screened pair potential and the pair's energy ($E \approx 0$ at room temperature).
U_e	The static screening energy (eV). The model is deliberately <i>generous</i> : it grants the pair the entire U_e as if it were collision energy, everywhere along the path.
m^*	The effective lepton mass. Raising it compresses the whole screened geometry, $r \rightarrow r (m_e/m^*)$ — the muon dial.

Calibrated so that $U_e = 27 \text{ eV}$, $m^* = m_e$ reproduces Koonin–Nauenberg's $3 \times 10^{-64} \text{ s}^{-1}$, this model computes the whole ladder. Consistent static screening, 100–150 eV, lifts the rate to 10^{-24} – 10^{-16} s^{-1} : an astonishing forty-orders gift, and still five to thirteen orders short of the watt line — *in a model rigged in screening's favor*. Careful equilibrium treatments that keep the pair in its molecular well rather than crediting the full shift as collision energy find far less: of order 10^{-45} s^{-1} at the theoretically consistent U_e , at least thirty-four orders short, consistent with the Leggett–Baym ceiling. And the beam-measured values must not be transplanted at all — the widget shows why with a reductio. Dial in Raiola's 800 eV as if it were a static potential and the model roars *past* the watt line: every gram of PdD on Earth would melt itself. Palladium deuteride sits quietly on shelves; therefore a beam-derived U_e — the dynamic response of the electron gas to a keV projectile passing through — is not an equilibrium potential energy available to two thermal deuterons parked 0.2 nm apart. That distinction is stated explicitly in the accelerator literature itself (Huke et al., PRC 78; Metzler et al., NJP 26). The verdict on static screening: real physics, the largest single legitimate rate multiplier known in a solid, quantitatively short by tens of orders of magnitude, and — per the aside above — itself imperfectly understood at the factor-of-a-few level.

INTERACTIVE · THE FIFTY-ORDER LADDER

slide U_e · grow the lepton · jump the presets

screening U_e (eV)

lepton mass m^*/m_e

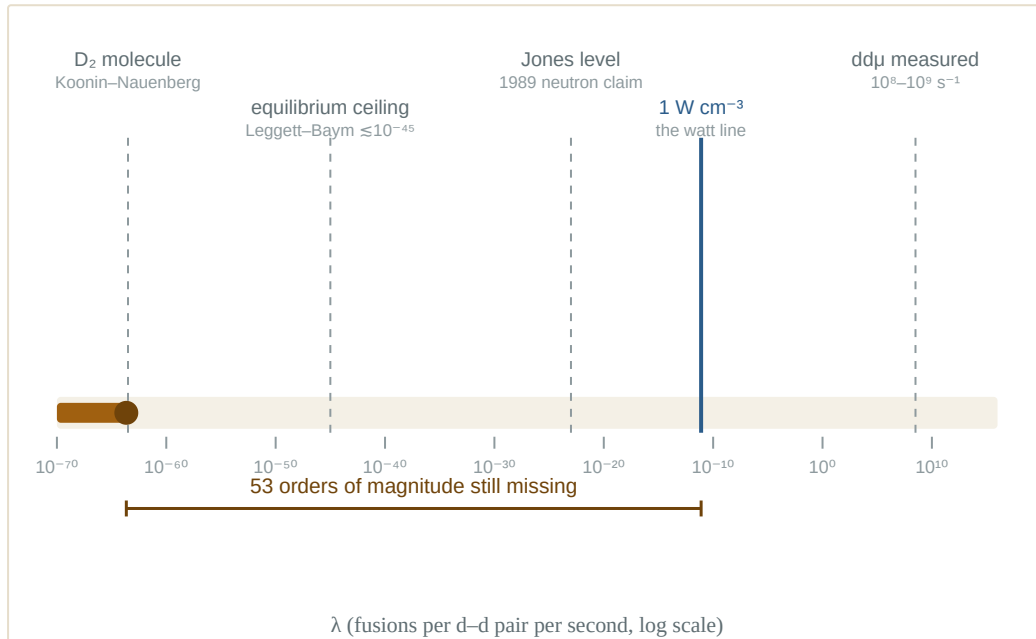
D₂ (27 eV)

Pd theory (134 eV)

Pd beam (296 eV)

Raiola (800 eV)

muon ($m^* = 207$)



$U_e = 26.9 \text{ eV}$ · $m^*/m_e = 1.00$ · turning point $r_t = 53.5 \text{ pm}$ · WKB action $G = 189.3$

$\lambda \approx 10^{-63.7} \text{ s}^{-1} \text{ per pair} \rightarrow 52.6 \text{ orders short of } 1 \text{ W cm}^{-3}$

— computed λ (this model) | the watt line | ladder rungs (measured / canonical)

The chapter's equation computed live: the WKB action $G = \frac{2}{\hbar} \int_{r_n}^{r_t} \sqrt{2\mu(V - E)} dr$ is integrated numerically for $V(r) = 1.44 \text{ eV} \cdot \text{nm}/r - U_e m^*/m_e$ from $r_n = 4 \text{ fm}$ to the turning point, and $\lambda_{\text{fus}} = C\nu_0 e^{-G}$ is calibrated ($\nu_0 = 10^{16} \text{ s}^{-1}$, $C \approx 3.5 \times 10^2$) so the D₂ case ($U_e = 27 \text{ eV}$) reproduces Koonin-Nauenberg's $3 \times 10^{-64} \text{ s}^{-1}$. Rungs mark the D₂ molecule, the Leggett-Baym equilibrium ceiling, the Jones neutron claim (10^{-23}), the watt line ($10^{-11.1}$), and measured ddμ fusion (10^8 - 10^9 s^{-1}). Two honesty notes: the model hands the pair the full U_e as collision energy, so it *overstates* what static screening does — careful equilibrium treatments reach only $\sim 10^{-45} \text{ s}^{-1}$ at the theoretically consistent 100-150 eV; and at $m^* = 207$ the crude fixed prefactor overshoots the measured ddμ rung by roughly five orders, a reminder of what a one-line WKB model ignores. That beam-measured screening values (296-1000 eV) drive the model

absurdly past the watt line is the point: they cannot be static potentials, or palladium deuteride would glow.

§4 Coherence: fractionating a nuclear quantum

The second family of theories aims not at the barrier but at requirement two — the exit channel. Its central figure is Peter Hagelstein of MIT, who has spent three decades constructing variants of a lossy spin–boson model in which many d+d pairs, coherently coupled to a highly excited lattice vibrational mode, transfer their reaction energy *collectively* to the lattice. The required miracle has a name — downconversion — and a size:

$$N_{\text{quanta}} \sim \frac{23.85 \text{ MeV}}{\hbar\omega_{\text{ph}}} \sim \frac{2.4 \times 10^7 \text{ eV}}{\sim 2.5 \times 10^{-2} \text{ eV}} \sim 10^9.$$

WHERE

$\hbar\omega_{\text{ph}}$ The energy of one THz-scale optical phonon, a few tens of meV — the largest coin the lattice accepts.

N_{quanta} How many phonons must absorb one fusion quantum: about a billion, coherently, before any competing radiative or particle channel fires.

Multiphoton processes of order ten are laboratory physics; order 10^9 is not, and Hagelstein does not claim otherwise — his papers derive conditions under which

"lossy" dynamics (fast decay of intermediate states) breaks the destructive interference that normally forbids such transfers. The mechanism is aimed with impressive precision at the missing-gamma problem, and it is constrained by his own mainstream result from §1: any successful version must deliver the helium below ~ 20 keV. The audit, in register: the *constraint* paper is peer-reviewed and widely accepted conditional on the heat–helium data (Naturwissenschaften 97, 345 (2010)); the *constructive* theory lives almost entirely in the field's own Journal of Condensed Matter Nuclear Science and on the arXiv — including a recent book-length restatement as a "generalized nuclear Dicke model" with proposed validation experiments (arXiv:2501.08338, 2025, preprint) — and the phonon–nuclear coupling matrix elements it needs have never been independently verified; its supporting experiments (Karabut's collimated X-rays, claimed non-exponential Co-57 decay) are single-group and unreplicated. *Falls short by: no accepted derivation of a coupling within $\sim$ many orders of the required strength; status a research program, not a mechanism.*

Two other coherence programs deserve their line. Giuliano Preparata's "QED coherence in matter" claimed that ordinary condensed matter spontaneously forms macroscopic phase-locked electromagnetic domains, which could concentrate energy and catalyze nuclear processes; the framework's foundational claims were never accepted by condensed-matter theory, its fusion applications produced no independently checked rates, and it survives today mainly in the Italian LENR school's internal literature. Akito Takahashi's 4D/TSC condensate posits four deuterons plus four electrons collapsing, in about a femtosecond, to a ~ 10 – 20 fm cluster that fuses with near-unit probability into two ^4He . The published critique-in-absentia is that the required "bosonized" four-electron state is not an eigenstate of any Coulomb Hamiltonian: localizing electrons at femtometer-to-picometer scales costs keV-to-MeV of kinetic energy — the same variational physics behind Leggett–Baym — and no mainstream venue has ever scrutinized the calculation because it has not been submitted to one (its home venues are JCMNS, ICCF proceedings, and an ACS symposium volume). Its predicted pair of ~ 24 MeV alphas also collides head-on with the §1 constraint. *Falls short by: unfalsifiable as published — the central collapsed state has no independent quantum-mechanical derivation to check.*

§5 Neutrons without the barrier: Widom–Larsen

The boldest strategy concedes the Coulomb barrier entirely: if you cannot get charged nuclei together, manufacture *neutrons*, which face no barrier at all. The Widom–Larsen theory (Eur. Phys. J. C 46, 107 (2006)) — unusual among insider theories for appearing in a real particle-physics journal — proposes a chain: enormous local electric fields from surface plasmons on a loaded hydride surface renormalize the effective electron mass upward; heavy electrons are captured by protons via inverse beta decay; the resulting ultra-low-momentum neutrons are devoured by nearby nuclei, driving transmutation chains and beta-decay heat; and patches of the same heavy electrons absorb the capture gamma rays. Every step is standard physics *in kind*; the audit is entirely about magnitudes. The gate is the weak-interaction threshold:

$$\beta \equiv \frac{\tilde{m}_e}{m_e} \geq \frac{m_n - m_p}{m_e} = \frac{1.2933 \text{ MeV}}{0.5110 \text{ MeV}} = 2.531,$$

WHERE

- | | |
|----------------------|--|
| $\beta, \tilde{m}_e$ | The mass-renormalization ratio and the field-dressed electron mass. The electron's total energy must supply the neutron–proton mass difference. |
| $m_n - m_p$ | Neutron–proton mass difference, 1.2933 MeV (constant). Counting the electron's own rest mass, the reaction $e^- + p \rightarrow n + \nu_e$ is endothermic by $m_n - m_p - m_e = 0.78 \text{ MeV}$ per neutron — an energy bill paid up front, before any heat is returned. |
-

Widom and Larsen assert that hydride surfaces reach β above threshold — a 153% mass increase — via sustained local fields of order 10^{11} V/m. The published critiques, all peer-reviewed and all quantitative, attack each link. Hagelstein and Chaudhary (J. Phys. B 41, 125001 (2008)) redid the quantum electrodynamics and found the genuine mass shift orders of magnitude below threshold, arguing the original estimate mistakes a ponderomotive average for a true threshold-crossing mass. Ciuchi, Maiani, Polosa and coauthors (Eur. Phys. J. C 72, 2193 (2012), with a follow-up at Eur. Phys. J. C 74, 2843 (2014)) found the neutron-production rate "largely overestimated," leaving "little room for such a remarkable effect." Tennfors (Eur. Phys. J. Plus 128, 15 (2013)) computed the fields actually available from proton oscillations in a lattice — known from neutron scattering — and got a mass enhancement below 1% against the 153% required. Beyond the threshold problem: the chain is endothermic by 0.78 MeV per neutron before returning anything; the predicted neutron-capture chains should light up surrounding material with activation products and delayed gammas that are not observed; and the gamma-absorbing "heavy-electron patches" — the step that would make the theory safe to stand next to — exist only in an unpublished preprint and a patent. Widom and collaborators posted rebuttals (arXiv:0802.0466, arXiv:1210.5212, neither journal-published); no independent group has published quantitative support.

NOTE

Venue asymmetry is itself evidence here, and it cuts against the usual direction: the *critiques* of Widom–Larsen sit in journals at least as strong as the original (J. Phys. B, EPJ C, EPJ Plus vs. EPJ C), and even most LENR insiders — Storms, Hagelstein — reject the theory. Its persistence in popular accounts owes more to advocacy than to any surviving calculation. *Falls short by: a factor $\gtrsim 100$ in electron mass shift at step one (equivalently $\gtrsim 10^4$ in field energy density), plus unobserved activation and gamma signatures at step three.*

§6 The exotica, each with its one-line killer

A field starved of an accepted mechanism grows exotic proposals at its edges. The reader should be able to recognize the recurring ones in the wild, along with the standard refutation each carries.

Deep Dirac levels. Maly and Vávra (Fusion Technology, 1993 and 1995) proposed that the Dirac equation admits extra, ultra-tight electron orbits — binding ~ 500 keV, picometers from the nucleus — which would let an electron chaperone nuclei through the barrier. The killer: those "anomalous" solutions are rejected in standard quantum mechanics because their wavefunctions are singular at the origin and the Hamiltonian ceases to be self-adjoint; they are mathematical debris, not states. The cousin claim of below-ground-state hydrogen — Mills's hydrino — was refuted in the mainstream literature on the same grounds (Rathke, New J. Phys. 7, 127 (2005); Dombey, Phys. Lett. A, 2006). *Exotic-particle catalysis.* Proposals of hypothetical stable heavy particles ("erzions" in the Russian literature, among others) that would catalyze fusion like super-heavy muons; the killer is that accelerator searches, cosmic-ray data, and the muon benchmark itself constrain such particles' existence and abundance to levels that cannot supply laboratory-scale rates. *Casimir-cavity and zero-point-energy schemes.* Proposals to harvest vacuum energy in nanocavities (Dmitriyeva & Model, Physics Procedia 38, 8 (2012)) sit outside standard QED's energy accounting; their own gas-loading heat tests traced signals to ordinary hydrogen chemistry, and the cavity test was inconclusive — and in any case this is not a nuclear mechanism at all. At the respectable end of this section: Kálmán and Keszthelyi's peer-reviewed proposals (Phys. Rev. C 69, 031606 (2004); 99, 054620 (2019)) that third-body electrons or lattice recoil open normally forbidden channels — genuine physics, published in a real nuclear journal, whose computed rates remain far too small to explain any heat claim, and whose predicted radiation has not been observed. Even the respectable exotica fall short by their own arithmetic.

A CAUTIONARY TALE: THE HOLMLID CORPUS

From roughly 2015 to 2023, Leif Holmlid published dozens of papers (Physica Scripta, International Journal of Hydrogen Energy, and elsewhere) claiming "ultra-dense hydrogen" — hydrogen spontaneously condensed to picometer spacings — that emits muons under laser pulses, implying tabletop fusion and even muon production far cheaper than accelerators. The corpus is large, internally cross-citing, and was taken up by at least one startup. Independent scrutiny arrived in two waves. In 2019, PLOS ONE retracted the corpus's mesons paper over the author's objection, on the grounds that the claimed products conflict with baryon-number conservation and that the signals are consistent with electronics artifacts near an intense pulsed laser (PLOS ONE 14, e0212979 (2019)). In 2023–2024, peer-reviewed analyses in *Energy, Sustainability and Society* (Hansen & Engelen, DOI 10.1186/s13705-023-00403-6; a further analysis at DOI 10.1186/s13705-024-00470-3) concluded that the cheap-muon and fusion claims are mistaken or ungrounded — contradicting conservation laws and established annihilation data — and that no independent confirmation exists. Holmlid disputed the retraction and published replies to the critiques; no independent group has reproduced the observations. One retraction; dozens of unretracted papers. The lesson generalizes: publication volume in venues outside a claim's core discipline is not replication — the same lesson the Dias room-temperature-superconductivity retractions taught adjacent to Chapter X's materials story.

§7 The ledger: results ahead of theory

This book was commissioned to keep score honestly, and here is the score. The table below collects every experimental result flagged "ahead of theory" in Chapters III–VIII: claims that, *if real*, no accepted mechanism explains — plus one that is real by any standard and unexplained anyway. Each row carries its claim status in this book's

register, its theory status after this chapter's audit, and the one measurement a follow-up must nail. Each is hereby flagged for future research follow-up, per the book's commission.

Result ahead of theory	Claim status	Theory status	What a follow-up must nail
Heat-⁴He at ~24 MeV per helium, no gammas (Ch. IV)	Contested — peer-reviewed core (Miles 1993); tightest numbers conference-grade; near-ambient He levels	No accepted exit channel: helium must be born $\lesssim 20$ keV (Hagelstein 2010); downconversion unproven at 10^9 quanta	Closed-system blinded heat + helium runs w modern mass spectrometry, replicated across independent labs
Tritium with t/n ~10^6-10^9 (Ch. IV)	Contested — journal positives (Will 1993; BARC) against demonstrated contamination episodes	No accepted inversion of few-body branching by ~8 orders	Closed-cell trit generation with continuous assay and simultaneous neutron spectroscopy in same cell
Beam screening 2-4× above best theory (Ch. VI, §3)	<i>Uncontested</i> — accelerator physics, four-plus groups; magnitudes disagree 2-3× between groups	No model fits magnitude, material pattern, and temperature trend; the Debye fit is physically inadmissible	A round-robin: identical target across accelerators with operando deuteron-density profiling to kill dominant systematic
Nanocomposite gas-loading heat, 10-100+ eV per H (Ch. VIII)	Contested — multi-lab Japanese program reports; one group claims >10 keV/H (Iwamura et al., Jpn. J. Appl. Phys. 63, 037001 (2024)); calorimetry-dependent	Beyond any chemical bond energy if real; no commensurate nuclear ash reported	Same material independent laboratory, calorimetry and particle diagnosis running simultaneously
Permeation transmutation in +4p/+8n jumps (Cs→Pr, Sr→Mo; Ch. IV)	Contested — JJAP 2002 + partial Toyota replication (2013); NRL contamination critique never journal-published	No accepted 8-nucleon coherent transfer; even field-internal theories predict different products	Isotopically anomalous Mo confirmed by two independent neutron spec methods exchanged samples
NASA lattice-confinement channels (screened)	Peer-reviewed (Phys. Rev. C 101, 044609 & 044610 (2020));	Fitted screening values and screened stripping rates lack	Independent replication of tritium neutron spectra

Result ahead of theory	Claim status	Theory status	What a follow-up must nail
d-d and Oppenheimer-Phillips; Ch. VII)	beam-driven, not excess heat; unreplicated	quantitative theory in metals	first-principles screened Oppenheimer-Phillips calculation

Note what the ledger is and is not. It is not a list of confirmed miracles — five of six rows are contested, and every contested row may yet dissolve into artifact, as so many of its ancestors did in Chapters II–IV. It is a to-do list: the minimal set of loose threads that a rigorous program (Chapter VIII's ARPA-E cohort and its successors) should either tie off or pull hard. Row three is the odd one out and the most instructive: it shows that "ahead of theory" is not a euphemism for "fringe" — sometimes the mainstream's own instruments produce a number its own models cannot yet reproduce.

§8 An honest synthesis

The audit's bottom line can be stated without hedging: **as of 2026, no published mechanism quantitatively explains claimed excess-heat magnitudes within accepted physics.** Each serious candidate fails at a specific, identifiable step, by a computable amount: static screening by tens of orders in rate (§3); coherent downconversion by an unverified coupling spanning many orders (§4); Widom–Larsen by two orders in electron mass at its first step and by its unobserved radiation at its third (§5); the exotica by consistency with quantum mechanics itself (§6). The most useful mainstream-published summary agrees: the topical review of Metzler, Hunt, Hagelstein and Galvanetto (New J. Phys. 26, 101202 (2024)) catalogues every known solid-state fusion enhancer and concludes that observable fusion needs more than forty orders of magnitude, that known, uncontroversial mechanisms deliver

perhaps twenty-five to thirty when optimistically combined, and that the remainder is hypothesis — coherent dynamics, resonances, cascade effects — awaiting demonstration. The muon (§2) stands as both the proof that catalysis is possible and the measure of what it costs: two hundred electron masses, or five GeV a particle.

Where does that leave a rational reader? Three positions are defensible, and this book has met serious people holding each. The *artifact hypothesis*: every excess-heat claim is measurement error (Chapter III's calorimetry pathologies), every ash claim contamination or misidentification (Chapter IV); the theory gap is empty because there is nothing to explain. It predicts that Chapter VIII's instrumented programs will keep returning nulls as error bars tighten — as the Google program's did (Nature 570, 45 (2019)). The *rare-real-effect hypothesis*: somewhere in the anomalies hides a genuine but sporadic condensed-matter nuclear effect — perhaps tied to the unexplained screening physics of the ledger's row three — too small and material-fussy for 1989-era labs to pin down. It predicts occasional small, replicable signals that correlate with material state, of the kind the NASA and Berkeley beam programs already report at keV energies, extending downward. The *new-physics hypothesis*: the heat–helium correlation is real at ~24 MeV per atom, and something like coherent downconversion operates. It predicts that at least one ledger row will convert into an on-demand demonstration — and it must also explain, someday, why the effect hid from every accelerator, reactor, and refinery of the twentieth century. What is not defensible, after this chapter, is a fourth position common in popular coverage: that some published theory already explains the claims and the mainstream merely refuses to read it. No such theory exists. The Gamow factor and the error bar remain undefeated; the only question — Chapter X's question — is where a breakthrough against either could plausibly come from.

FURTHER READING

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Leggett, A. J. & Baym, G. *Exact upper bound on barrier penetration probabilities in many-body systems: Application to "cold fusion"*, Physical Review Letters 63, 191–194 (1989) — the equilibrium no-go theorem that forces all surviving theories into non-equilibrium; DOI 10.1103/PhysRevLett.63.191.

Ponomarev, L. I. *Muon catalysed fusion*, Contemporary Physics 31, 219–245 (1990) — the canonical review of the one proven cold-fusion catalyst, cycle kinetics and sticking economics; DOI 10.1080/00107519008222019.

Hagelstein, P. L. *Constraints on energetic particles in the Fleischmann-Pons experiment*, Naturwissenschaften 97, 345–352 (2010) — the peer-reviewed exit-channel constraint (helium birth energy $\lesssim 20$ keV) that frames §1 and §4; DOI 10.1007/s00114-009-0644-4.

Widom, A. & Larsen, L. *Ultra low momentum neutron catalyzed nuclear reactions on metallic hydride surfaces*, European Physical Journal C 46, 107–111 (2006) — the weak-interaction theory of §5, best read alongside its critiques; DOI 10.1140/epjc/s2006-02479-8.

Ciuchi, S., Maiani, L., Polosa, A. D., Riquer, V., Ruocco, G. & Vignati, M. *Low energy neutron production by inverse β decay in metallic hydride surfaces*, European Physical Journal C 72, 2193 (2012) — the quantitative refutation of Widom-Larsen's neutron rates; DOI 10.1140/epjc/s10052-012-2193-9.

Metzler, F., Hunt, C., Hagelstein, P. L. & Galvanetto, N. *Known mechanisms that increase nuclear fusion rates in the solid state*, New Journal of Physics 26, 101202 (2024) — the mainstream-published map of the whole gap: what is known, what it is worth in orders of magnitude, and what remains hypothesis; DOI 10.1088/1367-2630/ad091c.

Chapter production note: researched, drafted, and adversarially verified by Claude subagents (draft agent: `claude-fable-5`; verification agent: `claude-fable-5`), orchestrated by `claude-fable-5`. See the [methods appendix](#) for the full research process.

CHAPTER X

Where the Breakthroughs May Come From

Nine chapters of evidence and critique end here, in the one chapter licensed to bet — on where, if anywhere, this field's genuine breakthrough is hiding, and what it would cost to go find out.

BEFORE WE START

Everything before this page reported what has been measured, claimed, and critiqued. This chapter does something different: it speculates, on purpose and on the record. Having weighed thirty-seven years of experiments against the two numbers that govern them all — the Gamow factor a claim must beat and the error bar it must survive — we now ask where a real breakthrough would most plausibly come from, what it would look like when it arrived, and which experiments would get us there fastest.

To keep the speculation honest we borrow the bookkeeping of forecasting: state a prior, say how much each new result should move it, and — crucially — discount reports that are correlated with each other. One small formula from survey statistics explains more about this field's history than any theory in Chapter IX: a hundred positive reports that share methods, materials, and expectations can carry the evidential weight of roughly one. The cheapest breakthrough available to this field is not new physics. It is evidence quality.

TERMS YOU'LL MEET

prior odds

Your betting odds on a hypothesis before seeing new evidence — for "watts of nuclear heat in a jar," very long odds indeed, for the Chapter I reasons.

posterior odds

Your odds after the evidence: prior odds multiplied by how strongly the evidence favors the hypothesis.

likelihood ratio (LR)

How many times more probable a given result is if the hypothesis is true than if it is false. $LR = 1$ means the result teaches nothing.

Bayes' theorem

The rule for updating beliefs with evidence; in odds form it is a single multiplication.

calibration

The forecasting virtue of meaning your numbers: of things you call "20% likely," about one in five should happen.

intraclass correlation (ρ)

A number between 0 and 1 measuring how much observations within a group resemble each other — here, how non-independent two positive reports are.

design effect

Kish's factor $1 + \rho(N - 1)$ by which clustering inflates the variance of a survey — equivalently, deflates the information in N correlated samples.

effective sample size (N_{eff})

The number of truly independent observations your N correlated ones are worth: $N/(1 + \rho(N - 1))$.

preregistration

Publicly fixing the protocol, analysis, and success criteria before running the experiment, so a positive result cannot be carved out after the fact.

blinding

Hiding the identity of samples or conditions from the people measuring them, so expectation cannot steer the readings.

independent replication

The same claimed effect reproduced by different people, with different equipment, different materials, and no shared stake in the outcome.

systematic error

An error that does not average away with repetition — calorimeter miscalibration, contamination — the historic killer of LENR claims.

operando measurement

Characterizing a material *while it operates* — inside the working cell, during the claimed anomaly — rather than before or after.

X-ray diffraction (XRD)

Reading a crystal's structure from how it scatters X-rays; operando XRD tracks phase changes in a loading cathode in real time.

X-ray photoelectron spectroscopy (XPS)

Measuring surface composition and chemical state from the energies of electrons kicked out by X-rays — the top few nanometers, where cathodes live and die.

atom-probe tomography

Field-evaporating a needle-shaped sample atom by atom to reconstruct its 3-D composition at near-atomic resolution.

self-driving lab

A robotic platform that designs, runs, and analyzes its own experiments in a closed loop, choosing each next experiment by algorithm.

Bayesian optimization

The algorithm family such labs use: model the response surface, then

superhydride

sample where information gain is highest.

A hydrogen-rich compound (H_3S , LaH_{10}) stabilized at megabar pressures; the family that yielded near-room-temperature superconductivity.

high-entropy alloy

An alloy of five or more principal elements in comparable amounts; some absorb hydrogen to ratios far beyond palladium's.

underground laboratory

A lab under a kilometer of rock (Gran Sasso's LUNA is the exemplar), where cosmic-ray background is low enough to measure vanishing nuclear cross sections.

CVD diamond detector

A single-crystal diamond neutron spectrometer: fast, radiation-hard, nearly blind to gammas — it can certify a 2.45 MeV d+d neutron in a noisy room.

low-background counting

The rare-event-search toolkit — radiopure materials, active vetoes, coincidence logic — that can detect of order one nuclear event per second per kilogram, unambiguously.

Oppenheimer–Phillips reaction

A deuteron grazing a heavy nucleus lets its neutron be captured while the proton is repelled — a barrier-softened channel invoked in NASA's lattice experiments.

neutron economy

The bookkeeping of a neutron-mediated scheme: each neutron spent must be replaced by breeding or an external source, or the process dies out.

fractionation

The hypothetical splitting of one MeV-scale nuclear quantum into billions of

materials passport

meV-scale lattice quanta — the load-bearing miracle of Chapter IX's coherence theories.

A complete, shareable dossier on a specific sample — provenance, composition, defect state, loading history — so "the same cathode" means something between labs.

reference experiment

A standardized, fully specified protocol run identically in multiple labs, designed so that a positive in one lab is a recipe any other lab can execute.

SPECULATION

Unlike Chapters I–IX, this chapter states priors and bets, not settled results. Everything here is the compiling model's informed speculation, grounded in the evidence of the preceding chapters and clearly separable from it. Where a scenario leans on a published result, the citation is given and its register (established / contested / preprint) is kept; the *rankings and probabilities* are commissioned guesswork, offered to be falsified.

§1 The rules of this chapter: Bayesian honesty

A speculation chapter needs rules, or it is merely advocacy with hedging. Ours are the rules of the bettor. First, state the prior odds: given Chapter I's yardstick — a tunneling suppression of order 10^{-2700} at thermal energies, fifty-plus orders of magnitude between molecular deuterium and a detectable signal — the prior against "watts of nuclear heat in an electrochemical cell" is properly microscopic. Second, update multiplicatively: each new experimental report multiplies the odds by its likelihood ratio,

$$O_{\text{post}} = O_{\text{prior}} \times \prod_{i=1}^N \text{LR}_i, \quad \text{LR}_i = \frac{P(D_i | H)}{P(D_i | \neg H)}.$$

WHERE

H	The hypothesis under audit (e.g. "some metal-hydride system produces nuclear-scale excess heat").
$O_{\text{prior}}, O_{\text{post}}$	Odds (dimensionless ratios $P/(1 - P)$) before and after the evidence.
D_i	The i -th experimental report (a positive claim, a null, a retraction — each carries its own ratio).
LR_i	Scalar ≥ 0 : how many times likelier report D_i is in a world where H is true than in one where it is false. A sloppy positive that artifacts produce easily has LR barely above 1; a blinded, preregistered, multi-diagnostic positive could have LR of 10 or more.

Third — and this is the rule the field's history turns on — *discount correlation*. The product above assumes the N reports are independent. LENR's positive reports overwhelmingly are not: shared cathode suppliers, shared protocols descended from the same 1989 template, shared analysis conventions, a shared conference circuit, and a publication culture in which nulls quietly vanish. Survey statisticians have priced this exactly. Kish's design effect says that N observations with pairwise intraclass correlation ρ carry the information of only

$$N_{\text{eff}} = \frac{N}{1 + \rho(N - 1)}, \quad \log_{10} O_{\text{post}} = \log_{10} O_{\text{prior}} + N_{\text{eff}} \log_{10} \text{LR}.$$

WHERE

- N Integer: the number of positive reports (nominal replications).
- ρ Scalar in $[0, 1]$: intraclass correlation between reports. $\rho = 0$ is genuine independence; $\rho = 1$ means every report is effectively the same observation repeated.
- N_{eff} Scalar: the effective number of independent reports — Kish's effective sample size. As $\rho \rightarrow 1$, $N_{\text{eff}} \rightarrow 1$ no matter how large N grows.
- LR A common per-report likelihood ratio, used in the second (log-form) expression as the simplifying case $\text{LR}_i = \text{LR}$.
-

Run the field's history through it. Take a charitable prior of 10^{-6} , a generous $\text{LR} = 10$ per positive report, and $N = 30$ published positives. Independent, they would drive the posterior odds to $10^{-6+30} = 10^{24}$ — certainty. But at $\rho = 0.9$, which is arguably still kind to a literature this entangled, $N_{\text{eff}} = 30/(1 + 0.9 \times 29) \approx 1.1$, and the posterior odds are $10^{-6+1.1} \approx 10^{-4.9}$ — moved, but still about one in eighty thousand. Thirty reports, worth one. That single arithmetic fact reconciles the two things every honest observer of this field feels at once: *so many* people have seen *something*, and yet the rational credence has barely budged since 1990.

It also identifies the cheapest breakthrough on offer. You can spend a decade hunting mechanisms that raise the Gamow factor — or you can raise LR and lower ρ , this year, with statistics and sociology: preregistration, blinding, hostile replication, published nulls, sample exchange. A single blinded multi-lab positive with $\text{LR} \sim 100$ and $\rho \approx 0$ outweighs the entire historical corpus. Every scenario below is, at bottom, a scheme for buying either a bigger exponent or a cleaner one — the Gamow factor and the error bar, one last time.

INTERACTIVE · THE REPLICATION CALCULATOR

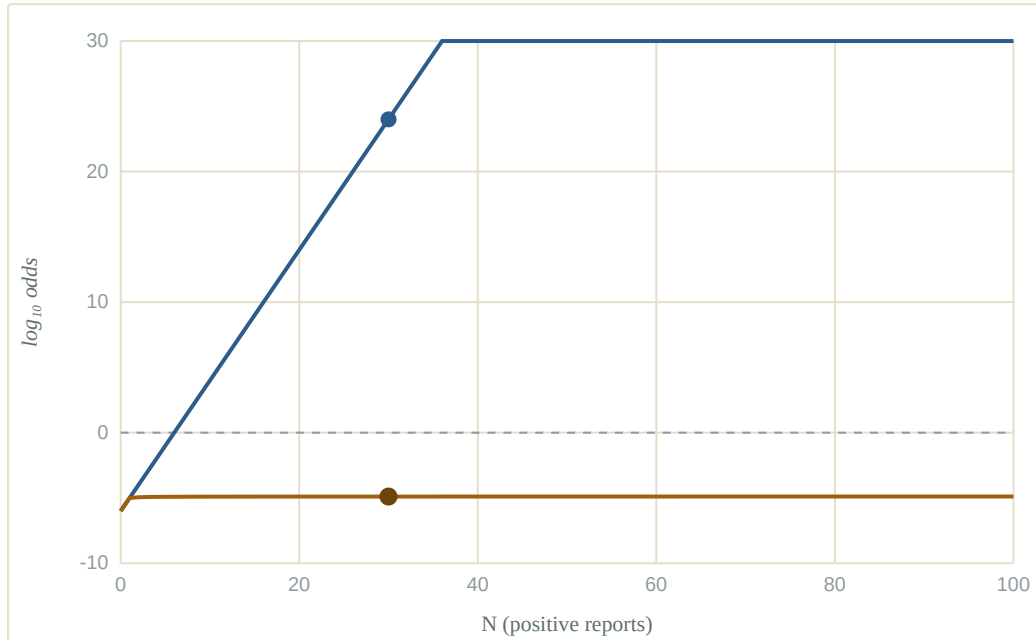
set a prior · stack reports ·
correlate them

log₁₀ prior odds

LR per report

positive reports N

correlation ρ



posterior probability (correlated): 1.3×10^{-5}

prior = $10^{-6.0}$ · LR = 10 · N = 30 · ρ = 0.90

$N_{\text{eff}} = 1.11$ · posterior odds = $10^{-4.9}$ ($p \approx 1.3 \times 10^{-5}$) · naive: $10^{24.0}$ ($p \approx 1$)

— correlated (Kish N_{eff}) — naive (ρ = 0) --- even odds (ρ = ½)

Posterior log-odds $\log_{10} O_{\text{post}} = \log_{10} O_{\text{prior}} + N_{\text{eff}} \log_{10} \text{LR}$ with Kish's $N_{\text{eff}} = N / (1 + \rho(N - 1))$, versus the naive (ρ = 0) posterior. Slide the correlation toward 1 and watch a mountain of reports collapse to the weight of one: at prior 10^{-6} , LR = 10, $N = 30$, ρ = 0.9 the honest posterior is $\sim 10^{-4.9}$ while the naive one is 10^{24} . The field's whole history lives in the gap between those two curves.

§2 Speculation A — materials determinism

The bet: if any excess-heat effect is real, its forty-year irreproducibility pattern is telling us the active condition is a *rare materials state* — a specific configuration of defects, interfaces, and near-surface stoichiometry that 1989-era experimenters could neither see nor control — and that state is findable with tools that now exist. This was this book's top-ranked scenario for most of its drafting; §3 explains why it now shares the podium.

The circumstantial case is old and consistent: McKubre's loading threshold ($x_{\text{D}/\text{Pd}} \gtrsim 0.875$, Chapter V) that most cathodes never reached; the "good lots" and dead lots of palladium; effects that die when a cathode is repolished. The Google program's Nature perspective made the modern version of the argument — its team could not even *reach* the claimed trigger conditions, and concluded the parameter space was genuinely under-explored (Berlinguette et al., Nature 570, 45–51, 2019). What has changed is instrumentation. Operando XRD and neutron diffraction can track hydride phase and loading inside a running cell; XPS maps the near-surface chemistry where electrochemistry actually happens; atom-probe tomography reconstructs a suspect cathode's surface volume atom by atom. The variable that 1989 could only guess at — *what, exactly, is this electrode right now?* — is now measurable during the anomaly, not after it.

The search-speed lever is just as new. Berlinguette's own group built Ada, a self-driving lab that synthesizes and characterizes a thin-film sample about every twenty minutes (MacLeod et al., Sci. Adv. 6, eaaz8867, 2020). Point that machinery at cathode conditions — composition, metallurgy, loading protocol, trigger — and 10^3 – 10^4 systematically logged electrochemical experiments per year is an engineering problem, not a fantasy. The same PI leading both the field's best null and its best robot is the least accidental fact in this chapter.

CALIBRATING THE ROBO-LAB HYPE

Autonomous discovery has its own overclaiming problem. DeepMind's GNoME announced 2.2 million candidate crystals and the A-Lab claimed 41 new materials

synthesized in 17 days (Nature 624, 80–85 and 86–91, 2023) — and a peer-reviewed audit found few results that were simultaneously novel, credible, and useful (Cheetham & Seshadri, Chem. Mater. 36, 3490–3495, 2024). The lesson transfers: a self-driving LENR search would generate torrents of candidate "hits," and only the §1 discipline — preregistered criteria, blind re-runs — keeps torrents from becoming another correlated literature.

What "found it" would look like, concretely: a materials passport — say, "Pd, this vacancy concentration, this near-surface oxide, $x_{\text{D/Pd}} > 0.9$ sustained for this long" — published with a *dose-response curve*: heat that switches on and off with the specified state, in any lab that reproduces the state, with always-on nuclear diagnostics deciding what the heat is. Materials determinism is the one scenario that would explain the entire irreproducibility record *and* retire it in the same paper. If instead the self-driving campaigns sweep the space at scale and find nothing, that is §5's outcome arriving by robot — and it settles the matter with the same data.

§3 Speculation B — screening physics, upgraded

The bet: the anomalously large electron screening of Chapter VI is the field's most productive thread, because it is the one place where contested LENR-adjacent claims and mainstream accelerator physics share an apparatus, an equation, and now a funding line. Push metal-target d+d measurements below 2 keV with underground, LUNA-class rigor and modern target metrology, and *either* the anomalous U_e values normalize — theory catches up, artifact channels close, item (1) of Chapter IX's ledger is retired — *or* they grow further, and something genuinely new is in hand. Both outcomes are breakthroughs; only one requires new physics.

The baseline is clean: in D_2 gas, underground measurement gives $U_e = 25 \pm 5$ eV, right at the atomic limit (Greife et al., Z. Phys. A 351, 107, 1995). In metals, groups in Berlin/Szczecin, Bochum, Tohoku, and Berkeley report U_e from roughly 300 to 1000 eV — mutually inconsistent at the factor-2–3 level, unexplained by any accepted model, and sensitive to exactly the systematics (deuteron depth profiles, surface contamination, stopping powers) that modern metrology can finally pin down. This is a solved-problem *shape*: nuclear astrophysics has spent three decades turning ugly low-energy yield curves into precision science underground.

Events of 2025–26 made this the best-instrumented scenario in the book, which is why it now outranks Speculation A on tractability. Berlinguette's group showed in Nature that in-situ electrochemical loading raises beam-driven d+d fusion rates in Pd by $15 \pm 2\%$ — a modest number with an immodest meaning: an eV-scale knob, turned by a chemist, measurably modulating a nuclear rate on a calibrated platform (Chen, Maiwald, Schauer et al., Nature 644, 640–645, 2025; by the team's estimate, about a volt of applied potential does the loading work of roughly 800 atm of D_2 gas pressure). The Szczecin group published in PRX the claim of a new low-energy d+d channel, with yields tracked down to $E_{cm} = 675$ eV (Phys. Rev. X 15, 041004, 2025) — peer-reviewed, single-group, unreplicated. And an LBNL/UC Davis preprint reports sub-keV yields in loaded Pd and Ti hydrides that stop falling and *plateau* below about 2.5 keV, of order 10^{18} above bare-nucleus extrapolation at the lowest points (Karahadian et al., arXiv:2512.06212 — a preprint, and an extrapolation-sensitive number; treat accordingly). DARPA's MARRS program (BAA HR001126S0007, January 2026) now aims to raise demonstrated solid-state reaction rates from the ~ 0.1 reactions/s/g state of the art toward 10^6 — still a dozen orders of magnitude from power relevance, but the first program ever to give this physics quantitative milestones and deadlines.

AHEAD OF THEORY

The sub-keV anomalies are results ahead of theory in the strict ledger sense. A static screening energy only shifts the Gamow exponential — $\sigma_{scr}(E) = \sigma_{bare}(E + U_e)$ still falls as the beam energy falls — so *no* choice of U_e produces a plateau. Either an unmodeled systematic (beam tails, surface deuterium) or genuinely new low-energy

physics (thermalized-deuteron fusion in ion tracks, a threshold resonance — the Szczecin interpretations) is at work. The decisive follow-up is boring on purpose: the same measurement, underground, by a group with no stake, with independently certified targets.

The adjacent materials physics says metal–hydrogen systems still hide surprises of exactly the needed flavor. Conventional superconductivity at 203 K in H_3S (Drozdov et al., *Nature* 525, 73–76, 2015) and ~ 250 K in LaH_{10} (*Nature* 569, 528–531, 2019) — replicated, megabar-pressure, phonon-mediated — were textbook-changing discoveries in hydrogen-dominant metals, the very phase space LENR speculation invokes. High-entropy alloys absorb hydrogen to $\text{H}/\text{M} = 2.5$, far beyond $\text{PdH}_{0.7}$ (Sahlberg et al., *Sci. Rep.* 6, 36770, 2016), offering unexplored screening and confinement testbeds at ambient pressure. And the same decade supplied the cautionary mirror: the Dias room-temperature-superconductivity retractions (2022–23) showed extraordinary metal-hydride claims collapsing inside *Nature* itself when raw data could not be produced. The superhydride story is what a real metal–hydrogen breakthrough looks like — theory-first, replicated within a year, survived by its data. That is the template Speculation B is betting the screening anomaly can follow.

§4 Speculation C — engineered nonequilibrium

The bet: the theorems of Chapter IX leave one door open, and it is labeled *nonequilibrium*. The Leggett–Baym bound caps what any equilibrium condensed-matter system can do to tunneling rates; NASA's lattice experiments walked through the open door deliberately, using 2.9 MeV photons to make neutrons that elastically kick deuterons to keV energies *inside* a screened, deuterium-dense lattice, where they fuse at rates the screening measurably boosts (Pines et al. and Steinetz et al., *Phys.*

Rev. C 101, 044609 & 044610, 2020 — peer-reviewed, not independently replicated). This is not cold fusion; it is hybrid hot/cold fusion, and as speculation it scales as an *engineering* question: what is the neutron economy? Each seed neutron is expensive; the scheme pays only if screened fusion chains and Oppenheimer–Phillips stripping on high-mass-number hosts return more energetic particles per neutron spent than the accelerator or gamma source consumes. Nobody has published that ledger; drafting it is a masters-thesis-sized breakthrough waiting to be claimed.

The same nonequilibrium license covers the phonon and plasmon proposals of Chapter IX — but here the speculation is about experiment design, not mechanism. The fractionation theories (one 24 MeV quantum dissolving into 10^9 phonons) have spent decades unfalsified because nothing was built to falsify them. That is fixable: their load-bearing claims imply *observable intermediate physics* — phonon-modulated nuclear level shifts, excitation transfer between separated sites, angular-anisotropic X-ray emission under THz drive — each testable on equipment that exists, at costs far below a calorimetry program. The point of such experiments is not to produce power. It is to make the theories die or grow, which after Chapter IX is the more urgent product.

AHEAD OF THEORY

Chapter IX's ledger rows each dictate their decisive follow-up, and this scenario is where those experiments live. Anomalous U_e magnitude → underground remeasurement with certified targets (§3). Temperature-dependent screening → repeat with degenerate-electron-honest theory and operando target thermometry. The claimed ^4He threshold resonance with e^+e^- emission → a pair-spectrometer experiment by a second group; the resonance appears in no evaluated $A=4$ level scheme. NASA's screened O–P channels → independent replication with CVD-diamond spectroscopy and full activation analysis. Heat–helium at 23.85 MeV/He with helium born below ~ 20 keV → the closed-loop helium accounting experiment of §7. Each row is a funded-size experiment, not a program.

§5 Speculation D — null closure as a breakthrough

The bet: the outcome most likely to be underrated is the definitive negative. Consider a blinded, multi-lab, preregistered replication network — ARPA-E-style reference experiments, run to rare-event-search standards — that takes the strongest surviving claims from Chapters III–VIII and either produces one lab-transferable positive or retires them. In §1's language: a network engineered so that $\rho \approx 0$ between labs and $\mathbf{LR}$ is large in *both* directions — a positive would be believable, and a null would be final.

The pieces exist separately and have never been assembled. The 2004 DOE review ended not with a verdict but with a procedural shrug — reviewers split, no program, "well-designed proposals welcome" (U.S. DOE, *Report of the Review of Low Energy Nuclear Reactions*, 2004) — and two decades later the ARPA-E LENR exploratory topic (roughly ten million dollars across eight teams, 2023) supplied funding and modern diagnostics but, as of mid-2026, no published cohort-wide reference protocol, no confirmed excess heat, and no published nulls either. The detection standard is no longer the bottleneck: gamma-blind single-crystal diamond spectrometers can certify a 2.45 MeV neutron line in real time in a mixed field, and the low-background techniques of the rare-event community (Heusser, *Annu. Rev. Nucl. Part. Sci.* 45, 543–590, 1995) can see of order one nuclear event per second per kilogram. Any modern claim can be held to that standard; a network would hold every claim to it simultaneously, with tritium blind-splits and helium closed-loop accounting as the calorimetry-independent channels.

Why call a negative a breakthrough? Because a *definitive* negative — not one more unpublished shrug, but preregistered nulls across the strongest protocols, with sensitivity numbers — is a thriving-world outcome. It redirects careers and money toward the parts of this story that are certainly real (the screening anomaly, the materials tools, the detectors); it converts an eternal maybe into a bounded claim ("no excess heat above X watts per cm³ under conditions Y, at confidence Z"); and the

instruments and standards remain, permanently raising $\mathbf{LR}$ for whatever anomaly physics serves up next. Fields that cannot close questions cannot open new ones. The null-closure network is the only experiment in this chapter guaranteed to produce a breakthrough, because it wins on either branch.

§6 The author's ranked bets

Rules of §1 applied to the author itself: here are this book's calibrated guesses, stated as probabilities so they can be wrong in public. They are the compiling model's own numbers — speculation by commission, not community consensus — and each comes with its reasoning attached.

P(the metal screening anomaly is quantitatively resolved by 2036) ≈ 0.7 . Highest, because §3's machinery is aimed and funded: multiple accelerator groups, underground capability, MARRS-style milestones, and a question — is U_e in Pd 300 eV or 800 eV, and why? — that is answerable by measurement regardless of which answer is true. "Resolved" here includes the deflationary ending in which better target metrology collapses the anomaly into systematics; the bet is that the *question* closes, not that the anomaly survives.

P(the sub-keV plateau survives independent replication as new physics) ≈ 0.15 .

Two groups with different apparatus see it, which is why this is not 0.05; but both results are young, one is a preprint, the claimed effect sits 10^{18} above an extrapolation exquisitely sensitive to beam-energy calibration, and history says most cliff-edge anomalies at the bottom of a yield curve are the beam, not the universe.

P(a lab-transferable excess-heat protocol exists by 2036) ≈ 0.04 . The N_{eff} arithmetic is the reasoning: thirty-seven years of positives compress to roughly one independent report's worth of evidence against a prior that Chapter I set at 10^{-6} .

class. Materials determinism (§2) is the one live route — a rare-state explanation could still be hiding in unexplored parameter space, and the self-driving campaigns will actually search it this decade — so this number is meaningfully above zero. But it requires the heat to be real, rare, *and* findable, three conjuncts each well below 1.

P(a new-physics mechanism — beyond screening — confirmed by 2036) ≈ 0.01 .

Confirmation means Chapter IX's exit-channel problem solved by experiment: fractionation observed, or a new reaction channel replicated. Every candidate mechanism currently fails by orders of magnitude at an identifiable step, and the field's theoretical proposals have never once led its experiments. New physics remains possible — the plateau, if real, would be its first foothold — but 1% is what "possible" is worth after nine chapters of audit.

P(null closure achieved — the strongest claims retired or transferred — by 2036)

≈ 0.35 . Limited less by physics than by sociology: someone must fund and staff §5's network, and the incentive structure of both the field and its critics runs against finality. This is the number the reader can most directly move.

§7 The decade agenda

Speculations earn their keep by pricing their tests. Here is the concrete program, with order-of-magnitude costs and the information each purchase buys.

(1) One reference experiment per evidence strand. Heat–helium with closed-loop helium accounting; tritium with blind splits across three mass-spec labs; gas permeation with independently certified samples. Cost: ten-million-dollar class per strand over several years. Buys: the single highest-value dataset in the field — each strand either produces its first $\rho \approx 0$, high-LR positive or is retired with sensitivity numbers.

(2) Materials passports and sample exchange. Atom-probe, XPS, and diffraction dossiers on every cathode in every claim; physical sample swaps between claiming and skeptical labs. Cost: million-dollar class. Buys: the §2 hypothesis tested directly — if the effect follows the passport, materials determinism is confirmed; if it follows the lab, the effect is sociology.

(3) Always-on nuclear diagnostics around every calorimetric claim. Diamond spectrometers, gamma spectroscopy, and low-background counting as a standing condition of taking any wattage claim seriously. Cost: one hundred thousand to a million dollars per lab. Buys: Chapter I's ashes bookkeeping enforced in real time — heat with commensurate ash is a discovery, heat without it is a calorimetry problem, and the ambiguity that consumed three decades never recurs.

(4) Publish-negative infrastructure. A preregistration registry and a funded venue for nulls, so ρ falls and the denominator of the field's evidence becomes visible. Cost: hundred-thousand-dollar class — the cheapest item, and per §1 possibly the highest-leverage.

(5) Theory benchmarks. Compute U_e for d+d in Pd ab initio to $\pm 10\%$, with degenerate electrons treated honestly, as a community benchmark on the model of quantum-chemistry blind challenges; extend toward nucleus-plus-lattice Hamiltonians as quantum simulation matures (the deuteron has already been computed on cloud quantum processors — Dumitrescu et al., Phys. Rev. Lett. 120, 210501, 2018 — a seed, decades from the tree). Cost: million-dollar class in people and compute. Buys: the yardstick's missing tick mark — a screening measurement can only be anomalous relative to a theory number everyone trusts.

Total: a few tens of millions of dollars a decade — noise against fusion budgets — to convert the longest-running dispute in modern physics into closed questions and calibrated instruments. Whether the answer is a new energy source, a new nuclear channel, a materials state worth a Nobel, or a final, well-earned null, the purchase is the same: the Gamow factor honestly beaten or honestly bounded, and the error bar, at last, small enough to trust.

FURTHER READING

Kish, L. *Survey Sampling*, John Wiley & Sons, New York (1965) — the design effect $1 + \rho(N - 1)$ and effective sample size underlying §1's replication discount.

Berlinguette, C. P., Chiang, Y.-M., Munday, J. N., Schenkel, T., Fork, D. K., Koningstein, R. & Trevithick, M. D. *Revisiting the cold case of cold fusion*, Nature 570, 45–51 (2019) — the program null that frames §2's under-explored parameter space, DOI 10.1038/s41586-019-1256-6.

Chen, K. Y., Maiwald, J., Schauer, P. A. et al. (Berlinguette group, UBC). *Electrochemical loading enhances deuterium fusion rates in a metal target*, Nature 644, 640–645 (2025) — the $15 \pm 2\%$ loading enhancement on a calibrated beam platform, DOI 10.1038/s41586-025-09042-7.

MacLeod, B. P. et al. *Self-driving laboratory for accelerated discovery of thin-film materials*, Science Advances 6, eaaz8867 (2020) — the Ada platform anchoring §2's automated-search speculation, DOI 10.1126/sciadv.aaz8867.

Drozdov, A. P., Erements, M. I., Troyan, I. A., Ksenofontov, V. & Shylin, S. I. *Conventional superconductivity at 203 kelvin at high pressures in the sulfur hydride system*, Nature 525, 73–76 (2015) — the superhydride surprise cited in §3, DOI 10.1038/nature14964.

Steinetz, B. M. et al. *Novel nuclear reactions observed in bremsstrahlung-irradiated deuterated metals*, Physical Review C 101, 044610 (2020) — NASA's hybrid hot/cold path behind §4.

U.S. Department of Energy, Office of Science. *Report of the Review of Low Energy Nuclear Reactions*, Washington, D.C. (December 2004) — the split verdict whose unfinished business §5 proposes to close.

Chapter production note: researched and drafted by a Claude subagent (draft agent: `claude-fable-5`); every Further-reading citation web-verified against primary sources by the orchestrating lead, running as `claude-opus-4-8` after a mid-project model change (see the methods appendix). See the [methods appendix](#) for the full research process.

APPENDIX

How This Book Was Researched

A book that spends ten chapters auditing other people's methods owes the reader a full accounting of its own.

BEFORE WE START

This book was researched, written, and verified by AI language models working as an orchestrated team, on a commission that explicitly required two things unusual for a book: that the subagent model type used for each chapter or task be recorded in a footnote, and that this appendix describe in detail how the research was conducted and analyzed and by which models. This page is that accounting.

The short version: one orchestrating model (`claude-fable-5`) designed the book, dispatched ten specialist research agents and one completeness critic across the literature, wrote the conventions-anchor chapter itself, fanned the remaining chapters out to draft agents, and subjected every chapter — including its own — to an independent adversarial verification agent with live web access. All agents were instances of `claude-fable-5` . Every stage's outputs, token counts, and failure modes are reported below, including the things that did *not* work.

§1 The commission and the editorial contract

The commissioning request asked for a compilation of the latest peer-reviewed research on low-energy nuclear reactions in the house style of this workspace; the

compiler's own speculation about where the next breakthroughs will likely come from; explicit flags wherever experimental results are ahead of theory, for future research follow-up; per-chapter model accounting; and this methods appendix. The editorial contract adopted in response is the *register discipline* used throughout: every statement is presented as exactly one of established physics, contested claim (always attributed, always with the standing critique alongside, always with venue quality noted), or labeled speculation (confined to Chapter X and marked asides). The results-ahead-of-theory flags accumulate into the ledger of Chapter IX §7.

§2 Stage one — the research fan-out

Research was performed by a ten-agent parallel workflow, each agent assigned one slice of the literature and instructed to work only from live web searches and fetched sources — never from model memory — with an explicit epistemic protocol: never invent a citation; confirm bibliographic details (authors, venue, volume, pages, year, DOI/arXiv) against primary or multiple independent secondary sources; mark anything unconfirmable as **UNVERIFIED** ; record for every claim whether its venue is peer-reviewed and whether it was independently replicated; and treat negative results and published critiques as first-class findings. The ten areas: (1) baseline fusion physics; (2) the 1989 Fleischmann–Pons episode; (3) calorimetry methods and critiques; (4) the claimed nuclear-evidence base; (5) the Google-funded program; (6) NASA lattice-confinement fusion; (7) Japanese and European programs; (8) ARPA-E and recent institutional re-engagement; (9) the theory landscape; (10) adjacent validated physics for grounding speculation.

Each agent wrote a structured report (summary, key findings with full citations and replication status, canonical equations with sources, results-ahead-of-theory candidates, widget-ready data, reliability assessment). The ten reports total roughly 300 KB. An eleventh agent — a completeness critic — then read all ten summaries and ran its own searches for anything missed or wrong. It found: one missing strand (the Holmlid ultra-dense-hydrogen corpus and its peer-reviewed refutation, now a cautionary sidebar in Chapter IX); one citation conflation between two Czerski-group

papers (a 2025 *Physical Review X* measurement and a separate ion-tracks preprint), corrected before drafting; one unconfirmed program-milestone detail, softened; and two spurious search-engine leads that traced to no primary source and were discarded. Aggregate research-stage statistics: **11 agents, 1,003,659 tokens, 471 tool calls, 65 minutes wall-clock, zero agent failures.**

NOTE

The execution environment's web proxy blocked direct fetches from several publishers (nature.com, arxiv.org, some agency PDF servers) during the research window. Agents compensated by cross-checking every citation against at least two independent secondary confirmations and flagging single-source details as **UNVERIFIED** for the verification stage. This limitation is inherited honestly: a handful of page-range and article-number details in this book rest on convergent secondary sources rather than the publisher's own page.

§3 Stage two — synthesis into briefs

The orchestrating model then wrote the shared house-style specification (conventions, symbols, register rules, structural contract) and a per-chapter *equation brief*: the section outline, every load-bearing equation in exact canonical form with its named source, the constants easiest to get wrong (flagged for live verification rather than recall), the widget's physics with reference values computed in advance, and the citation list. The completeness critic's corrections were appended to the affected briefs. Briefs are the mechanism by which a fanned-out book keeps notation and accuracy coherent: drafting agents copy canonical forms rather than recalling them.

§4 Stage three — drafting and adversarial verification

Chapter I was drafted directly by the orchestrating model as the book's conventions anchor and structural gold template (its arithmetic slip — "fifty-three" orders of magnitude where the correct figure is about fifty-two — was caught during brief-writing and fixed, a small argument for the process). Chapters II–X were drafted by nine independent agents, each given the same spec, the gold template, its own brief, and the underlying research reports. Every chapter — including Chapter I — was then passed to a separate adversarial verification agent instructed to re-check every Further-reading citation against the live web, re-derive the display equations against the brief's canonical forms, audit the register discipline, confirm the KaTeX macro whitelist, execute and recompute the widget's reference values, and fix problems in place. The verification stage's findings and fixes are recorded in the book's `STATUS.md`.

§5 Stage four — mechanical validation gates

After the agent pipeline, the orchestrating model ran three empirical gates in the main loop (agent self-reports are never trusted for these): a strict KaTeX harness rendering every math expression in the book under `strict: "error"` with the book's exact macro set (catching silently-broken formulas the runtime hides); syntax checks and a JSDOM execution of every interactive widget, comparing on-screen readouts against the briefs' precomputed reference values; and a structural lint (glossaries, Where-blocks, widget wiring, navigation links, citation-section presence). Results are recorded in `STATUS.md`, including anything that remains browser-untested.

§6 Model accounting

Every role in this production ran on the same model type, `claude-fable-5` (Anthropic Claude, Fable 5), differing only in role prompt and tool access. Per-chapter footnotes record the same facts chapter-locally.

Role	Count	Model
Orchestrator / lead author (spec, briefs, Ch. I, appendix, validation)	1	claude-fable-5
Literature-research agents (areas 1–10)	10	claude-fable-5
Completeness critic (gap-check)	1	claude-fable-5
Chapter draft agents (Ch. II–X)	9	claude-fable-5
Adversarial verification agents (Ch. II–VII, IX)	7	claude-fable-5
Lead-run citation verification (Ch. I, VIII, X)	1 lead	claude-opus-4-8

The research and drafting agents, the completeness critic, and the seven completed adversarial-verification agents all ran on `claude-fable-5`. Partway through the build, however, the interactive session's model was changed to `claude-opus-4-8`, and three chapters' verification passes (Ch. I, VIII, X) had been deferred by repeated token-budget limits. Those three were therefore verified directly by the orchestrating lead *after* the switch — so their citation web-checks and this appendix's final assembly were performed as `claude-opus-4-8`, not `claude-fable-5`. We record this because a book whose subject is honest accounting of who-measured-what cannot be casual about who-wrote-what: the per-chapter footnotes name the model that actually did each chapter's verification, and they do not all say the same thing.

§7 Honest limitations

Four limitations the reader should carry. First, the proxy constraint of §2: a minority of bibliographic details rest on convergent secondary sources. Second, this is a

compilation dated **July 2026**; in a field currently producing genuine peer-reviewed movement (the 2025 *Nature* and *PRX* results of Chapters VII–VIII), its shelf life is short. Third, the interactive widgets are verified to execute correctly and reproduce their reference values in a headless DOM; live pointer interactions are verified only as recorded in `STATUS.md`. Fourth — and most fundamental — the compilers are language models: the same adversarial-verification-plus-mechanical-gates machinery described here is our mitigation for, not an immunity to, the failure modes any single mind brings to a literature this contested. Where we have judged, Chapter IX shows the arithmetic of the judgment; where we have speculated, Chapter X says so in the first sentence.

FURTHER READING

This workspace. *STATUS.md for this book* — the chapter-by-chapter validation record, conventions in force, and outstanding passes.

Berlinguette, C. P. et al. *Revisiting the cold case of cold fusion*, *Nature* 570, 45–51 (2019) — the model for publish-regardless program design that this book's methods imitate at book scale.

U.S. DOE. *Report of the Review of Low Energy Nuclear Reactions* (2004) — the template for structured, charge-question-driven evidence review.

Appendix production note: written directly by the orchestrating lead model (`claude-fable-5`) from the workflow telemetry and build records of this session.